

1510.06699: formula evidence

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Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00001	1	0.755	$\{ \}^{\{1,2, \mathrm{a}\}}$	1,2,a	A.N. Lasenby ^{1, 2, a)} and M.P. Hobson ^{1, b)}
1510.06699_F00002	1	0.755	$\{ \}^{\{1, \mathrm{~b}\}}$	1, b	A.N. Lasenby ^{1, 2, a)} and M.P. Hobson ^{1, b)}
1510.06699_F00003	1	0.969	$\{ \}^{\{1\}}$	1)	¹⁾ <i>Astrophysics Group, Cavendish Laboratory, JJ Thomson Avenue, Cambridge CB3 0HE, UK</i>
1510.06699_F00004	1	0.979	$\{ \}^{\{2\}}$	2)	²⁾ <i>Kavli Institute for Cosmology, Madingley Road, Cambridge CB3 0HA, UK</i>
1510.06699_F00005	1	0.947	$\{ \}^{\{\underline{1}\}}$	<u>1</u>	Following an initial attempt by Utiyama ¹ , the idea
1510.06699_F00006	1	0.998	φ	φ	(or fields) φ with energy-momentum and, in general,
1510.06699_F00007	1	0.798	$\{ \}^{\{\text{a} \} \}$	a)	^{a)} Electronic mail: a.n.lasenby@mrao.cam.ac.uk
1510.06699_F00008	1	0.997	$\{ \}^{\{\text{b} \} \}$	b)	^{b)} Electronic mail: mph@mrao.cam.ac.uk
1510.06699_F00009	1	1.000	$S_{\mathrm{M}}=$	$S_{\mathrm{M}}=$	field dynamics are described by a matter action $S_{\mathrm{M}}=$
1510.06699_F00010	1	1.000	$\int L_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) d^4 x$	$\int L_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) d^4 x$	$\int L_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) d^4 x$ that is invariant under global Poincaré
1510.06699_F00011	1	1.000	h_{a}^{μ}	h_a^{μ}	new field variables h_a^{μ} and A_{μ}^{ab} , which are interpreted
1510.06699_F00012	1	1.000	$A^{\mathrm{a} \mathrm{b}}_{\mu}$	A_{μ}^{ab}	new field variables h_a^{μ} and A_{μ}^{ab} , which are interpreted
1510.06699_F00013	1	0.990	$\mathcal{D}_{\mathrm{a}} \varphi$	$\mathcal{D}_a \varphi$	a covariant derivative $\mathcal{D}_a \varphi$, and the matter action in the
1510.06699_F00014	1	1.000	$S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_{\mathrm{a}} \varphi) d^4 x$	$S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi) d^4 x$	to obtain $S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi) d^4 x$ (where the factor
1510.06699_F00015	1	1.000	$h \equiv \operatorname{det}(\mathcal{h}_{\mathrm{a}}^{\mu})$	$h \equiv \det(h_a^{\mu})$	containing $h \equiv \det(h_a^{\mu})$ is required to make the inte-
1510.06699_F00016	2	1.000	$[\mathcal{D}_{\mathrm{a}}, \mathcal{D}_{\mathrm{b}}] \varphi$	$[\mathcal{D}_a, \mathcal{D}_b] \varphi$	tering $[\mathcal{D}_a, \mathcal{D}_b] \varphi$. This procedure yields two field strength
1510.06699_F00017	2	1.000	$\mathcal{R}^{\mathrm{a} \mathrm{b}}_{\mathrm{cd}}(h, A, \partial A)$	$\mathcal{R}_{cd}^{ab}(h, A, \partial A)$	tensors: $\mathcal{R}_{cd}^{ab}(h, A, \partial A)$ corresponding to the gauge field
1510.06699_F00018	2	0.999	$\mathcal{T}^{\mathrm{a}}_{\mathrm{bc}}(h, \partial h, A)$	$\mathcal{T}_{bc}^a(h, \partial h, A)$	Poincaré group; and $\mathcal{T}_{bc}^a(h, \partial h, A)$ corresponding to
1510.06699_F00019	2	1.000	$S_{\mathrm{G}}=\int h^{-1} L_{\mathrm{G}}(\mathcal{R}^{\mathrm{a} \mathrm{b}}_{\mathrm{cd}}, \mathcal{T}^{\mathrm{a}}_{\mathrm{bc}}) d^4 x$	$S_{\mathrm{G}}=\int h^{-1} L_{\mathrm{G}}(\mathcal{R}_{cd}^{ab}, \mathcal{T}_{bc}^a) d^4 x$	$S_{\mathrm{G}}=\int h^{-1} L_{\mathrm{G}}(\mathcal{R}_{cd}^{ab}, \mathcal{T}_{bc}^a) d^4 x$, where the Lagrangian
1510.06699_F00020	2	1.000	L_{G}	L_{G}	L_{G} is a scalar depending on the two field strengths.
1510.06699_F00021	2	1.000	$\tau^{\mathrm{k}}_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_{\mathrm{a}}^{\mu}$	$\tau^k_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_a^{\mu}$	energy-momentum $\tau^k_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_a^{\mu}$ and spin-angular-
1510.06699_F00022	2	1.000	$\sigma_{\mathrm{ab}}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta A_{\mu}^{\mathrm{ab}}$	$\sigma_{ab}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta A_{\mu}^{ab}$	momentum $\sigma_{ab}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta A_{\mu}^{ab}$ of the matter field act
1510.06699_F00023	2	1.000	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_{\mathrm{a}} \varphi)$	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi)$	as sources, where $\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi)$. In contrast to
1510.06699_F00024	2	1.000	$\mathcal{R} \equiv \mathcal{R}^{\mathrm{ab}}_{\mathrm{ab}}$	$\mathcal{R} \equiv \mathcal{R}_{ab}^{ab}$	derivatives of the gauge fields, namely $\mathcal{R} \equiv \mathcal{R}_{ab}^{ab}$. In-
1510.06699_F00025	2	1.000	$L_{\mathrm{G}} \propto \mathcal{R}$	$L_{\mathrm{G}} \propto \mathcal{R}$	deed, Kibble chose simply to set $L_{\mathrm{G}} \propto \mathcal{R}$, which defines
1510.06699_F00026	2	0.733	$\{ \}^{\{\underline{5}\}}$	<u>5</u>	transformations has been criticised by Hehl et al. ⁵ , who
1510.06699_F00027	2	1.000	$\delta \varphi \equiv \varphi'(\boldsymbol{x}) - \varphi(\boldsymbol{x})$	$\delta \varphi \equiv \varphi'(x') - \varphi(x)$	ble considered the ‘total’ variation $\delta \varphi \equiv \varphi'(x') - \varphi(x)$ of
1510.06699_F00028	2	1.000	$\delta_0 \varphi \equiv \varphi'(\boldsymbol{x}) - \varphi(\boldsymbol{x})$	$\delta_0 \varphi \equiv \varphi'(x) - \varphi(x)$	ation $\delta_0 \varphi \equiv \varphi'(x) - \varphi(x)$. The latter approach is closer
1510.06699_F00029	2	1.000	$\delta_0 \varphi$	$\delta_0 \varphi$	matter field transformation $\delta_0 \varphi$, the partial derivative in
1510.06699_F00030	2	0.988	$\{ \}^{\{6,9\}}$	6,9	theory in Minkowski spacetime ^{6,9} , in the same way as the
1510.06699_F00031	2	0.991	$\{ \}^{\{\underline{10}\}}$	<u>10</u>	see Blagojevic ¹⁰ , in which many of the issues above are
1510.06699_F00032	2	1.000	$\mathcal{R}^{\mathrm{ab}}_{\mathrm{cd}}$	$\mathcal{R}_{cd}^{ab}$	sors $\mathcal{R}_{cd}^{ab}$ and $\mathcal{T}_{bc}^a$, and their inclusion in L_{G} cannot
1510.06699_F00033	2	1.000	$\mathcal{T}^{\mathrm{a}}_{\mathrm{bc}}$	$\mathcal{T}_{bc}^a$	sors $\mathcal{R}_{cd}^{ab}$ and $\mathcal{T}_{bc}^a$, and their inclusion in L_{G} cannot
1510.06699_F00034	2	1.000	$\{ \}^{\{11,12\}}$	11,12	adding the linear Hilbert-Einstein term ^{11,12} . Hence, one
1510.06699_F00035	2	1.000	$\kappa=8 \pi G / c^4$	$\kappa=8 \pi G / c^4$	where $\kappa=8 \pi G / c^4$ is Einstein’s gravitational constant,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00036	2	■ 1.000	\Lambda	Λ	Λ is a cosmological constant, a is a dimensionless free
1510.06699_F00037	2	■ 1.000	a	a	Λ is a cosmological constant, a is a dimensionless free
1510.06699_F00038	3	■ 1.000	\alpha_{i}	α_i	in which the α_i and β_i are also dimensionless free parame-
1510.06699_F00039	3	■ 1.000	\beta_{i}	β_i	in which the α_i and β_i are also dimensionless free parame-
1510.06699_F00040	3	■ 0.362	{ }^{\underline{12,14}}-\underline{16}	$\underline{12,14-16}$	investigated previously ^{12,14-16} and may, in principle, be
1510.06699_F00041	3	■ 1.000	\alpha_{1}, \alpha_{3}	α_1, α_3	alised Gauss–Bonnet identity to set one of α_1 , α_3 or α_6
1510.06699_F00042	3	■ 1.000	\alpha_{6}	α_6	alised Gauss–Bonnet identity to set one of α_1 , α_3 or α_6
1510.06699_F00043	3	■ 0.999	c=\hbar=1	$c=\hbar=1$	in the form (1) so that, in natural units ($c=\hbar=1$,
1510.06699_F00044	3	■ 0.944	{ }^{-4}	-4	(length) ^{−4} , and hence the corresponding action S_G is di-
1510.06699_F00045	3	■ 0.944	S_{\mathrm{G}}	S_G	(length) ^{−4} , and hence the corresponding action S_G is di-
1510.06699_F00046	3	■ 1.000	\kappa	κ	dimensionless; note that in natural units κ has dimensions
1510.06699_F00047	3	■ 0.986	{ }^{-2}	-2	of (length) ² and Λ has dimensions of (length) ^{−2} .
1510.06699_F00048	3	■ 0.585	{ }^{19,20}	19,20	suffer from Ostrogradsky’s instability ^{19,20} , wherein the
1510.06699_F00049	3	■ 1.000	b^a{a}{ }_{\mu}	$b^a{}_{\mu}$	$b^a{}_{\mu}$ of the inverse h -field) and $A^{ab}{}_{\mu}$, the ‘curvature’ and
1510.06699_F00050	3	■ 1.000	h	h	$b^a{}_{\mu}$ of the inverse h -field) and $A^{ab}{}_{\mu}$, the ‘curvature’ and
1510.06699_F00051	3	■ 1.000	b^a{a}{ }_{\emptyset}	$b^a{}_{\emptyset}$	derivatives of $b^a{}_{\emptyset}$ and $A^{ab}{}_{\emptyset}$; thus 10 of the field equa-
1510.06699_F00052	3	■ 1.000	A^a b{ }_{\emptyset}	$A^{ab}{}_{\emptyset}$	derivatives of $b^a{}_{\emptyset}$ and $A^{ab}{}_{\emptyset}$; thus 10 of the field equa-
1510.06699_F00053	3	■ 1.000	b^a{a}{ }_{\alpha}	$b^a{}_{\alpha}$	remaining 30 variables $b^a{}_{\alpha}$ and $A^{ab}{}_{\alpha}$ ($\alpha=1,2,3$), one
1510.06699_F00054	3	■ 1.000	A^a b{ }_{\alpha}(\alpha=1,2,3)	$A^{ab}{}_{\alpha}(\alpha=1,2,3)$	remaining 30 variables $b^a{}_{\alpha}$ and $A^{ab}{}_{\alpha}$ ($\alpha=1,2,3$), one
1510.06699_F00055	3	■ 1.000	\mathcal{R}+\mathcal{T}^2	$\mathcal{R}+\mathcal{T}^2$	so-called $\mathcal{R}+\mathcal{T}^2$ theory, for which $L_G=\kappa^{-1}(a\mathcal{R}+L_{\mathcal{T}^2})$.
1510.06699_F00056	3	■ 1.000	L_{\mathrm{G}}=\kappa^{-1}\left(a \mathcal{R}+L_{\mathcal{T}^2}\right)	$L_G=\kappa^{-1}(a\mathcal{R}+L_{\mathcal{T}^2})$	so-called $\mathcal{R}+\mathcal{T}^2$ theory, for which $L_G=\kappa^{-1}(a\mathcal{R}+L_{\mathcal{T}^2})$.
1510.06699_F00057	3	■ 1.000	A	A	namics imposes an algebraic relation between the A -field
1510.06699_F00058	3	■ 1.000	L_{\mathcal{R}^2}	$L_{\mathcal{R}^2}$	in $L_{\mathcal{R}^2}$, in which case the ‘torsion’ becomes a propagating
1510.06699_F00059	3	■ 1.000	\mathcal{R}+\mathcal{R}^2	$\mathcal{R}+\mathcal{R}^2$	field. Consequently, a number of $\mathcal{R}+\mathcal{R}^2$ -type theories
1510.06699_F00060	3	■ 0.970	\mathcal{R}^2+\mathcal{T}^2	$\mathcal{R}^2+\mathcal{T}^2$	have been investigated, as have $\mathcal{R}^2+\mathcal{T}^2$ theories ²¹ . It
1510.06699_F00061	3	■ 0.986	\mathcal{R}+\mathcal{R}^2+\mathcal{T}^2	$\mathcal{R}+\mathcal{R}^2+\mathcal{T}^2$	$\mathcal{R}+\mathcal{R}^2+\mathcal{T}^2$ theories is given by Obukhov et al. ¹² . The
1510.06699_F00062	3	■ 1.000	\mathcal{R}	$\mathcal{R}$	if L_G consists of $\mathcal{R}$ and $\mathcal{T}^2$ terms, then it contains dimen-
1510.06699_F00063	3	■ 1.000	\mathcal{T}^2	$\mathcal{T}^2$	if L_G consists of $\mathcal{R}$ and $\mathcal{T}^2$ terms, then it contains dimen-
1510.06699_F00064	3	■ 1.000	\mathcal{R}^2	$\mathcal{R}^2$	quantisation. On the other hand, the $\mathcal{R}^2$ terms contain
1510.06699_F00065	3	■ 1.000	{ }^{\underline{25}}	$\underline{25}$	can eliminate such higher derivatives from the theory ²⁵ ,
1510.06699_F00066	3	■ 0.881	{ }^{\underline{26}}	$\underline{26}$	well-defined initial value problem ²⁶ . Nonetheless, the
1510.06699_F00067	3	■ 0.995	a, \alpha_{i}	a, α_i	that the parameters a , α_i and β_i in the Lagrangian L_G
1510.06699_F00068	3	■ 0.924	{ }^{\underline{27}}	$\underline{27}$	obey certain conditions ²⁷ .
1510.06699_F00069	4	■ 1.000	h_a^{\mu}	h_a^{μ}	Scale-invariant gauge theories of grav
1510.06699_F00070	4	■ 1.000	L_{\mathrm{G}}^{\left(2\right)}	$L_G^{(2)}$	the proposed group, observational laboratory, or the
1510.06699_F00071	4	■ 1.000	k	k	²⁾ Kavli Institute for Cosmology, Madingley Road, Ca
1510.06699_F00072	4	■ 1.000	J^P	J^P	Poincare invariance to include invariance under .
1510.06699_F00073	4	■ 1.000	k^a	k^a	transformations which allow for a more freedom
1510.06699_F00074	4	■ 0.998	J^P=2^{+}	$J^P=2^{+}$	in Minkowski spacetime. Our approach therefore
1510.06699_F00075	4	■ 1.000	J^P=2^{\pm}, 1^{\pm}, 0^{\pm}	$J^P=2^{\pm}, 1^{\pm}, 0^{\pm}$	scale-invariant Poincaré gauge theory (PGT) and
1510.06699_F00076	4	■ 1.000	2(5+3+1)=18	$2(5+3+1)=18$	scale-invariant Poincaré gauge theory (PGT) and
1510.06699_F00077	4	■ 1.000	S	S	and Weyl–Cartan connections, respectively. In a
1510.06699_F00078	4	■ 1.000	k^2	k^2	general invariance properties than those imposed
1510.06699_F00079	4	■ 1.000	-R/\left(k^2-m^2\right)	$-R/(k^2-m^2)$	WGT by considering an ‘extended’ form for the
1510.06699_F00080	4	■ 1.000	R>0	$R>0$	dilations, which includes its ‘normal’ transforma
1510.06699_F00081	4	■ 1.000	m^2 \geq 0	$m^2\geq 0$	Weyl gauge theory (eWGT) has a number of inte
1510.06699_F00082	4	■ 1.000	{ }^{18,33}	18,33	weyl gauge theory (eWGT) has a number of inte
1510.06699_F00083	4	■ 1.000	J^P=1^{-}	$J^P=1^{-}$	show that they are special cases of WGT and ev
					fundamental interactions, the gauge principle has proved
					very successful. The electromagnetic, weak and strong
					it is natural to consider the gauge group description of
					gravitation. In the gravitational case, however, in place
					it is natural to consider the gauge group description of
					gravitation. In the gravitational case, however, in place
					of gauging an internal compact semi-simple Lie symmetry
					Following an initial attempt by Utiyama ³ , the idea
					of gauging the Poincaré group was fully developed by
					Kibble ⁴ (and also considered by Sciama ⁵). The phys-
					ical model envisaged in Kibble’s approach is an un-

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1510.06699_F00123	5	■ 0.803	$\phi^{\{2\}} \mathcal{R}$	$\phi^2 \mathcal{R}$	ample, terms proportional to $\phi^2 \mathcal{R}$ or $\phi^2 L_{\mathcal{T}+2}$ are Weyl-
1510.06699_F00124	5	■ 0.803	$\phi^{\{2\}} L_{\mathcal{T}+2}$	$\phi^2 L_{\mathcal{T}+2}$	ample, terms proportional to $\phi^2 \mathcal{R}$ or $\phi^2 L_{\mathcal{T}+2}$ are Weyl-
1510.06699_F00125	5	■ 1.000	$\{ \}^{\{54,55\}}$	54,55	been widely investigated ^{54,55} , whereas the phenomeno-
1510.06699_F00126	5	■ 0.999	$\phi^{\{2\}} L_{\mathcal{T}+2}$	$\phi^2 L_{\mathcal{T}+2}$	logical value of the $\phi^2 L_{\mathcal{T}+2}$ terms has received relatively
1510.06699_F00127	5	■ 0.969	$\{ \}^{\{56,57\}}$	56,57	little attention ^{56,57} . In any case, the resulting total La-
1510.06699_F00128	5	■ 1.000	ψ	ψ	tation for a free Dirac field ψ , which has Weyl weight
1510.06699_F00129	5	■ 1.000	$w(\psi)=w(\bar{\psi})=-\frac{3}{2}$	$w(\psi) = w(\bar{\psi}) = -\frac{3}{2}$	$w(\psi) = w(\bar{\psi}) = -\frac{3}{2}$. This action is not scale-invariant
1510.06699_F00130	5	■ 1.000	$m \bar{\psi} \psi$	$m \bar{\psi} \psi$	owing to the mass term $m \bar{\psi} \psi$, so it appears that one re-
1510.06699_F00131	5	■ 0.983	$\{ \}^{\{10\}} \zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu$	$^{10} \zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu$	vergence of the dilation current ¹⁰ $\zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu$. In
1510.06699_F00132	6	■ 1.000	$m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$	$m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$	replacement $m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$ in the Dirac action, where μ
1510.06699_F00133	6	■ 1.000	μ	μ	replacement $m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$ in the Dirac action, where μ
1510.06699_F00134	6	■ 1.000	$\mu \phi$	$\mu \phi$	is a dimensionless parameter but $\mu \phi$ has the dimensions
1510.06699_F00135	6	■ 1.000	G	G	‘constant’ G to vary with cosmic epoch. This idea can
1510.06699_F00136	6	■ 1.000	$\phi = \phi_0$	$\phi = \phi_0$	stein gauge. Setting $\phi = \phi_0$ in the equation of motion
1510.06699_F00137	6	■ 1.000	$m = \mu \phi_0$	$m = \mu \phi_0$	tation as a massive field with $m = \mu \phi_0$. It is usually
1510.06699_F00138	6	■ 1.000	w	w	$h_a{}^\mu$ and $A^{ab}{}_\mu$ transform covariantly with weights of w ,
1510.06699_F00139	6	■ 1.000	$\mathcal{R}_{abcd}$	$\mathcal{R}_{abcd}$	strength (or ‘curvature’) $\mathcal{R}_{abcd}$ and translational gauge
1510.06699_F00140	6	■ 1.000	$\mathcal{T}_{abc}$	$\mathcal{T}_{abc}$	field strength (or ‘torsion’) $\mathcal{T}_{abc}$ are treated in a more
1510.06699_F00141	6	■ 0.617	$\{ \}^{\{59-62,89\}}$	$^{59-62,89}$	so is the total action) ^{59-62,89} .
1510.06699_F00142	6	■ 1.000	Y_4	Y_4	Weyl–Cartan Y_4 spacetime (although we do give a brief
1510.06699_F00143	7	■ 1.000	$\widehat{Y}_4$	$\widehat{Y}_4$	a new spacetime geometry (that we term $\widehat{Y}_4$), which is
1510.06699_F00144	7	■ 1.000	$\mathcal{M}$	$\mathcal{M}$	We begin with a Minkowski spacetime $\mathcal{M}$, labelled
1510.06699_F00145	7	■ 1.000	x^μ	x^μ	using Cartesian inertial coordinates x^μ , on which the
1510.06699_F00146	7	■ 1.000	$\varphi(x)$	$\varphi(x)$	gravitational) fields $\varphi(x)$ is described by a matter action
1510.06699_F00147	7	■ 0.959	e_μ	e_μ	e_μ , which are tangents to the coordinate curves and for
1510.06699_F00148	7	■ 0.997	$e_\mu \cdot e_\nu = \eta_{\mu\nu}$	$e_\mu \cdot e_\nu = \eta_{\mu\nu}$	Cartesian inertial coordinates satisfy $e_\mu \cdot e_\nu = \eta_{\mu\nu}$, where
1510.06699_F00149	7	■ 1.000	$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$	$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$	$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$. At each point of $\mathcal{M}$, we can
1510.06699_F00150	7	■ 1.000	$\hat{e}_a(x)$	$\hat{e}_a(x)$	mal tetrad $\hat{e}_a(x)$, such that $\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$. In global
1510.06699_F00151	7	■ 1.000	$\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$	$\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$	mal tetrad $\hat{e}_a(x)$, such that $\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$. In global
1510.06699_F00152	7	■ 1.000	$e_\mu(x)$	$e_\mu(x)$	$e_\mu(x)$, i.e. $\hat{e}_a = \delta_a^\mu e_\mu$. We will adopt the common con-
1510.06699_F00153	7	■ 1.000	$\hat{e}_a = \delta_a^\mu e_\mu$	$\hat{e}_a = \delta_a^\mu e_\mu$	$e_\mu(x)$, i.e. $\hat{e}_a = \delta_a^\mu e_\mu$. We will adopt the common con-
1510.06699_F00154	7	■ 1.000	$\text{SO}(3,1)$	$\text{SO}(3,1)$	$\text{SO}(3,1)$ by $\text{SL}(2, C)$, so that spinor fields can be
1510.06699_F00155	7	■ 1.000	$\text{SL}(2, C)$	$\text{SL}(2, C)$	$\text{SO}(3,1)$ by $\text{SL}(2, C)$, so that spinor fields can be
1510.06699_F00156	7	■ 1.000	$\{\varphi^a(x)\}$	$\{\varphi^a(x)\}$	ter field components $\{\varphi^a(x)\}$ simply carry a generic Ro-
1510.06699_F00157	8	■ 0.999	$S(\Lambda)$	$S(\Lambda)$	where $S(\Lambda)$ is the matrix corresponding to the element Λ
1510.06699_F00158	8	■ 1.000	$\Lambda^\mu{}_\nu = \delta^\mu_\nu$	$\Lambda^\mu{}_\nu = \delta^\mu_\nu$	formation, i.e. with $\Lambda^\mu{}_\nu = \delta^\mu_\nu$ and $a^\mu = 0$ in (6) and
1510.06699_F00159	8	■ 1.000	$a^\mu = 0$	$a^\mu = 0$	formation, i.e. with $\Lambda^\mu{}_\nu = \delta^\mu_\nu$ and $a^\mu = 0$ in (6) and
1510.06699_F00160	8	■ 1.000	e^ρ	e^ρ	tor e^ρ . Adopting global inertial coordinates, such that
1510.06699_F00161	8	■ 0.999	$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$	$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$	$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$, the metric must therefore transform
1510.06699_F00162	8	■ 1.000	$\eta_{\mu\nu} \rightarrow \gamma_{\mu\nu} = e^{2\rho} \eta_{\mu\nu}$	$\eta_{\mu\nu} \rightarrow \gamma_{\mu\nu} = e^{2\rho} \eta_{\mu\nu}$	according to $\eta_{\mu\nu} \rightarrow \gamma_{\mu\nu} = e^{2\rho} \eta_{\mu\nu}$. Hence, under this
1510.06699_F00163	8	■ 1.000	$w=2$	$w = 2$	active dilation the metric has a Weyl weight of $w = 2$.
1510.06699_F00164	8	■ 1.000	$x'^\mu = e^\rho x^\mu$	$x'^\mu = e^\rho x^\mu$	ordinate transformation $x'^\mu = e^\rho x^\mu$, which ‘shrinks’ the
1510.06699_F00165	8	■ 1.000	$\gamma'_{\mu\nu} = e^{-2\rho} \gamma_{\mu\nu} = \eta_{\mu\nu}$	$\gamma'_{\mu\nu} = e^{-2\rho} \gamma_{\mu\nu} = \eta_{\mu\nu}$	coordinate transformation, $\gamma'_{\mu\nu} = e^{-2\rho} \gamma_{\mu\nu} = \eta_{\mu\nu}$ and so,
1510.06699_F00166	8	■ 1.000	$e^{4\rho}$	$e^{4\rho}$	transformation (6) (for which the Jacobian equals $e^{4\rho}$), a

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1510.06699_F00167	8	■ 1.000	m	m	m , one has the Klein–Gordon Lagrangian
1510.06699_F00168	8	■ 1.000	m=0	$m = 0$	dilation invariant only for $m = 0$ and provided the Weyl
1510.06699_F00169	8	■ 1.000	$\bar{\psi} \psi$	$\bar{\psi} \psi$	pretation of $\bar{\psi} \psi$ as a number density. As mentioned in the
1510.06699_F00170	8	■ 1.000	A_{μ}	A_{μ}	Finally, we consider the electromagnetic field A_{μ}
1510.06699_F00171	8	■ 1.000	$F_{\mu\nu} \equiv \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$	$F_{\mu\nu} \equiv \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$	where $F_{\mu\nu} \equiv \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$ and J^{μ} is some source cur-
1510.06699_F00172	8	■ 1.000	J^{μ}	J^{μ}	where $F_{\mu\nu} \equiv \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$ and J^{μ} is some source cur-
1510.06699_F00173	8	■ 1.000	w_J	w_J	where w and w_J are the Weyl weights of A_{μ} and J_{μ}
1510.06699_F00174	8	■ 1.000	J_{μ}	J_{μ}	where w and w_J are the Weyl weights of A_{μ} and J_{μ}
1510.06699_F00175	8	■ 1.000	$w(A_{\mu}) = -1$	$w(A_{\mu}) = -1$	(9), provided that $w(A_{\mu}) = -1$ and $w(J_{\mu}) = -3$. It is
1510.06699_F00176	8	■ 1.000	$w(J_{\mu}) = -3$	$w(J_{\mu}) = -3$	(9), provided that $w(A_{\mu}) = -1$ and $w(J_{\mu}) = -3$. It is
1510.06699_F00177	8	■ 1.000	$J^{\mu} \equiv \bar{\psi} \gamma^{\mu} \psi$	$J^{\mu} \equiv \bar{\psi} \gamma^{\mu} \psi$	worth noting that the Dirac current $J^{\mu} \equiv \bar{\psi} \gamma^{\mu} \psi$ satisfies
1510.06699_F00178	9	■ 1.000	a^{μ}	a^{μ}	the four translation parameters a^{μ} and the dilation pa-
1510.06699_F00179	9	■ 1.000	ρ	ρ	rameter ρ into eleven independent arbitrary functions of
1510.06699_F00180	9	■ 1.000	x	x	x . This leads to a complete decoupling of the translation
1510.06699_F00181	9	■ 1.000	x'	x'	lation parts. Thus, it is customary to view x' as arising
1510.06699_F00182	9	■ 1.000	$\Lambda(x)$	$\Lambda(x)$	local Lorentz rotation $\Lambda(x)$ of the orthonormal tetrad
1510.06699_F00183	9	■ 0.997	$\rho(x)$	$\rho(x)$	$\hat{e}_a(x)$, or local (physical) dilation $\rho(x)$, at each point.
1510.06699_F00184	9	■ 1.000	$\Lambda^a{}_b(x)$	$\Lambda^a{}_b(x)$	Lorentz rotations $\Lambda^a{}_b(x)$. This means that any spinor,
1510.06699_F00185	9	■ 1.000	$\partial_{\mu} \varphi$	$\partial_{\mu} \varphi$	field. Consequently, the partial derivative, $\partial_{\mu} \varphi$, is co-
1510.06699_F00186	9	■ 1.000	$\eta^{ab} = \text{diag}(1, -1, -1, -1)$	$\eta_{ab} = \eta^{ab} = \text{diag}(1, -1, -1, -1)$	ric $\eta_{ab} = \eta^{ab} = \text{diag}(1, -1, -1, -1)$, whereas Greek in-
1510.06699_F00187	9	■ 1.000	$\gamma_{\mu\nu} = e_{\mu} \cdot e_{\nu}$	$\gamma_{\mu\nu} = e_{\mu} \cdot e_{\nu}$	ric $\gamma_{\mu\nu} = e_{\mu} \cdot e_{\nu}$ and its inverse $\gamma^{\mu\nu}$ corresponding to
1510.06699_F00188	9	■ 1.000	$\gamma^{\mu\nu}$	$\gamma^{\mu\nu}$	ric $\gamma_{\mu\nu} = e_{\mu} \cdot e_{\nu}$ and its inverse $\gamma^{\mu\nu}$ corresponding to
1510.06699_F00189	9	■ 1.000	$\hat{e}^a = \eta^{ab} \hat{e}_b$	$\hat{e}^a = \eta^{ab} \hat{e}_b$	by $\hat{e}^a = \eta^{ab} \hat{e}_b$ and $e^{\mu} = \gamma^{\mu\nu} e_{\nu}$, respectively. Moreover,
1510.06699_F00190	9	■ 1.000	$e^{\mu} = \gamma^{\mu\nu} e_{\nu}$	$e^{\mu} = \gamma^{\mu\nu} e_{\nu}$	by $\hat{e}^a = \eta^{ab} \hat{e}_b$ and $e^{\mu} = \gamma^{\mu\nu} e_{\nu}$, respectively. Moreover,
1510.06699_F00191	9	■ 1.000	$\gamma_{\mu\nu}$	$\gamma_{\mu\nu}$	metric $\gamma_{\mu\nu}$ of the Minkowski spacetime obeys the trans-
1510.06699_F00192	9	■ 1.000	(Λ, ρ)	(Λ, ρ)	two steps. Firstly, one defines the (Λ, ρ) -covariant deriva-
1510.06699_F00193	9	■ 1.000	$D_{\mu} \varphi(x)$	$D_{\mu} \varphi(x)$	where $D_{\mu} \varphi(x)$ is the Λ -covariant derivative introduced
1510.06699_F00194	9	■ 1.000	$A^{ab}{}_{\mu}(x)$	$A^{ab}{}_{\mu}(x)$	of complex conjugation. Most importantly, $A^{ab}{}_{\mu}(x)$ is
1510.06699_F00195	9	■ 1.000	$B_{\mu}(x)$	$B_{\mu}(x)$	tions and $B_{\mu}(x)$ is the gauge-field corresponding to local
1510.06699_F00196	9	■ 1.000	$\Sigma_{ab} = -\Sigma_{ab}$	$\Sigma_{ab} = -\Sigma_{ab}$	dilations. The matrices $\Sigma_{ab} = -\Sigma_{ab}$ are the generators of
1510.06699_F00197	9	■ 0.483	$(2, C)$	$(2, C)$	the $\text{SL}(2, C)$ representation $\text{S}(\Lambda)$ to which $\varphi(x)$ belongs,
1510.06699_F00198	9	■ 0.483	(Λ)	(Λ)	the $\text{SL}(2, C)$ representation $\text{S}(\Lambda)$ to which $\varphi(x)$ belongs,
1510.06699_F00199	9	■ 1.000	Σ_{ab}	Σ_{ab}	Given the antisymmetry of the generator matrices Σ_{ab} ,
1510.06699_F00200	9	■ 1.000	$A^{ab}{}_{\mu} =$	$A^{ab}{}_{\mu} =$	to share this antisymmetry property, such that $A^{ab}{}_{\mu} =$
1510.06699_F00201	9	■ 0.890	$-A^{ba}{}_{\mu}$	$-A^{ba}{}_{\mu}$	$-A^{ba}{}_{\mu}$. The aim of introducing the A and B gauge fields
1510.06699_F00202	9	■ 0.890	B	B	$-A^{ba}{}_{\mu}$. The aim of introducing the A and B gauge fields
1510.06699_F00203	9	■ 1.000	$w=0$	$w = 0$	local dilations (i.e. has Weyl weight $w = 0$).
1510.06699_F00204	10	■ 1.000	$\partial_{\mu}^* \varphi(x)$	$\partial_{\mu}^* \varphi(x)$	It is straightforward to show that $\partial_{\mu}^* \varphi(x)$ itself also trans-
1510.06699_F00205	10	■ 0.999	$D_{\mu}^* \varphi$	$D_{\mu}^* \varphi$	derivative, linearly related to $D_{\mu}^* \varphi$, by introducing a new,
1510.06699_F00206	10	■ 1.000	$h_a{}^{\mu}(x)$	$h_a{}^{\mu}(x)$	‘translational’ gauge field $h_a{}^{\mu}(x)$ that transforms via
1510.06699_F00207	10	■ 1.000	$w = -1$	$w = -1$	Lorentz four-vector, and has Weyl weight $w = -1$ under
1510.06699_F00208	10	■ 1.000	$b^a{}_{\mu}(x)$	$b^a{}_{\mu}(x)$	$b^a{}_{\mu}(x)$ (which has Weyl weight $w = 1$) and satisfies
1510.06699_F00209	10	■ 1.000	$w=1$	$w = 1$	$b^a{}_{\mu}(x)$ (which has Weyl weight $w = 1$) and satisfies
1510.06699_F00210	10	■ 1.000	$\partial \varphi$	$\partial \varphi$	mations in the same way as $\partial \varphi$ does under global Weyl
1510.06699_F00211	10	■ 1.000	$x \rightarrow x'$	$x \rightarrow x'$	$w = -4$. However, since $x \rightarrow x'$ is a GCT, we need a

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1510.06699_F00212	10	■ 0.997	$L_{\mathrm{M}}(\varphi; \mathcal{D}^*\varphi)$	$L_{\mathrm{M}}(\varphi; \mathcal{D}^*\varphi)$	$L_{\mathrm{M}}(\varphi; \mathcal{D}^*\varphi)$ by a factor which transforms by the appro-
1510.06699_F00213	10	■ 1.000	$w=4$	$w = 4$	weight $w = 4$. From the transformation properties of the
1510.06699_F00214	10	■ 1.000	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi)$	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi)$	by the Lagrangian density $\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi)$. We
1510.06699_F00215	10	■ 0.999	h, A	h, A	note that the gravitational gauge fields h, A and B con-
1510.06699_F00216	10	■ 0.613	$L_{\mathcal{T}*2}$	$L_{\mathcal{T}*2}$	weight $w = -4$ and so may be added to L_{M} (where $L_{\mathcal{T}*2}$
1510.06699_F00217	10	■ 0.999	D_{μ}^*	D_{μ}^*	choice, one may extend the (Λ, ρ) -covariant derivative D_{μ}^*
1510.06699_F00218	10	■ 1.000	$\left(\Sigma_{cd}^1\right)^a{}_b = (\delta_c^a \eta_{db} - \delta_d^a \eta_{cb})$	$(\Sigma_{cd}^1)^a{}_b = (\delta_c^a \eta_{db} - \delta_d^a \eta_{cb})$	where $(\Sigma_{cd}^1)^a{}_b = (\delta_c^a \eta_{db} - \delta_d^a \eta_{cb})$ are the generators of the
1510.06699_F00219	10	■ 0.509	C	C	der GCT, in addition to $\mathrm{SL}(2, C)$ behaviour. In this case,
1510.06699_F00220	11	■ 1.000	$\{ \}^{\{0\}} \nabla_{\mu}^* \equiv \nabla_{\mu} + w B_{\mu}, {}^0\Gamma^{\lambda}{}_{\mu\nu} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} + \partial_{\nu} \gamma_{\mu\rho} -$	${}^0\nabla_{\mu}^* \equiv {}^0\nabla_{\mu} + w B_{\mu}, {}^0\Gamma^{\lambda}{}_{\mu\nu} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} + \partial_{\nu} \gamma_{\mu\rho} -$	where ${}^0\nabla_{\mu}^* \equiv {}^0\nabla_{\mu} + w B_{\mu}, {}^0\Gamma^{\lambda}{}_{\mu\nu} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} + \partial_{\nu} \gamma_{\mu\rho} -$
1510.06699_F00221	11	■ 0.709	$\partial_{\rho} \gamma_{\mu\nu}$	$\partial_{\rho} \gamma_{\mu\nu}$	$\partial_{\rho} \gamma_{\mu\nu})$ is the metric connection corresponding to the
1510.06699_F00222	11	■ 0.991	$\mathrm{X}^{\rho}{}_{\sigma}$	$\mathrm{X}^{\rho}{}_{\sigma}$	and $\mathrm{X}^{\rho}{}_{\sigma}$ are the $\mathrm{GL}(4, R)$ generator matrices appropriate
1510.06699_F00223	11	■ 0.991	$\operatorname{GL}(4, R)$	$\mathrm{GL}(4, R)$	and $\mathrm{X}^{\rho}{}_{\sigma}$ are the $\mathrm{GL}(4, R)$ generator matrices appropriate
1510.06699_F00224	11	■ 1.000	$\{ \}^{\{0\}} \Gamma^{\sigma}{}_{\rho\mu}$	${}^0\Gamma^{\sigma}{}_{\rho\mu}$	spacetime, the connection coefficients ${}^0\Gamma^{\sigma}{}_{\rho\mu}$ are not dy-
1510.06699_F00225	11	■ 1.000	$\{ \}^{\{0\}} \nabla_{\mu}$	${}^0\nabla_{\mu}$	spacetime, meaning that the derivatives ${}^0\nabla_{\mu}$ and ${}^0\nabla_{\nu}$
1510.06699_F00226	11	■ 1.000	$\{ \}^{\{0\}} \nabla_{\nu}$	${}^0\nabla_{\nu}$	spacetime, meaning that the derivatives ${}^0\nabla_{\mu}$ and ${}^0\nabla_{\nu}$
1510.06699_F00227	11	■ 1.000	$\boldsymbol{h}$	$\boldsymbol{h}$	rank-2 tensor field, which we will denote by $\boldsymbol{h}$, defined on
1510.06699_F00228	11	■ 1.000	$h_a{}^{\mu} \equiv \boldsymbol{h}(\hat{e}_a, e^{\mu})$	$h_a{}^{\mu} \equiv \boldsymbol{h}(\hat{e}_a, e^{\mu})$	by $h_a{}^{\mu} \equiv \boldsymbol{h}(\hat{e}_a, e^{\mu})$. Similarly, the inverse of $\boldsymbol{h}$, which
1510.06699_F00229	11	■ 1.000	$\boldsymbol{b}$	$\boldsymbol{b}$	we denote by $\boldsymbol{b}$, is also a rank-2 tensor field defined on
1510.06699_F00230	11	■ 1.000	$b^a{}_{\mu} \equiv \boldsymbol{b}(\hat{e}^a, e_{\mu})$	$b^a{}_{\mu} \equiv \boldsymbol{b}(\hat{e}^a, e_{\mu})$	considered so far are given by $b^a{}_{\mu} \equiv \boldsymbol{b}(\hat{e}^a, e_{\mu})$. Other
1510.06699_F00231	11	■ 1.000	$\boldsymbol{J}$	$\boldsymbol{J}$	field $\boldsymbol{J}$ defined on the Minkowski spacetime, which may
1510.06699_F00232	11	■ 1.000	$\boldsymbol{J} = J^{\mu} e_{\mu} = J_{\mu} e^{\mu}$	$\boldsymbol{J} = J^{\mu} e_{\mu} = J_{\mu} e^{\mu}$	basis or its dual as $\boldsymbol{J} = J^{\mu} e_{\mu} = J_{\mu} e^{\mu}$. Replacing the
1510.06699_F00233	11	■ 1.000	$\mathcal{J}_a \equiv h_a{}^{\mu} J_{\mu} = h_{a\mu} J^{\mu}$	$\mathcal{J}_a \equiv h_a{}^{\mu} J_{\mu} = h_{a\mu} J^{\mu}$	$\boldsymbol{h}$, we may obtain the quantities $\mathcal{J}_a \equiv h_a{}^{\mu} J_{\mu} = h_{a\mu} J^{\mu}$ or
1510.06699_F00234	11	■ 1.000	$\mathcal{J}^a \equiv h^a{}_{\mu} J^{\mu} = h^{a\mu} J_{\mu}$	$\mathcal{J}^a \equiv h^a{}_{\mu} J^{\mu} = h^{a\mu} J_{\mu}$	$\mathcal{J}^a \equiv h^a{}_{\mu} J^{\mu} = h^{a\mu} J_{\mu}$. Adopting the viewpoint that $\boldsymbol{h}$ is
1510.06699_F00235	11	■ 0.967	$\mathcal{J} = \boldsymbol{h}(\cdot, \boldsymbol{J})^{67}$	$\mathcal{J} = \boldsymbol{h}(\cdot, \boldsymbol{J})^{67}$	basis of a <i>new</i> vector field $\mathcal{J} = \boldsymbol{h}(\cdot, \boldsymbol{J})^{67}$. It is also worth
1510.06699_F00236	11	■ 0.820	$\boldsymbol{h}(\mathcal{J}, \cdot)$	$\boldsymbol{h}(\mathcal{J}, \cdot)$	the vector $\boldsymbol{h}(\mathcal{J}, \cdot)$, which has contravariant components
1510.06699_F00237	11	■ 1.000	$h_a{}^{\mu} \mathcal{J}^a$	$h_a{}^{\mu} \mathcal{J}^a$	in the coordinate basis given by $h_a{}^{\mu} \mathcal{J}^a$; these are the
1510.06699_F00238	11	■ 1.000	$\mathcal{J}$	$\mathcal{J}$	example, if we place $\mathcal{J}$ into the first ‘slot’ of $\boldsymbol{b}$, we re-
1510.06699_F00239	11	■ 0.690	$\boldsymbol{J} = \boldsymbol{b}(\mathcal{J}, \cdot)$	$\boldsymbol{J} = \boldsymbol{b}(\mathcal{J}, \cdot)$	cover the <i>original</i> vector $\boldsymbol{J} = \boldsymbol{b}(\mathcal{J}, \cdot)$, since its covari-
1510.06699_F00240	11	■ 1.000	$b^a{}_{\mu} \mathcal{J}_a = J_{\mu}$	$b^a{}_{\mu} \mathcal{J}_a = J_{\mu}$	ant components in the coordinate basis are $b^a{}_{\mu} \mathcal{J}_a = J_{\mu}$,
1510.06699_F00241	11	■ 1.000	$J_{\mu} = b_{a\mu} \mathcal{J}^a$	$J_{\mu} = b_{a\mu} \mathcal{J}^a$	$J_{\mu} = b_{a\mu} \mathcal{J}^a$, and their contravariant counterparts as
1510.06699_F00242	11	■ 1.000	$J^{\mu} = b_a{}^{\mu} \mathcal{J}^a = b^{a\mu} \mathcal{J}_a$	$J^{\mu} = b_a{}^{\mu} \mathcal{J}^a = b^{a\mu} \mathcal{J}_a$	$J^{\mu} = b_a{}^{\mu} \mathcal{J}^a = b^{a\mu} \mathcal{J}_a$. In all these cases, the compo-
1510.06699_F00243	11	■ 1.000	$\boldsymbol{b}(\cdot, \boldsymbol{J})$	$\boldsymbol{b}(\cdot, \boldsymbol{J})$	tion: consider, for example, the vector $\boldsymbol{b}(\cdot, \boldsymbol{J})$; this has
1510.06699_F00244	11	■ 1.000	$b^a{}_{\mu} J^{\mu}$	$b^a{}_{\mu} J^{\mu}$	$b^a{}_{\mu} J^{\mu}$, but these are <i>not</i> , in general, equal to $\mathcal{J}^a$.
1510.06699_F00245	11	■ 1.000	$\mathcal{J}^a$	$\mathcal{J}^a$	$b^a{}_{\mu} J^{\mu}$, but these are <i>not</i> , in general, equal to $\mathcal{J}^a$.
1510.06699_F00246	11	■ 0.999	$\boldsymbol{T}$	$\boldsymbol{T}$	indices: for example, if $\boldsymbol{T}$ is a rank-2 tensor field, then
1510.06699_F00247	11	■ 0.968	$\mathcal{T}$	$\mathcal{T}$	one obtains the <i>new</i> rank-2 tensor field $\mathcal{T}$ with covari-
1510.06699_F00248	11	■ 1.000	$\mathcal{T}_{ab} =$	$\mathcal{T}_{ab} =$	ant components (say) in the tetrad basis given by $\mathcal{T}_{ab} =$

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1510.06699_F00249	11	■ 1.000	$\mathrm{h}_{\{a\}}\{\}^{\{\mu\}}\mathrm{h}_{\{b\}}\{\}^{\{\nu\}}T_{\{\mu\}\{\nu\}}$	$h_a{}^\mu h_b{}^\nu T_{\mu\nu}$	$h_a{}^\mu h_b{}^\nu T_{\mu\nu}$. Using the reciprocity relations (28), one may
1510.06699_F00250	11	■ 1.000	$T_{\{\mu\}\{\nu\}}=b^{\{a\}}\{\}_{\{\mu\}}b^{\{b\}}\{\}_{\{\nu\}}\mathcal{T}_{\{a\}\{b\}}$	$T_{\mu\nu} = b^a{}_\mu b^b{}_\nu \mathcal{T}_{ab}$	therefore also write, for example, $T_{\mu\nu} = b^a{}_\mu b^b{}_\nu \mathcal{T}_{ab}$. One
1510.06699_F00251	11	■ 0.999	$\hat{\mathrm{e}}_{\{\boldsymbol{e}\}}\{a\}$	$\hat{e}_a$	frame vectors $\hat{e}_a$; in this case, the components $h_a{}^\mu J_\mu$,
1510.06699_F00252	11	■ 0.999	$\mathrm{h}_{\{a\}}\{\}^{\{\mu\}}J_{\{\mu\}}$	$h_a{}^\mu J_\mu$	frame vectors $\hat{e}_a$; in this case, the components $h_a{}^\mu J_\mu$,
1510.06699_F00253	11	■ 1.000	$J_{\{a\}}$	J_a	for example, are interpreted as the components J_a of the
1510.06699_F00254	12	■ 1.000	$R^{\{a\}\{b\}}\{\}_{\{\mu\}\{\nu\}}$	$R^{ab}{}_{\mu\nu}$	Both $R^{ab}{}_{\mu\nu}$ and $H_{\mu\nu}$ transform covariantly under GCT
1510.06699_F00255	12	■ 1.000	$H_{\{\mu\}\{\nu\}}$	$H_{\mu\nu}$	Both $R^{ab}{}_{\mu\nu}$ and $H_{\mu\nu}$ transform covariantly under GCT
1510.06699_F00256	12	■ 1.000	$\mathcal{D}_{\{a\}}^*\{\varphi\}^*\mathrel{\varphi}=\mathcal{D}_{\{a\}}^*\varphi$	$\mathcal{D}_a^* \varphi =$	two generalised covariant derivatives. Since $\mathcal{D}_a^* \varphi =$
1510.06699_F00257	12	■ 1.000	$\mathrm{h}_{\{a\}}\{\}^{\{\mu\}}\mathcal{D}_{\{\mu\}}^*\{\varphi\}^*\mathrel{\varphi}=\mathcal{D}_{\{\mu\}}^*\varphi$	$h_a{}^\mu \mathcal{D}_\mu^* \varphi$	$h_a{}^\mu \mathcal{D}_\mu^* \varphi$, this commutator differs from (38) by an addi-
1510.06699_F00258	12	■ 1.000	$\mathcal{R}^{\{a\}\{b\}}\{\}_{\{c\}\{d\}}\mathrel{\equiv}\mathrm{h}_{\{a\}}\{\}^{\{\mu\}}\mathrm{h}_{\{b\}}\{\}^{\{\nu\}}R^{\{a\}\{b\}}\{\}_{\{\mu\}\{\nu\}}R^{\{c\}\{d\}}\{\}_{\{\mu\}\{\nu\}}$	$\mathcal{R}^{ab}{}_{cd} \equiv h_a{}^\mu h_b{}^\nu R^{ab}{}_{\mu\nu}$	where $\mathcal{R}^{ab}{}_{cd} \equiv h_a{}^\mu h_b{}^\nu R^{ab}{}_{\mu\nu}$ and $\mathcal{H}_{cd} = h_c{}^\mu h_d{}^\nu H_{\mu\nu}$, and
1510.06699_F00259	12	■ 1.000	$\mathcal{H}_{\{c\}\{d\}}=\mathrm{h}_{\{c\}}\{\}^{\{\mu\}}\mathrm{h}_{\{d\}}\{\}^{\{\nu\}}H_{\{\mu\}\{\nu\}}$	$\mathcal{H}_{cd} = h_c{}^\mu h_d{}^\nu H_{\mu\nu}$	where $\mathcal{R}^{ab}{}_{cd} \equiv h_a{}^\mu h_b{}^\nu R^{ab}{}_{\mu\nu}$ and $\mathcal{H}_{cd} = h_c{}^\mu h_d{}^\nu H_{\mu\nu}$, and
1510.06699_F00260	12	■ 0.669	$\mathcal{B}_{\{a\}}=\mathrm{h}_{\{a\}}\{\}^{\{\mu\}}B_{\{\mu\}}$	$\mathcal{B}_a = h_a{}^\mu B_\mu$	where $\mathcal{B}_a = h_a{}^\mu B_\mu$. It is also convenient to define $\mathcal{T}_b^* \equiv$
1510.06699_F00261	12	■ 0.669	$\mathcal{T}_{\{b\}}^*\mathrel{\equiv}\mathcal{T}_b^*$	$\mathcal{T}_b^* \equiv$	where $\mathcal{B}_a = h_a{}^\mu B_\mu$. It is also convenient to define $\mathcal{T}_b^* \equiv$
1510.06699_F00262	12	■ 1.000	$\mathcal{T}^{\{a\}\{b\}}\{\}_{\{a\}\{b\}}=\mathcal{T}_{\{b\}}+3\mathcal{B}_{\{b\}}$	$\mathcal{T}^{*a}{}_{ba} = \mathcal{T}_b + 3\mathcal{B}_b$	$\mathcal{T}^{*a}{}_{ba} = \mathcal{T}_b + 3\mathcal{B}_b$, where $\mathcal{T}_b$ is the corresponding PGT
1510.06699_F00263	12	■ 1.000	$\mathcal{T}_{\{b\}}$	$\mathcal{T}_b$	$\mathcal{T}^{*a}{}_{ba} = \mathcal{T}_b + 3\mathcal{B}_b$, where $\mathcal{T}_b$ is the corresponding PGT
1510.06699_F00264	12	■ 1.000	$\mathcal{R}^{\{a\}\{b\}}\{\}_{\{c\}\{d\}},\mathcal{H}_{\{c\}\{d\}}$	$\mathcal{R}^{ab}{}_{cd}, \mathcal{H}_{cd}$	It is easy to show that $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{H}_{cd}$ and $\mathcal{T}^{*a}{}_{cd}$ trans-
1510.06699_F00265	12	■ 1.000	$\mathcal{T}^{\{a\}\{b\}}\{\}_{\{c\}\{d\}}$	$\mathcal{T}^{*a}{}_{cd}$	It is easy to show that $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{H}_{cd}$ and $\mathcal{T}^{*a}{}_{cd}$ trans-
1510.06699_F00266	12	■ 1.000	$w(\mathcal{R}^{\{a\}\{b\}}\{\}_{\{c\}\{d\}})=w(\mathcal{H}_{\{c\}\{d\}})=-2$	$w(\mathcal{R}^{ab}{}_{cd}) = w(\mathcal{H}_{cd}) = -2$	$w(\mathcal{R}^{ab}{}_{cd}) = w(\mathcal{H}_{cd}) = -2$ and $w(\mathcal{T}^{*a}{}_{cd}) = -1$ respec-
1510.06699_F00267	12	■ 1.000	$w(\mathcal{T}^{\{a\}\{b\}}\{\}_{\{c\}\{d\}})=-1$	$w(\mathcal{T}^{*a}{}_{cd}) = -1$	$w(\mathcal{R}^{ab}{}_{cd}) = w(\mathcal{H}_{cd}) = -2$ and $w(\mathcal{T}^{*a}{}_{cd}) = -1$ respec-
1510.06699_F00268	12	■ 1.000	$\mathcal{R}^{\{a\}\{b\}}\{\}_{\{c\}\{d\}},\mathcal{H}_{\{a\}\{b\}}$	$\mathcal{R}^{ab}{}_{cd}, \mathcal{H}_{ab}$	strengths $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{H}_{ab}$ and $\mathcal{T}^{*a}{}_{bc}$ are related to the <i>curva-</i>
1510.06699_F00269	12	■ 1.000	$A_{\{a\}\{b\}}\{\mu\}=\mathcal{A}_{\{a\}\{b\}}\mu$	$A_{ab\mu} =$	the relevant relationship is easily found to be $A_{ab\mu} =$
1510.06699_F00270	12	■ 1.000	$b^{\{c\}}\{\}_{\{\mu\}}\mathcal{A}_{\{a\}\{b\}}\{c\}$	$b^c{}_\mu \mathcal{A}_{abc}$	$b^c{}_\mu \mathcal{A}_{abc}$, where
1510.06699_F00271	12	■ 1.000	$\mathcal{T}_{\{a\}\{b\}}\{c\}^*\mathrel{\equiv}\mathcal{T}_{abc}^*$	$\mathcal{T}_{abc}^*$	equations of motion result in $\mathcal{T}_{abc}^*$ being independent of
1510.06699_F00272	12	■ 1.000	$\mathcal{A}_{\{a\}\{b\}}\{c\}$	$\mathcal{A}_{abc}$	$\mathcal{A}_{abc}$, (46) gives an explicit expression for the <i>A</i> -field in
1510.06699_F00273	12	■ 1.000	$\{\}^{\{0\}}\mathcal{A}_{\{a\}\{b\}}\{c\}^*\mathrel{\equiv}\frac{1}{2}(c_{\{a\}\{b\}}^*\{c\}^*+\mathcal{L}_{\{a\}\{b\}}^*\{c\}^*)$	${}^0\mathcal{A}_{abc}^* \equiv \frac{1}{2}(c_{abc}^* +$	in which we have defined the quantities ${}^0\mathcal{A}_{abc}^* \equiv \frac{1}{2}(c_{abc}^* +$
1510.06699_F00274	12	■ 0.506	$\mathcal{L}_{\{b\}\{c\}}^*\{a\}^*-\mathcal{L}_{\{c\}\{a\}}^*\{b\}^*\mathrel{\equiv}\mathcal{L}_{bca}^*-\mathcal{L}_{cab}^*)$	$c_{bca}^* - c_{cab}^*)$	$c_{bca}^* - c_{cab}^*)$ that, as indicated, are fully determined by the
1510.06699_F00275	12	■ 1.000	$\mathcal{K}_{\{a\}\{b\}}^*\{c\}^*\mathrel{\equiv}-\frac{1}{2}(\mathcal{T}_{abc}^*+\mathcal{T}_{bca}^*-\mathcal{T}_{cab}^*)$	$\mathcal{K}_{abc}^* \equiv -\frac{1}{2}(\mathcal{T}_{abc}^* + \mathcal{T}_{bca}^* - \mathcal{T}_{cab}^*)$	defined $\mathcal{K}_{abc}^* \equiv -\frac{1}{2}(\mathcal{T}_{abc}^* + \mathcal{T}_{bca}^* - \mathcal{T}_{cab}^*)$, which when re-
1510.06699_F00276	12	■ 0.982	$\{\}^{\{0\}}\mathcal{A}_{\{a\}\{b\}}\{c\}^*\mathrel{\equiv}\mathcal{A}_{abc}^*$	${}^0\mathcal{A}_{abc}^*$	that both ${}^0\mathcal{A}_{abc}^*$ and $\mathcal{K}_{abc}^*$ are antisymmetric in their first
1510.06699_F00277	12	■ 0.982	$\mathcal{K}_{\{a\}\{b\}}^*\{c\}^*\mathrel{\equiv}\mathcal{K}_{abc}^*$	$\mathcal{K}_{abc}^*$	that both ${}^0\mathcal{A}_{abc}^*$ and $\mathcal{K}_{abc}^*$ are antisymmetric in their first
1510.06699_F00278	12	■ 0.991	$\mathcal{A}_{\{a\}\{b\}}\{c\}^*\mathrel{\equiv}\mathcal{A}_{abc}^*$	$\mathcal{A}_{abc}^*$	${}^0\mathcal{A}_{abc}^*$ transform in the same way as $\mathcal{A}_{abc}^*$, whereas $\mathcal{K}_{abc}^*$
1510.06699_F00279	12	■ 0.961	$\{\}^{\{0\}}\mathcal{D}_{\{a\}}^*\mathrel{\equiv}\mathrm{h}_{\{a\}}\{\}^{\{\mu\}}\mathcal{D}_{\{\mu\}}^*\{\}^{\{0\}}\mathrel{\equiv}\mathcal{D}_a^{*0}$	${}^0\mathcal{D}_a^* \equiv h_a{}^\mu {}^0\mathcal{D}_\mu^*$	covariant derivative ${}^0\mathcal{D}_a^* \equiv h_a{}^\mu {}^0\mathcal{D}_\mu^*$, which consequently
1510.06699_F00280	12	■ 1.000	$\mathcal{D}_{\{a\}}^*\mathrel{\equiv}\mathcal{D}_a^*$	$\mathcal{D}_a^*$	torsion (and hence contortion) to zero, then $\mathcal{D}_a^*$ reduces
1510.06699_F00281	12	■ 1.000	$\{\}^{\{0\}}\mathcal{D}_{\{a\}}^*\mathrel{\equiv}{}^0\mathcal{D}_a^*$	${}^0\mathcal{D}_a^*$	to ${}^0\mathcal{D}_a^*$.
1510.06699_F00282	12	■ 1.000	$\mathcal{R}^{\{a\}\{b\}}\{\}_{\{c\}\{d\}},\mathcal{T}^{\{a\}\{b\}}\{\}_{\{c\}\{d\}}$	$\mathcal{R}^{ab}{}_{cd}, \mathcal{T}^{*a}{}_{bc}$	gauge field strengths $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{T}^{*a}{}_{bc}$ and $\mathcal{H}_{ab}$ in WGT.
1510.06699_F00283	13	■ 0.947	$\mathcal{T}_{\{a\}\{b\}}\{c\}^*\mathrel{\equiv}\mathcal{T}_{abc}^*$	$\mathcal{T}_{abc}^* =$	the ‘torsion’ is totally antisymmetric, such that $\mathcal{T}_{abc}^* =$
1510.06699_F00284	13	■ 0.990	$\epsilon_{\{a\}\{b\}}\{c\}\{d\}t^{\{d\}}$	$\epsilon_{abcd}t^d$	$\epsilon_{abcd}t^d$ for some vector t^d , then the second term on the
1510.06699_F00285	13	■ 0.990	$t^{\{d\}}$	t^d	$\epsilon_{abcd}t^d$ for some vector t^d , then the second term on the
1510.06699_F00286	13	■ 0.999	d	d	By contracting over the indices <i>a</i> and <i>d</i> in the ‘ <i>R</i> -
1510.06699_F00287	13	■ 1.000	b	b	Contracting again over <i>b</i> and <i>e</i> , one then finds the twice-
1510.06699_F00288	13	■ 1.000	e	e	Contracting again over <i>b</i> and <i>e</i> , one then finds the twice-
1510.06699_F00289	13	■ 0.770	$\mathcal{T}_{\{a\}}^*\mathrel{\equiv}\mathcal{T}_a^*$	$\mathcal{T}_a^*$	torsion trace $\mathcal{T}_a^*$ vanishes, which clearly includes the spe-

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1510.06699_F00290	13	■ 1.000	$\mathcal{H}$	$\mathcal{H}$	It is clear that the ‘ $\mathcal{H}$ -identity’ (53) has no non-trivial
1510.06699_F00291	13	■ 1.000	$\mathcal{R}_{abcd}, \mathcal{T}_{abc}^*$	$\mathcal{R}_{abcd}, \mathcal{T}_{abc}^*$	itational gauge fields. From $\mathcal{R}_{abcd}$, $\mathcal{T}_{abc}^*$ and $\mathcal{H}_{ab}$, one can
1510.06699_F00292	13	■ 1.000	$\mathcal{L}_{\mathrm{G}} \equiv h^{-1} L_{\mathrm{G}}$	$\mathcal{L}_{\mathrm{G}} \equiv h^{-1} L_{\mathrm{G}}$	sity $\mathcal{L}_{\mathrm{G}} \equiv h^{-1} L_{\mathrm{G}}$ has weight zero (i.e scale invari-
1510.06699_F00293	13	■ 1.000	$\mathcal{R}_{abcd}, \mathcal{H}_{ab}$	$\mathcal{R}_{abcd}, \mathcal{H}_{ab}$	ant). It is trivial to show that $\mathcal{R}_{abcd}$, $\mathcal{H}_{ab}$ and $\mathcal{T}_{abc}^*$
1510.06699_F00294	13	■ 1.000	$w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$	$w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$	weights $w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}_{abc}^*) = -1$
1510.06699_F00295	13	■ 1.000	$w(\mathcal{T}_{abc}^*) = -1$	$w(\mathcal{T}_{abc}^*) = -1$	weights $w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}_{abc}^*) = -1$
1510.06699_F00296	13	■ 0.988	$\{ \}^{\underline{70}}$	$\overline{70}$	respectively ⁷⁰ . Hence L_{G} can be quadratic in $\mathcal{R}_{abcd}$ and
1510.06699_F00297	13	■ 0.994	$L_{\mathcal{H}^2}$	$L_{\mathcal{H}^2}$	where the expressions for $L_{\mathcal{R}^2}$ and $L_{\mathcal{H}^2}$ are given in (4).
1510.06699_F00298	13	■ 1.000	$D \leq 4$	$D \leq 4$	contributes a total derivative to the action (in $D \leq 4$
1510.06699_F00299	13	■ 0.994	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^*$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^*$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^*$. However, since these terms depend upon the
1510.06699_F00300	13	■ 1.000	$\mathcal{L}_{\mathrm{G}}$	$\mathcal{L}_{\mathrm{G}}$	sity $\mathcal{L}_{\mathrm{G}}$. As an illustration, in Appendix D we give a
1510.06699_F00301	13	■ 1.000	$L_{\mathrm{G}} = L_{\mathcal{H}^2}$	$L_{\mathrm{G}} = L_{\mathcal{H}^2}$	$L_{\mathrm{G}} = L_{\mathcal{H}^2}$ and L_{M} contains a term proportional to $\phi^2 \mathcal{R}$.
1510.06699_F00302	13	■ 1.000	S_{T}	S_{T}	The total action S_{T} is simply the sum of the matter and
1510.06699_F00303	13	■ 1.000	$h, A, \partial A$	$h, A, \partial A$	$h, A, \partial A$ and ∂B (suppressing indices for brevity). Note,
1510.06699_F00304	13	■ 1.000	∂B	∂B	$h, A, \partial A$ and ∂B (suppressing indices for brevity). Note,
1510.06699_F00305	13	■ 1.000	$\varphi, \partial \varphi, h, A$	$\varphi, \partial \varphi, h, A$	φ induce a dependence on $\varphi, \partial \varphi, h, A$ and B . We will
1510.06699_F00306	13	■ 1.000	∂A	∂A	$\phi^2 \mathcal{R}$, then this brings an additional dependence on ∂A .
1510.06699_F00307	13	■ 0.901	$\phi^2 L_{\mathcal{T}^* \in}$	$\phi^2 L_{\mathcal{T}^* \in}$	Finally, if one includes terms of the generic form $\phi^2 L_{\mathcal{T}^* 2}$,
1510.06699_F00308	13	■ 1.000	∂h	∂h	then it produces an additional dependence on ∂h . Con-
1510.06699_F00309	14	■ 1.000	$\mathcal{L}_{\mathrm{T}}$	$\mathcal{L}_{\mathrm{T}}$	Since $\mathcal{L}_{\mathrm{T}}$ is at most quadratic in the field strength ten-
1510.06699_F00310	14	■ 1.000	$\mathcal{R}_{abcd}(h, A, \partial A), \mathcal{T}_{abc}^*(h, \partial h, A, B)$	$\mathcal{R}_{abcd}(h, A, \partial A), \mathcal{T}_{abc}^*(h, \partial h, A, B)$	sors $\mathcal{R}_{abcd}(h, A, \partial A)$, $\mathcal{T}_{abc}^*(h, \partial h, A, B)$ and $\mathcal{H}_{ab}(h, \partial B)$,
1510.06699_F00311	14	■ 1.000	$t^a{}_{\mu} \equiv$	$t^a{}_{\mu} \equiv$	In the free gravitational sector, we have defined $t^a{}_{\mu} \equiv$
1510.06699_F00312	14	■ 0.999	$\delta \mathcal{L}_{\mathrm{G}} / \delta h_a{}^{\mu} = \partial \mathcal{L}_{\mathrm{G}} / \partial h_a{}^{\mu}, s_{ab}{}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta A^{ab}{}_{\mu}$	$\delta \mathcal{L}_{\mathrm{G}} / \delta h_a{}^{\mu} = \partial \mathcal{L}_{\mathrm{G}} / \partial h_a{}^{\mu}, s_{ab}{}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta A^{ab}{}_{\mu}$	$\delta \mathcal{L}_{\mathrm{G}} / \delta h_a{}^{\mu} = \partial \mathcal{L}_{\mathrm{G}} / \partial h_a{}^{\mu}, s_{ab}{}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta A^{ab}{}_{\mu}$ and
1510.06699_F00313	14	■ 1.000	$j^{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta B_{\mu}$	$j^{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta B_{\mu}$	$j^{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta B_{\mu}$, respectively. In the matter sector,
1510.06699_F00314	14	■ 1.000	$\tau^a{}_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_a{}^{\mu}$	$\tau^a{}_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_a{}^{\mu}$	the energy-momentum $\tau^a{}_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_a{}^{\mu}$, spin-angular-
1510.06699_F00315	14	■ 0.692	$\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu} = \partial \mathcal{L}_{\mathrm{M}} / \partial B_{\mu}$	$\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu} = \partial \mathcal{L}_{\mathrm{M}} / \partial B_{\mu}$	$\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu} = \partial \mathcal{L}_{\mathrm{M}} / \partial B_{\mu}$ act as sources ⁷² . It is easy
1510.06699_F00316	14	■ 0.692	$\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu} = \partial \mathcal{L}_{\mathrm{M}} / \partial B_{\mu}$	$\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu} = \partial \mathcal{L}_{\mathrm{M}} / \partial B_{\mu}$	$\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu} = \partial \mathcal{L}_{\mathrm{M}} / \partial B_{\mu}$ act as sources ⁷² . It is easy
1510.06699_F00317	14	■ 1.000	$w = 1, w = 0$	$w = 1, w = 0$	$w = 1, w = 0$ and $w = 0$, respectively. It is, in fact, more
1510.06699_F00318	14	■ 1.000	$t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu}, s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu}$	$t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu}, s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu}$	where $t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu}$, $s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu}$ and $j^a \equiv j^{\mu} b^a{}_{\mu}$, and
1510.06699_F00319	14	■ 1.000	$j^a \equiv j^{\mu} b^a{}_{\mu}$	$j^a \equiv j^{\mu} b^a{}_{\mu}$	where $t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu}$, $s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu}$ and $j^a \equiv j^{\mu} b^a{}_{\mu}$, and
1510.06699_F00320	14	■ 0.992	$\bar{\partial} \mathcal{L}_{\mathrm{M}} / \partial \varphi \equiv [\partial \mathcal{L}_{\mathrm{M}}(\varphi, \mathcal{D}_{\mu}^* u, \dots) / \partial \varphi]_{u=\varphi}$	$\bar{\partial} \mathcal{L}_{\mathrm{M}} / \partial \varphi \equiv [\partial \mathcal{L}_{\mathrm{M}}(\varphi, \mathcal{D}_{\mu}^* u, \dots) / \partial \varphi]_{u=\varphi}$	where $\bar{\partial} \mathcal{L}_{\mathrm{M}} / \partial \varphi \equiv [\partial \mathcal{L}_{\mathrm{M}}(\varphi, \mathcal{D}_{\mu}^* u, \dots) / \partial \varphi]_{u=\varphi}$, so that φ
1510.06699_F00321	14	■ 1.000	$\mathcal{D}_{\mu}^* \varphi$	$\mathcal{D}_{\mu}^* \varphi$	and $\mathcal{D}_{\mu}^* \varphi$ are treated as independent variables, rather
1510.06699_F00322	14	■ 1.000	$\mathcal{L}_{\mathrm{M}} = h^{-1} L_{\mathrm{M}}$	$\mathcal{L}_{\mathrm{M}} = h^{-1} L_{\mathrm{M}}$	where we have defined $\mathcal{L}_{\mathrm{M}} = h^{-1} L_{\mathrm{M}}$ and $\bar{\partial} L_{\mathrm{M}} / \partial \varphi \equiv$
1510.06699_F00323	14	■ 1.000	$\bar{\partial} L_{\mathrm{M}} / \partial \varphi \equiv$	$\bar{\partial} L_{\mathrm{M}} / \partial \varphi \equiv$	where we have defined $\mathcal{L}_{\mathrm{M}} = h^{-1} L_{\mathrm{M}}$ and $\bar{\partial} L_{\mathrm{M}} / \partial \varphi \equiv$
1510.06699_F00324	14	■ 0.974	$[\partial L_{\mathrm{M}}(\varphi, \mathcal{D}_{\mu}^* u, \dots) / \partial \varphi]_{u=\varphi}$	$[\partial L_{\mathrm{M}}(\varphi, \mathcal{D}_{\mu}^* u, \dots) / \partial \varphi]_{u=\varphi}$	$[\partial L_{\mathrm{M}}(\varphi, \mathcal{D}_{\mu}^* u, \dots) / \partial \varphi]_{u=\varphi}$, so that φ and $\mathcal{D}_{\mu}^* \varphi$ are treated
1510.06699_F00325	14	■ 1.000	S_{M}	S_{M}	the minimal-coupling procedure applied to the action S_{M} ,
1510.06699_F00326	14	■ 0.999	$\{ \}^{9,74}$	$\{ \}^{9,74}$	equations directly ^{9,74} , which would be given by (67a) and

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1510.06699_F00327	14	■ 0.963	2, c	$2, C$	S_G under (infinitesimal) local Lorentz (or $SL(2, C)$) rota-
1510.06699_F00328	15	■ 0.718	(69 c)	(69c)	(69c)
1510.06699_F00329	15	■ 1.000	$w = -\frac{3}{2}$	$w = -\frac{3}{2}$	As we show in Section IIL, however, this interpretation
1510.06699_F00330	15	■ 1.000	$\psi \rightarrow \psi e^{i\alpha}$	$\psi \rightarrow \psi e^{i\alpha}$	that the equations
1510.06699_F00331	15	■ 1.000	γ	γ	where, on each γ -matrix, one has simply replaced the
1510.06699_F00332	15	■ 1.000	$\bar{\psi}$	$\bar{\psi}$	that, since both ψ and $\bar{\psi}$ have Weyl weight $w = -\frac{3}{2}$, the
1510.06699_F00333	15	■ 1.000	$\mathcal{D}_a$	$\mathcal{D}_a$	term in (71) can be replaced by its PGT counterpart $\mathcal{D}_a$,
1510.06699_F00334	15	■ 1.000	$\mathcal{D}_a^* \psi$	$\mathcal{D}_a^* \psi$	varying the action with respect to ψ . Since $\mathcal{D}_a^* \psi$ and $\mathcal{T}_a^*$
1510.06699_F00335	15	■ 1.000	$w = -\frac{5}{2}$	$w = -\frac{5}{2}$	$w = -\frac{5}{2}$ and $w = -1$ respectively, we see immediately
1510.06699_F00336	15	■ 1.000	$\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$	$\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$	directly by recalling that $\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$ and $\mathcal{T}_a =$
1510.06699_F00337	15	■ 1.000	$\mathcal{T}_a =$	$\mathcal{T}_a =$	directly by recalling that $\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$ and $\mathcal{T}_a =$
1510.06699_F00338	15	■ 1.000	$\mathcal{T}_a^* - 3 \mathcal{B}_a$	$\mathcal{T}_a^* - 3 \mathcal{B}_a$	$\mathcal{T}_a^* - 3 \mathcal{B}_a$ to recover (73). We note that (73) and (74) are
1510.06699_F00339	16	■ 1.000	$\mathcal{T}_a$	$\mathcal{T}_a$	density $\mathcal{L}_G$, but the expression for $\mathcal{T}_a$, or $\mathcal{T}_a^*$, will depend
1510.06699_F00340	16	■ 1.000	$\Sigma_{ab} = \frac{1}{4} [\gamma_a, \gamma_b]$	$\Sigma_{ab} = \frac{1}{4} [\gamma_a, \gamma_b]$	given by $\Sigma_{ab} = \frac{1}{4} [\gamma_a, \gamma_b]$, after a short calculation one
1510.06699_F00341	16	■ 0.945	$\mathcal{T}_{[abc]}^* = 0$	$\mathcal{T}_{[abc]}^* = 0$	of the Dirac field ψ . In other words, if $\mathcal{T}_{[abc]}^* = 0$ then
1510.06699_F00342	16	■ 1.000	$\tau = h^{-1} \mu \phi \bar{\psi} \psi$	$\tau = h^{-1} \mu \phi \bar{\psi} \psi$	ric and has the trace $\tau = h^{-1} \mu \phi \bar{\psi} \psi$, whereas the spin-
1510.06699_F00343	16	■ 1.000	$\sigma_{abc} = \sigma_{[abc]}$	$\sigma_{abc} = \sigma_{[abc]}$	that $\sigma_{abc} = \sigma_{[abc]}$. Consequently, all contractions of the
1510.06699_F00344	16	■ 1.000	$\mathcal{D}_a \rightarrow \mathcal{D}_a^*$	$\mathcal{D}_a \rightarrow \mathcal{D}_a^*$	ply making the replacement $\mathcal{D}_a \rightarrow \mathcal{D}_a^*$ throughout; the
1510.06699_F00345	16	■ 1.000	μ, ν, λ	μ, ν, λ	where μ, ν, λ and a are dimensionless constants (usually
1510.06699_F00346	16	■ 0.669	β_1, β_2	β_1, β_2	stants β_1, β_2 and β_3 in $L_{T^{*2}}$. Since none of the additional
1510.06699_F00347	16	■ 0.669	β_3	β_3	stants β_1, β_2 and β_3 in $L_{T^{*2}}$. Since none of the additional
1510.06699_F00348	16	■ 1.000	$F_{\mu\nu} \equiv$	$F_{\mu\nu} \equiv$	where the electromagnetic field strength tensor $F_{\mu\nu} \equiv$
1510.06699_F00349	16	■ 1.000	$\partial_\mu A_\nu - \partial_\nu A_\mu$	$\partial_\mu A_\nu - \partial_\nu A_\mu$	$\partial_\mu A_\nu - \partial_\nu A_\mu$ and J^μ is some source current density. We
1510.06699_F00350	16	■ 1.000	$A_\mu \rightarrow A_\mu + \partial_\mu \chi$	$A_\mu \rightarrow A_\mu + \partial_\mu \chi$	formation $A_\mu \rightarrow A_\mu + \partial_\mu \chi$, where χ is any scalar field,
1510.06699_F00351	16	■ 1.000	χ	χ	formation $A_\mu \rightarrow A_\mu + \partial_\mu \chi$, where χ is any scalar field,
1510.06699_F00352	16	■ 1.000	$\partial_\mu J^\mu = 0$	$\partial_\mu J^\mu = 0$	ity equation $\partial_\mu J^\mu = 0$, which embodies the conservation
1510.06699_F00353	16	■ 1.000	$\mathcal{A}_a = h_a^\mu A_\mu$	$\mathcal{A}_a = h_a^\mu A_\mu$	field $\mathcal{A}_a = h_a^\mu A_\mu$. Similarly, we introduce the covari-
1510.06699_F00354	16	■ 1.000	$\mathcal{J}_a = h_a^\mu J_\mu$	$\mathcal{J}_a = h_a^\mu J_\mu$	ant current density $\mathcal{J}_a = h_a^\mu J_\mu$. This leads us, however,
1510.06699_F00355	16	■ 1.000	$w(\mathcal{A}_a) = -1$	$w(\mathcal{A}_a) = -1$	weights $w(\mathcal{A}_a) = -1$ and $w(\mathcal{J}_a) = -3$; consequently, the
1510.06699_F00356	16	■ 1.000	$w(\mathcal{J}_a) = -3$	$w(\mathcal{J}_a) = -3$	weights $w(\mathcal{A}_a) = -1$ and $w(\mathcal{J}_a) = -3$; consequently, the
1510.06699_F00357	16	■ 1.000	$w(A_\mu) = 0$	$w(A_\mu) = 0$	fields carrying a Greek index have weights $w(A_\mu) = 0$
1510.06699_F00358	16	■ 1.000	$w(J_\mu) = -2$	$w(J_\mu) = -2$	and $w(J_\mu) = -2$.
1510.06699_F00359	16	■ 1.000	$\widehat{\mathcal{F}}_{ab}^* \equiv \mathcal{D}_a^* \mathcal{A}_b - \mathcal{D}_b^* \mathcal{A}_a$	$\widehat{\mathcal{F}}_{ab}^* \equiv \mathcal{D}_a^* \mathcal{A}_b - \mathcal{D}_b^* \mathcal{A}_a$	where we have defined $\widehat{\mathcal{F}}_{ab}^* \equiv \mathcal{D}_a^* \mathcal{A}_b - \mathcal{D}_b^* \mathcal{A}_a$. Since
1510.06699_F00360	16	■ 1.000	$w(\widehat{\mathcal{F}}_{ab}^*) = -2, \mathcal{L}_M$	$w(\widehat{\mathcal{F}}_{ab}^*) = -2, \mathcal{L}_M$	$w(\widehat{\mathcal{F}}_{ab}^*) = -2$, $\mathcal{L}_M$ has an overall Weyl weight of zero,
1510.06699_F00361	17	■ 1.000	$\mathcal{F}_{ab} \equiv h_a^\mu h_b^\nu F_{\mu\nu}$	$\mathcal{F}_{ab} \equiv h_a^\mu h_b^\nu F_{\mu\nu}$	where we have defined $\mathcal{F}_{ab} \equiv h_a^\mu h_b^\nu F_{\mu\nu}$, in which
1510.06699_F00362	17	■ 1.000	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $\mathcal{T}^{*c}_{ab}$ is
1510.06699_F00363	17	■ 1.000	$\mathcal{T}^{*c}_{ab}$	$\mathcal{T}^{*c}_{ab}$	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $\mathcal{T}^{*c}_{ab}$ is
1510.06699_F00364	17	■ 1.000	$\mathcal{F}_{ab}$	$\mathcal{F}_{ab}$	gauge field, as defined in (42). It is apparent that $\mathcal{F}_{ab}$
1510.06699_F00365	17	■ 1.000	$F_{\mu\nu}$	$F_{\mu\nu}$	ance of $F_{\mu\nu}$, whereas the second term on the RHS of (82)
1510.06699_F00366	17	■ 1.000	$\widehat{\mathcal{F}}_{ab}^*$	$\widehat{\mathcal{F}}_{ab}^*$	transformation, and so neither is $\widehat{\mathcal{F}}_{ab}^*$. Moreover, $\mathcal{F}_{ab}$ has
1510.06699_F00367	17	■ 1.000	$w(\mathcal{F}_{ab}) = -2$	$w(\mathcal{F}_{ab}) = -2$	a Weyl weight $w(\mathcal{F}_{ab}) = -2$. Thus, substituting (82)
1510.06699_F00368	17	■ 1.000	$\mathcal{L}_M$	$\mathcal{L}_M$	for $\mathcal{L}_M$ is identical to that obtained in PGT, as should be

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1510.06699_F00369	17	■ 0.999	$\delta \mathrm{L}_{\mathrm{EM}} / \delta A_{\mu} = 0$	$\delta \mathcal{L}_{\mathrm{EM}} / \delta A_{\mu} = 0$	is given by $\delta \mathcal{L}_{\mathrm{EM}} / \delta A_{\mu} = 0$, which is easily shown to be
1510.06699_F00370	17	■ 0.997	$\delta \mathrm{L}_{\mathrm{EM}} / \delta \mathcal{A}_a = 0$	$\delta \mathcal{L}_{\mathrm{EM}} / \delta \mathcal{A}_a = 0$	equivalent to $\delta \mathcal{L}_{\mathrm{EM}} / \delta \mathcal{A}_a = 0$, and in manifestly WGT-
1510.06699_F00371	17	■ 1.000	$\mathrm{D}_{\mathrm{a}} \mathcal{F}^{ac} = (\mathcal{D}_a^* +$ $\mathrm{D}_{\mathrm{a}})^{ac}$	$\mathcal{D}_a \mathcal{F}^{ac} = (\mathcal{D}_a^* +$ $\mathcal{D}_a)^{ac}$	is easily checked directly by noting that $\mathcal{D}_a \mathcal{F}^{ac} = (\mathcal{D}_a^* +$ $\mathcal{D}_a)^{ac}$
1510.06699_F00372	17	■ 1.000	$2 \mathcal{B}_a \mathcal{F}^{ac}$	$2 \mathcal{B}_a \mathcal{F}^{ac}$	$2 \mathcal{B}_a \mathcal{F}^{ac}$ and $\mathcal{T}_a = \mathcal{T}_a^* - 3 \mathcal{B}_a$, which on substitution
1510.06699_F00373	17	■ 1.000	$\mathcal{T}_a = \mathcal{T}_a^* - 3 \mathcal{B}_a$	$\mathcal{T}_a = \mathcal{T}_a^* - 3 \mathcal{B}_a$	$2 \mathcal{B}_a \mathcal{F}^{ac}$ and $\mathcal{T}_a = \mathcal{T}_a^* - 3 \mathcal{B}_a$, which on substitution
1510.06699_F00374	17	■ 1.000	$\mathcal{J}^a = 0$	$\mathcal{J}^a = 0$	angular-momentum tensors. Setting $\mathcal{J}^a = 0$ in (83), one
1510.06699_F00375	17	■ 1.000	$\sigma_{ab}^c = 0$	$\sigma_{ab}^c = 0$	whereas $\sigma_{ab}^c = 0$ immediately. We see that τ^a_b is the
1510.06699_F00376	17	■ 1.000	τ^a_b	τ^a_b	whereas $\sigma_{ab}^c = 0$ immediately. We see that τ^a_b is the
1510.06699_F00377	17	■ 1.000	τ_{ab}	τ_{ab}	gravity. Moreover, τ_{ab} is symmetric and traceless, as
1510.06699_F00378	17	■ 1.000	$\mathcal{J}^a = q \bar{\psi} \gamma^a \psi$	$\mathcal{J}^a = q \bar{\psi} \gamma^a \psi$	current density of the Dirac field is simply $\mathcal{J}^a = q \bar{\psi} \gamma^a \psi$,
1510.06699_F00379	17	■ 1.000	$w(\mathcal{J}^a) = -3$	$w(\mathcal{J}^a) = -3$	which has the required Weyl weight $w(\mathcal{J}^a) = -3$. Thus,
1510.06699_F00380	17	■ 1.000	$\phi \neq 0$	$\phi \neq 0$	(provided $\phi \neq 0$), which is known as the Einstein gauge.
1510.06699_F00381	17	■ 1.000	$\phi(x) = 1$	$\phi(x) = 1$	Although it is most common simply to set $\phi(x) = 1$, this
1510.06699_F00382	17	■ 1.000	$\phi(x) = \phi_0$	$\phi(x) = \phi_0$	can be misleading and so we instead set $\phi(x) = \phi_0$, where
1510.06699_F00383	17	■ 1.000	ϕ_0	ϕ_0	ϕ_0 is a constant with the same units as ϕ . Adopting the
1510.06699_F00384	18	■ 0.638	(91 c)	(91c)	(91c)
1510.06699_F00385	18	■ 1.000	$\widehat{\mathcal{A}}^{ab}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{ab}_\mu$	$\widehat{\mathcal{A}}^{ab}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{ab}_\mu$	ables $\widehat{\mathcal{A}}^{ab}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{ab}_\mu$ and $\widehat{\mathcal{B}}_a \equiv \widehat{h}_a{}^\mu \widehat{B}_\mu$.
1510.06699_F00386	18	■ 1.000	$\widehat{\mathcal{B}}_a \equiv \widehat{h}_a{}^\mu \widehat{B}_\mu$	$\widehat{\mathcal{B}}_a \equiv \widehat{h}_a{}^\mu \widehat{B}_\mu$	ables $\widehat{\mathcal{A}}^{ab}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{ab}_\mu$ and $\widehat{\mathcal{B}}_a \equiv \widehat{h}_a{}^\mu \widehat{B}_\mu$.
1510.06699_F00387	18	■ 1.000	$\mathcal{D}_a^* \phi = -(\phi^2 / \phi_0) \widehat{\mathcal{B}}_a$	$\mathcal{D}_a^* \phi = -(\phi^2 / \phi_0) \widehat{\mathcal{B}}_a$	the scalar field ϕ may be written as $\mathcal{D}_a^* \phi = -(\phi^2 / \phi_0) \widehat{\mathcal{B}}_a$,
1510.06699_F00388	18	■ 1.000	$\widehat{\chi} = (\phi / \phi_0)^{-w} \chi$	$\widehat{\chi} = (\phi / \phi_0)^{-w} \chi$	where $\widehat{\chi} = (\phi / \phi_0)^{-w} \chi$ and we have defined the derivative
1510.06699_F00389	18	■ 0.996	$\widehat{\mathcal{D}}_a \equiv \widehat{h}_a{}^\mu \mathcal{D}_\mu$	$\widehat{\mathcal{D}}_a \equiv \widehat{h}_a{}^\mu \mathcal{D}_\mu$	operators $\widehat{\mathcal{D}}_a \equiv \widehat{h}_a{}^\mu \mathcal{D}_\mu$ and $\widehat{\mathcal{D}}_a^* = \widehat{\mathcal{D}}_a + w \widehat{\mathcal{B}}_a$, the latter of
1510.06699_F00390	18	■ 0.996	$\widehat{\mathcal{D}}_a^* = \widehat{\mathcal{D}}_a + w \widehat{\mathcal{B}}_a$	$\widehat{\mathcal{D}}_a^* = \widehat{\mathcal{D}}_a + w \widehat{\mathcal{B}}_a$	operators $\widehat{\mathcal{D}}_a \equiv \widehat{h}_a{}^\mu \mathcal{D}_\mu$ and $\widehat{\mathcal{D}}_a^* = \widehat{\mathcal{D}}_a + w \widehat{\mathcal{B}}_a$, the latter of
1510.06699_F00391	18	■ 1.000	$(\phi / \phi_0)^{1-w}$	$(\phi / \phi_0)^{1-w}$	all multiplicative factor $(\phi / \phi_0)^{1-w}$, the scale-invariant
1510.06699_F00392	18	■ 1.000	$\widehat{\mathcal{D}}_a^* \widehat{\chi}$	$\widehat{\mathcal{D}}_a^* \widehat{\chi}$	quantity $\widehat{\mathcal{D}}_a^* \widehat{\chi}$ has the <i>same</i> functional dependency on $\widehat{\chi}$,
1510.06699_F00393	18	■ 1.000	$\widehat{\chi}$	$\widehat{\chi}$	quantity $\widehat{\mathcal{D}}_a^* \widehat{\chi}$ has the <i>same</i> functional dependency on $\widehat{\chi}$,
1510.06699_F00394	18	■ 0.999	$\widehat{h}_a{}^\mu, \widehat{A}^{ab}_\mu$	$\widehat{h}_a{}^\mu, \widehat{A}^{ab}_\mu$	$\widehat{h}_a{}^\mu, \widehat{A}^{ab}_\mu$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}_a^* \chi$ does on χ , $h_a{}^\mu$,
1510.06699_F00395	18	■ 0.999	$\widehat{B}_\mu$	$\widehat{B}_\mu$	$\widehat{h}_a{}^\mu, \widehat{A}^{ab}_\mu$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}_a^* \chi$ does on χ , $h_a{}^\mu$,
1510.06699_F00396	18	■ 0.999	$\mathcal{D}_a^* \chi$	$\mathcal{D}_a^* \chi$	$\widehat{h}_a{}^\mu, \widehat{A}^{ab}_\mu$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}_a^* \chi$ does on χ , $h_a{}^\mu$,
1510.06699_F00397	18	■ 0.999	$\chi, h_a{}^\mu$	$\chi, h_a{}^\mu$	$\widehat{h}_a{}^\mu, \widehat{A}^{ab}_\mu$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}_a^* \chi$ does on χ , $h_a{}^\mu$,
1510.06699_F00398	18	■ 1.000	$\mathcal{R}^{ab}_{cd} = (\phi / \phi_0)^2 \widehat{\mathcal{R}}^{ab}_{cd}, \mathcal{T}^{*a}_{bc} = (\phi / \phi_0) \widehat{\mathcal{T}}^{*a}_{bc}$	$\mathcal{R}^{ab}_{cd} = (\phi / \phi_0)^2 \widehat{\mathcal{R}}^{ab}_{cd}, \mathcal{T}^{*a}_{bc} = (\phi / \phi_0) \widehat{\mathcal{T}}^{*a}_{bc}$	ten as $\mathcal{R}^{ab}_{cd} = (\phi / \phi_0)^2 \widehat{\mathcal{R}}^{ab}_{cd}, \mathcal{T}^{*a}_{bc} = (\phi / \phi_0) \widehat{\mathcal{T}}^{*a}_{bc}$ and
1510.06699_F00399	18	■ 1.000	$\mathcal{H}_{ab} = (\phi / \phi_0)^2 \widehat{\mathcal{H}}_{ab}$	$\mathcal{H}_{ab} = (\phi / \phi_0)^2 \widehat{\mathcal{H}}_{ab}$	$\mathcal{H}_{ab} = (\phi / \phi_0)^2 \widehat{\mathcal{H}}_{ab}$, where each quantity with a caret is
1510.06699_F00400	18	■ 1.000	$h^{-1} = (\phi / \phi_0)^{-4} \widehat{h}^{-1}$	$h^{-1} = (\phi / \phi_0)^{-4} \widehat{h}^{-1}$	Noting that $h^{-1} = (\phi / \phi_0)^{-4} \widehat{h}^{-1}$, one is thus led to the
1510.06699_F00401	18	■ 1.000	$\varphi, h_a{}^\mu, A^{ab}_\mu$	$\varphi, h_a{}^\mu, A^{ab}_\mu$	Thus, if χ represents φ , $h_a{}^\mu$, A^{ab}_μ or B_μ , one may im-
1510.06699_F00402	18	■ 1.000	$\delta \mathcal{L}_T / \delta \widehat{\chi} = 0$	$\delta \mathcal{L}_T / \delta \widehat{\chi} = 0$	motion $\delta \mathcal{L}_T / \delta \widehat{\chi} = 0$ has the <i>same</i> functional form as
1510.06699_F00403	18	■ 1.000	$\delta \mathcal{L}_T / \delta \chi _{\phi=\phi_0} = 0$	$\delta \mathcal{L}_T / \delta \chi _{\phi=\phi_0} = 0$	(the corresponding term in) $\delta \mathcal{L}_T / \delta \chi _{\phi=\phi_0} = 0$, but with
1510.06699_F00404	18	■ 1.000	$\delta \mathcal{L}_T / \delta \phi = \delta \mathcal{L}_M / \delta \phi$	$\delta \mathcal{L}_T / \delta \phi = \delta \mathcal{L}_M / \delta \phi$	variables, one begins by noting that $\delta \mathcal{L}_T / \delta \phi = \delta \mathcal{L}_M / \delta \phi$,

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1510.06699_F00405	18	■ 1.000	$\backslash\mathrm{delta}\backslash\mathrm{mathcal{L}}_{\backslash\mathrm{mathrm{T}}}\backslash\mathrm{delta}\backslash\mathrm{varphi}=\mathrm{delta}\backslash\mathrm{mathcal{L}}_{\backslash\mathrm{mathrm{M}}}\backslash\mathrm{delta}\backslash\mathrm{varphi}=0$	$\delta\mathcal{L}_T/\delta\varphi = \delta\mathcal{L}_M/\delta\varphi = 0$	$\delta\mathcal{L}_T/\delta\varphi = \delta\mathcal{L}_M/\delta\varphi = 0$ and set $\phi = \phi_0$ in the result-
1510.06699_F00406	18	■ 1.000	$\backslash\mathrm{partial}\backslash\mathrm{mathcal{L}}_{\backslash\mathrm{mathrm{M}}}\backslash\mathrm{partial}\backslash\mathrm{widehat{B}}_{\backslash\mathrm{mu}}$	$\partial\mathcal{L}_M/\partial\widehat{B}_\mu$	may replace $\partial\mathcal{L}_M/\partial\widehat{B}_\mu$ by $\delta\mathcal{L}_M/\delta\widehat{B}_\mu$ in the above equa-
1510.06699_F00407	18	■ 1.000	$\backslash\mathrm{delta}\backslash\mathrm{mathcal{L}}_{\backslash\mathrm{mathrm{M}}}\backslash\mathrm{delta}\backslash\mathrm{widehat{B}}_{\backslash\mathrm{mu}}$	$\delta\mathcal{L}_M/\delta\widehat{B}_\mu$	may replace $\partial\mathcal{L}_M/\partial\widehat{B}_\mu$ by $\delta\mathcal{L}_M/\delta\widehat{B}_\mu$ in the above equa-
1510.06699_F00408	19	■ 1.000	$\mathrm{m}\rightarrow\mu\phi$	$m \rightarrow \mu\phi$	making the replacement $m \rightarrow \mu\phi$, as in (70). One begins
1510.06699_F00409	19	■ 1.000	$\mathrm{frac{1}{2}}$	$\frac{1}{2}$	by constructing the appropriate action for a spin- $\frac{1}{2}$ point
1510.06699_F00410	19	■ 1.000	$\mathrm{p}_{\backslash\mathrm{mu}}\rightarrow\mathrm{p}_{\backslash\mathrm{a}}\equiv\mathrm{h}_{\backslash\mathrm{a}}^{\backslash\mathrm{mu}}\mathrm{p}_{\backslash\mathrm{mu}}$	$p_\mu \rightarrow p_a \equiv h_a{}^\mu p_\mu$	needs only to make the replacements $p_\mu \rightarrow \underline{p}_a \equiv h_a{}^\mu p_\mu$,
1510.06699_F00411	19	■ 1.000	$\mathrm{dot{x}}^{\backslash\mathrm{mu}}\rightarrow\mathrm{v}^{\backslash\mathrm{a}}\equiv\mathrm{b}^{\backslash\mathrm{a}}_{\backslash\mathrm{mu}}\mathrm{dot{x}}^{\backslash\mathrm{mu}}$	$\dot{x}^\mu \rightarrow v^a \equiv b^a{}_\mu \dot{x}^\mu$	$\dot{x}^\mu \rightarrow v^a \equiv b^a{}_\mu \dot{x}^\mu$ and $m \rightarrow \mu\phi$, which gives ⁷⁵
1510.06699_F00412	19	■ 1.000	$\mathrm{p}^{\backslash\mathrm{a}}(\backslash\mathrm{lambda})$	$p^a(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00413	19	■ 1.000	$\mathrm{v}^{\backslash\mathrm{a}}(\backslash\mathrm{lambda})$	$v^a(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00414	19	■ 1.000	$\mathrm{e}(\backslash\mathrm{lambda})$	$e(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00415	19	■ 1.000	$\backslash\mathrm{lambda}$	λ	along the worldline, which is parameterised by λ . We
1510.06699_F00416	19	■ 1.000	$\mathrm{w}\mathrm{left}(\mathrm{p}^{\backslash\mathrm{a}}\mathrm{right})=-1,\mathrm{w}\mathrm{left}(\mathrm{v}^{\backslash\mathrm{a}}\mathrm{right})=0,\mathrm{w}(\mathrm{e})=1,\mathrm{w}(\backslash\mathrm{lambda})=1$	$w(p^a) = -1, w(v^a) = 0, w(e) = 1, w(\lambda) = 1$	(96) are $w(p^a) = -1, w(v^a) = 0, w(e) = 1, w(\lambda) = 1$ and
1510.06699_F00417	19	■ 1.000	$\mathrm{p}^{\backslash\mathrm{2}}\equiv\mathrm{p}_{\backslash\mathrm{a}}\mathrm{p}^{\backslash\mathrm{a}}$	$p^2 \equiv p_a p^a$	where $p^2 \equiv p_a p^a$ and the quantities $c^c{}_{ab}$ are the PGT
1510.06699_F00418	19	■ 1.000	$\mathrm{c}^{\backslash\mathrm{c}}_{\backslash\mathrm{a}\backslash\mathrm{b}}$	$c^c{}_{ab}$	where $p^2 \equiv p_a p^a$ and the quantities $c^c{}_{ab}$ are the PGT
1510.06699_F00419	19	■ 1.000	$\mathrm{D}^{\backslash\mathrm{c}}_{\backslash\mathrm{c}}$	${}^0\mathcal{D}_c^*$	where ${}^0\mathcal{D}_c^*$ is the reduced WGT covariant derivative, de-
1510.06699_F00420	19	■ 1.000	$\mathrm{e}=1/\mathrm{mu}\phi$	$e = 1/(\mu\phi)$	one chooses $e = 1/(\mu\phi)$, in which case $v^2 = 1$ and λ cor-
1510.06699_F00421	19	■ 1.000	$\mathrm{v}^{\backslash\mathrm{2}}=1$	$v^2 = 1$	one chooses $e = 1/(\mu\phi)$, in which case $v^2 = 1$ and λ cor-
1510.06699_F00422	19	■ 1.000	tau	τ	responds to the proper time τ along the worldline; in this
1510.06699_F00423	19	■ 0.348	$\mathrm{76-78}$	76–78	terms of the reduced PGT-covariant derivative as ^{76–78}
1510.06699_F00424	19	■ 0.996	$\mathrm{A}^{\backslash\mathrm{a}\backslash\mathrm{b}}_{\backslash\mathrm{mu}}$	${}^0A^{ab}{}_\mu$	Appendix B), the quantities ${}^0A^{ab}{}_\mu$ appearing in ${}^0\mathcal{D}_b$ con-
1510.06699_F00425	19	■ 0.996	$\mathrm{D}_{\backslash\mathrm{b}}$	${}^0\mathcal{D}_b$	Appendix B), the quantities ${}^0A^{ab}{}_\mu$ appearing in ${}^0\mathcal{D}_b$ con-
1510.06699_F00426	19	■ 0.906	$\mathrm{v}^{\backslash\mathrm{b}}\mathrm{D}_{\backslash\mathrm{b}}\mathrm{v}_{\backslash\mathrm{a}}=0$	$v^b\mathcal{D}_b v_a = 0$	the equation of motion $v^b\mathcal{D}_b v_a = 0$). We note that
1510.06699_F00427	19	■ 1.000	$\mathrm{79,80}$	^{79,80}	this conclusion differs from some previous studies ^{79,80} ,
1510.06699_F00428	19	■ 1.000	$\mathrm{q}_{\backslash\mathrm{W}}\mathrm{H}_{\backslash\mathrm{a}\backslash\mathrm{b}}\mathrm{v}^{\backslash\mathrm{b}}$	$(q_W/m)\mathcal{H}_{ab}v^b$	$(q_W/m)\mathcal{H}_{ab}v^b$, where q_W is the particle’s Weyl charge
1510.06699_F00429	19	■ 1.000	$\mathrm{q}_{\backslash\mathrm{W}}$	q_W	$(q_W/m)\mathcal{H}_{ab}v^b$, where q_W is the particle’s Weyl charge
1510.06699_F00430	19	■ 1.000	$\mathrm{mu}=0$	$\mu = 0$	By setting $\mu = 0$ in the action (96) and choosing the
1510.06699_F00431	19	■ 1.000	$\mathrm{e}=1$	$e = 1$	einbein $e = 1$, such that $v^2 = 0$, one finds that the
1510.06699_F00432	19	■ 1.000	$\mathrm{v}^{\backslash\mathrm{2}}=0$	$v^2 = 0$	einbein $e = 1$, such that $v^2 = 0$, one finds that the
1510.06699_F00433	20	■ 1.000	$\mathrm{p}_{\backslash\mathrm{a}},\mathrm{v}^{\backslash\mathrm{a}}$	$\widehat{p}_a, \widehat{v}^a$	where $\widehat{p}_a, \widehat{v}^a$ and $\widehat{e}$ are to be considered as functions of
1510.06699_F00434	20	■ 1.000	e	$\widehat{e}$	where $\widehat{p}_a, \widehat{v}^a$ and $\widehat{e}$ are to be considered as functions of
1510.06699_F00435	20	■ 0.990	$\backslash\mathrm{widehat{\lambda}}$	$\widehat{\lambda}$	$\widehat{\lambda}$, such that for example $\widehat{p}_a(\widehat{\lambda}) \equiv \widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$. It is then
1510.06699_F00436	20	■ 0.990	$\backslash\mathrm{widehat{p}}_{\backslash\mathrm{a}}(\backslash\mathrm{widehat{\lambda}})\equiv\backslash\mathrm{widehat{h}}_{\backslash\mathrm{a}}^{\backslash\mathrm{mu}}\mathrm{p}_{\backslash\mathrm{mu}}(\backslash\mathrm{lambda}(\backslash\mathrm{widehat{\lambda}}))$	$\widehat{p}_a(\widehat{\lambda}) \equiv \widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$	$\widehat{\lambda}$, such that for example $\widehat{p}_a(\widehat{\lambda}) \equiv \widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$. It is then
1510.06699_F00437	20	■ 1.000	$\backslash\mathrm{widehat{\epsilon}}=\mathrm{left}[1-\backslash\mathrm{widehat{\lambda}}\mathrm{left(d}\mathrm{x}^{\backslash\mathrm{mu}}/\mathrm{d}\backslash\mathrm{widehat{\lambda}}\mathrm{right})\backslash\mathrm{partial}_{\backslash\mathrm{mu}}\mathrm{ln}\mathrm{left(\phi/\phi_0right)}\mathrm{right}]\backslash\mathrm{widehat{e}}$	$\widehat{\epsilon} = \left[1 - \widehat{\lambda} \left(dx^\mu/d\widehat{\lambda}\right) \partial_\mu \ln(\phi/\phi_0)\right] \widehat{e}$	bein $\widehat{\epsilon} = [1 - \widehat{\lambda}(dx^\mu/d\widehat{\lambda})\partial_\mu \ln(\phi/\phi_0)]\widehat{e}$. Thus, in terms
1510.06699_F00438	20	■ 1.000	$\backslash\mathrm{widehat{\lambda}},\backslash\mathrm{widehat{p}}_{\backslash\mathrm{a}},\backslash\mathrm{widehat{v}}^{\backslash\mathrm{a}}$	$\widehat{\lambda}, \widehat{p}_a, \widehat{v}^a$	of the scale-invariant variables $\widehat{\lambda}, \widehat{p}_a, \widehat{v}^a$ and $\widehat{\epsilon}$, respec-
1510.06699_F00439	20	■ 1.000	$\backslash\mathrm{widehat{\epsilon}}$	$\widehat{\epsilon}$	of the scale-invariant variables $\widehat{\lambda}, \widehat{p}_a, \widehat{v}^a$ and $\widehat{\epsilon}$, respec-
1510.06699_F00440	20	■ 1.000	$\backslash\mathrm{lambda},\mathrm{p}_{\backslash\mathrm{a}},\mathrm{v}^{\backslash\mathrm{a}}$	λ, p_a, v^a	it does in terms of the original variables λ, p_a, v^a and e
1510.06699_F00441	20	■ 0.999	$\backslash\mathrm{widehat{h}}_{\backslash\mathrm{a}}^{\backslash\mathrm{mu}}$	$\widehat{h}_a{}^\mu$	counterpart $\widehat{h}_a{}^\mu$ or $\widehat{b}_\mu^a$. Thus, the calculation presented
1510.06699_F00442	20	■ 0.999	$\backslash\mathrm{widehat{b}}_{\backslash\mathrm{mu}}^{\backslash\mathrm{a}}$	$\widehat{b}_\mu^a$	counterpart $\widehat{h}_a{}^\mu$ or $\widehat{b}_\mu^a$. Thus, the calculation presented
1510.06699_F00443	20	■ 1.000	$\mathrm{D}_{\backslash\mathrm{a}}$	${}^0\widehat{\mathcal{D}}_a \equiv$	where we have defined the derivative operator ${}^0\widehat{\mathcal{D}}_a \equiv$

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1510.06699_F00444	20	■0.977	$\widehat{h}_{a\{ }^{\mu 0} \mathcal{D}_{\mu}$	$\widehat{h}_a^{\mu 0} \mathcal{D}_{\mu}$	$\widehat{h}_a^{\mu} {}^0 \mathcal{D}_{\mu}$.
1510.06699_F00445	20	■1.000	$\{ }^{\wedge\{0} \mathcal{D}_{\mu\}^{\wedge\{*}$	${}^0 D_{\mu}^*$	variant derivative operator ${}^0 D_{\mu}^*$ in (48), to which the
1510.06699_F00446	20	■0.837	$\mathcal{T}_{a\ b\ c\}^{\wedge\{*} \equiv 0$	$\mathcal{T}_{abc}^* \equiv 0$	$\mathcal{T}_{abc}^* \equiv 0$. Hence such quantities may be written entirely
1510.06699_F00447	20	■1.000	$\{ }^{\wedge\{0} \mathcal{R}^{\{*} \ a\ b\} \}_{\mu \nu} \left(h, \partial h, \partial^2 h, B, \partial B \right)$	${}^0 R^{*ab}{}_{\mu\nu} (h, \partial h, \partial^2 h, B, \partial B)$	or ${}^0 R^{*ab}{}_{\mu\nu} (h, \partial h, \partial^2 h, B, \partial B)$, which is again given by
1510.06699_F00448	20	■1.000	$\{ }^{\wedge\{0} \mathcal{A}^{\{*} \ a\ b\} \}_{\mu}$	${}^0 A^{*ab}{}_{\mu}$	the formula (39), but with $A^{ab}{}_{\mu}$ replaced by ${}^0 A^{*ab}{}_{\mu}$, and
1510.06699_F00449	20	■0.998	$\{ }^{\wedge\{0} \mathcal{R}^{\{*} \ a\ b\} \}_{c\ d} = h_{c\{ }^{\wedge\{0} \mathcal{R}^{\{*} \ a\ b\} \}_{\mu} h_{d\{ }^{\wedge\{0} \mathcal{R}^{\{*} \ a\ b\} \}_{\mu \nu}$	${}^0 \mathcal{R}^{*ab}{}_{cd} = h_c{}^{\mu} h_d{}^{\mu} {}^0 R^{*ab}{}_{\mu\nu}$	where ${}^0 \mathcal{R}^{*ab}{}_{cd} = h_c{}^{\mu} h_d{}^{\mu} {}^0 R^{*ab}{}_{\mu\nu}$. Unlike (41), the com-
1510.06699_F00450	20	■1.000	$\{ }^{\wedge\{0} \mathcal{T}^{\{*} \ a\} \}_{b\ c} \equiv$	${}^0 \mathcal{T}^{*a}{}_{bc} \equiv$	lational field strength of the h gauge field, since ${}^0 \mathcal{T}^{*a}{}_{bc} \equiv$
1510.06699_F00451	20	■1.000	$h_{b\{ }^{\wedge\{0} \mathcal{R}^{\{*} \ a\ b\} \}_{c\} h_{c\{ }^{\wedge\{0} \mathcal{R}^{\{*} \ a\ b\} \}_{d\} \left(\{ }^{\wedge\{0} \mathcal{D}_{\mu\}^{\wedge\{*} \ b^{\wedge\{a\} \}_{\nu} \{ }^{\wedge\{0} \mathcal{D}_{\nu\}^{\wedge\{*} \ b^{\wedge\{a\} \}_{\mu} \right) = 0$	$h_b{}^{\mu} h_c{}^{\nu} ({}^0 D_{\mu}^* b^a{}_{\nu} - {}^0 D_{\nu}^* b^a{}_{\mu}) = 0$	$h_b{}^{\mu} h_c{}^{\nu} ({}^0 D_{\mu}^* b^a{}_{\nu} - {}^0 D_{\nu}^* b^a{}_{\mu}) = 0$. As one might expect, the
1510.06699_F00452	20	■0.913	$\mathcal{D}_{a\}^{\wedge\{*} \rightarrow \mathcal{D}_{a\}^{\wedge\{*}, \mathcal{R}_{abcd} \rightarrow \mathcal{R}_{abcd}^*$	$\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*, \mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^*$	with the replacements $\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*, \mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^*$ and
1510.06699_F00453	20	■0.982	$\mathcal{T}_{a\ b\ c\}^{\wedge\{*} \rightarrow \{ }^{\wedge\{0} \mathcal{T}_{a\ b\ c\}^{\wedge\{*} \equiv 0$	$\mathcal{T}_{abc}^* \rightarrow {}^0 \mathcal{T}_{abc}^* \equiv 0$	$\mathcal{T}_{abc}^* \rightarrow {}^0 \mathcal{T}_{abc}^* \equiv 0$.
1510.06699_F00454	20	■1.000	$\{ }^{\wedge\{0} \mathcal{R}^{\{*} \ a\ b\} \}_{c\ d}$	${}^0 \mathcal{R}^{*ab}{}_{cd}$	${}^0 \mathcal{R}^{*ab}{}_{cd}$.
1510.06699_F00455	20	■1.000	$\mathcal{L}_{\mathrm{G}} = h^{-1} L_{\mathrm{G}}$	$\mathcal{L}_{\mathrm{G}} = h^{-1} L_{\mathrm{G}}$	density $\mathcal{L}_{\mathrm{G}} = h^{-1} L_{\mathrm{G}}$, where by analogy with (58),
1510.06699_F00456	21	■0.516	$\{ }^{\wedge\{0} \mathcal{R}_{\text{abcd}}^{\wedge\{*}$	${}^0 \mathcal{R}_{abcd}^*$	dimensions), since ${}^0 \mathcal{R}_{abcd}^*$ and its contractions also sat-
1510.06699_F00457	21	■0.999	$\mathcal{R}_{a\ b\ c\ d\} \rightarrow \{ }^{\wedge\{0} \mathcal{R}_{a\ b\ c\ d\}^{\wedge\{*}$	$\mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^*$	$\mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^*$. Thus, one can set to zero any one of the
1510.06699_F00458	21	■0.996	$\mathcal{D}_{a\}^{\wedge\{*} \rightarrow \mathcal{D}_{a\}^{\wedge\{0}$	$\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*$	$\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*$ and $\mathcal{T}_a^* \rightarrow {}^0 \mathcal{T}_a^* \equiv 0$. One should also note,
1510.06699_F00459	21	■0.996	$\mathcal{T}_{a\}^{\wedge\{*} \rightarrow \{ }^{\wedge\{0} \mathcal{T}_{a\}^{\wedge\{*} \equiv 0$	$\mathcal{T}_a^* \rightarrow {}^0 \mathcal{T}_a^* \equiv 0$	$\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*$ and $\mathcal{T}_a^* \rightarrow {}^0 \mathcal{T}_a^* \equiv 0$. One should also note,
1510.06699_F00460	21	■0.756	$\mathcal{D}_{a\}^{\wedge\{*} \rightarrow \mathcal{D}_{a\}^{\wedge\{*}, \mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^*$	$\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*, \mathcal{T}_{abc} \rightarrow {}^0 \mathcal{T}_{abc}^* \equiv 0$	with the replacements $\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*, \mathcal{T}_{abc} \rightarrow {}^0 \mathcal{T}_{abc}^* \equiv 0$ and
1510.06699_F00461	21	■1.000	$s_{a\ b\ c\} \equiv 0 \equiv \sigma_{a\ b\ c\}$	$s_{abc} \equiv 0 \equiv \sigma_{abc}$	$s_{abc} \equiv 0 \equiv \sigma_{abc}$. Furthermore, by performing analogous
1510.06699_F00462	21	■0.975	$\mathcal{T}_{a\ b\ c\}^{\wedge\{*} = 0$	$\mathcal{T}_{abc}^* = 0$	condition $\mathcal{T}_{abc}^* = 0$ in the field equations of WGT. In the
1510.06699_F00463	21	■1.000	$\{ }^{\wedge\{82-84}$	$82-84$	only for the Lovelock action ⁸²⁻⁸⁴ , which coincides with
1510.06699_F00464	21	■1.000	$D=4$	$D=4$	term) in $D=4$ dimensions.
1510.06699_F00465	21	■0.741	$\{ }^{\wedge\{\underline{85}\}$	85	frame vectors $e_{\mu}(x)$ are related by ⁸⁵
1510.06699_F00466	21	■0.968	$\hat{e}^{\{\boldsymbol{a}\}}(x)$	$\hat{e}^a(x)$	sis vectors $\hat{e}^a(x)$ and $e^{\mu}(x)$ in each set. For any other
1510.06699_F00467	21	■0.968	$\boldsymbol{e}^{\{\mu\}}(x)$	$\boldsymbol{e}^{\mu}(x)$	sis vectors $\hat{e}^a(x)$ and $e^{\mu}(x)$ in each set. For any other
1510.06699_F00468	21	■0.769	$J_{\mu} \boldsymbol{e}^{\{\mu\}}$	$J_{\mu} \boldsymbol{e}^{\mu}$	vector $\boldsymbol{J}$, written in the coordinate basis as (say) $J_{\mu} \boldsymbol{e}^{\mu}$,
1510.06699_F00469	21	■1.000	$J_{a\} = h_{a\{ }^{\wedge\{0} J_{\mu\} \}_{\mu}$	$J_a = h_a{}^{\mu} J_{\mu}$	one then identifies the quantities $J_a = h_a{}^{\mu} J_{\mu}$, for ex-
1510.06699_F00470	21	■1.000	$h_{a\{ }^{\wedge\{0} J^{\wedge\{a\}} = J^{\wedge\{0} \mu\}$	$h_a{}^{\mu} J^a = J^{\mu}$	that $h_a{}^{\mu} J^a = J^{\mu}$ and $b^a{}_{\mu} J^{\mu} = J^a$. Moreover, any con-
1510.06699_F00471	21	■1.000	$b^{\wedge\{a\} \{ }^{\wedge\{0} \mu\} \}_{\mu} J^{\wedge\{0} \mu\} = J^{\wedge\{0} a\}$	$b^a{}_{\mu} J^{\mu} = J^a$	that $h_a{}^{\mu} J^a = J^{\mu}$ and $b^a{}_{\mu} J^{\mu} = J^a$. Moreover, any con-
1510.06699_F00472	21	■1.000	$g_{\mu \nu}$	$g_{\mu\nu}$	a more general spacetime with metric $g_{\mu\nu}$. Conversely,
1510.06699_F00473	21	■1.000	$h^{\wedge\{-1\}} = \sqrt{-g}$	$h^{-1} = \sqrt{-g}$	From (115), one also finds that $h^{-1} = \sqrt{-g}$ (where we
1510.06699_F00474	22	■1.000	$w \left(g_{\mu \nu} \right) = 2$	$w \left(g_{\mu\nu} \right) = 2$	have Weyl weights $w(g_{\mu\nu}) = 2$ and $w(h_a{}^{\mu}) = -1$ respec-
1510.06699_F00475	22	■1.000	$w \left(h_{a\{ }^{\wedge\{0} \mu\} \right) = -1$	$w \left(h_a{}^{\mu} \right) = -1$	have Weyl weights $w(g_{\mu\nu}) = 2$ and $w(h_a{}^{\mu}) = -1$ respec-
1510.06699_F00476	22	■1.000	$w \left(\eta_{a\ b} \right) = 0$	$w \left(\eta_{ab} \right) = 0$	tively, and so (115) and (116) imply that $w(\eta_{ab}) = 0$, as
1510.06699_F00477	22	■1.000	$e_{\{a\} \{ }^{\wedge\{0} \mu\}$	$e_a{}^{\mu}$	$h_a{}^{\mu}$ and $b^a{}_{\mu}$ as $e_a{}^{\mu}$ and $e^a{}_{\mu}$, respectively.
1510.06699_F00478	22	■1.000	$e^{\wedge\{a\} \{ }^{\wedge\{0} \mu\}$	$e^a{}_{\mu}$	$h_a{}^{\mu}$ and $b^a{}_{\mu}$ as $e_a{}^{\mu}$ and $e^a{}_{\mu}$, respectively.
1510.06699_F00479	22	■1.000	$x + \delta x$	$x + \delta x$	and $x + \delta x$, which is accompanied by a local change in
1510.06699_F00480	22	■1.000	$J^{\wedge\{a\}$	J^a	for some vector J^a of weight w is defined as
1510.06699_F00481	22	■1.000	$J^{\wedge\{a\}}(x)$	$J^a(x)$	which is required to compare vectors $J^a(x)$ and $J^a(x + \delta x)$

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1510.06699_F00482	22	■ 1.000	$J^{\mathbf{a}}(x+\delta x)$	$J^a(x+\delta x)$	which is required to compare vectors $J^a(x)$ and $J^a(x+\delta x)$
1510.06699_F00483	22	■ 1.000	$\hat{\mathbf{e}}_{\mathbf{a}}(x+\delta x)$	$\hat{e}_a(x+\delta x)$	tetrad frames $\hat{e}_a(x)$ and $\hat{e}_a(x+\delta x)$ respectively. Hence,
1510.06699_F00484	22	■ 1.000	∂_{μ}^*	∂_{μ}^*	where we have used the partial derivative operator ∂_{μ}^*
1510.06699_F00485	22	■ 1.000	$\eta_{\mathbf{a} \mathbf{b}}$	η_{ab}	the Lorentz metric η_{ab} at each point. Then demanding
1510.06699_F00486	22	■ 1.000	$A^{\mathbf{a} \mathbf{b}}\{ \}_{\mathbf{\mu}}=-A^{\mathbf{b} \mathbf{a}}\{ \}_{\mathbf{\mu}}$	$A^{ab}_{\mu}=-A^{ba}_{\mu}$	antisymmetric, i.e. $A^{ab}_{\mu}=-A^{ba}_{\mu}$, as previously.
1510.06699_F00487	22	■ 0.528	$\nabla_{\mathbf{\mu}}^*=\partial_{\mathbf{\mu}}^*+\Gamma^{\mathbf{\sigma}}_{\mathbf{\rho} \mathbf{\mu}}\mathbf{X}^{\mathbf{\rho}}_{\mathbf{\sigma}}$ $\rho \mathrm{~}\mu \mathrm{~}\mathbf{X}^{\mathbf{\rho}}_{\mathbf{\sigma}} \mathrm{~}\mathbf{X}^{\mathbf{\rho}}_{\mathbf{\sigma}}$	$\nabla_{\mu}^*=\partial_{\mu}^*+\Gamma^{\sigma}_{\rho\mu}\mathbf{X}^{\rho}_{\sigma}$	where $\nabla_{\mu}^*=\partial_{\mu}^*+\Gamma^{\sigma}_{\rho\mu}\mathbf{X}^{\rho}_{\sigma}$ and $\mathbf{X}^{\rho}_{\sigma}$ are the $\mathrm{GL}(4,R)$ gen-
1510.06699_F00488	22	■ 1.000	$\Delta_{\mathbf{\mu}}^*$	Δ_{μ}^*	of the field to which Δ_{μ}^* is applied. If a field ψ carries
1510.06699_F00489	22	■ 1.000	$\nabla_{\mathbf{\mu}}^*\psi=\partial_{\mathbf{\mu}}^*\psi$	$\nabla_{\mu}^*\psi=\partial_{\mu}^*\psi$	only Latin indices, then $\nabla_{\mu}^*\psi=\partial_{\mu}^*\psi$ and so $\Delta_{\mu}^*\psi=D_{\mu}^*\psi$;
1510.06699_F00490	22	■ 1.000	$\Delta_{\mathbf{\mu}}^*\psi=D_{\mathbf{\mu}}^*\psi$	$\Delta_{\mu}^*\psi=D_{\mu}^*\psi$	only Latin indices, then $\nabla_{\mu}^*\psi=\partial_{\mu}^*\psi$ and so $\Delta_{\mu}^*\psi=D_{\mu}^*\psi$;
1510.06699_F00491	22	■ 1.000	$D_{\mathbf{\mu}}^*\psi=\partial_{\mathbf{\mu}}^*\psi$	$D_{\mu}^*\psi=\partial_{\mu}^*\psi$	$D_{\mu}^*\psi=\partial_{\mu}^*\psi$ and so $\Delta_{\mu}^*\psi=\nabla_{\mu}^*\psi$. When acting on an ob-
1510.06699_F00492	22	■ 1.000	$\Delta_{\mathbf{\mu}}^*\psi=\nabla_{\mathbf{\mu}}^*\psi$	$\Delta_{\mu}^*\psi=\nabla_{\mu}^*\psi$	$D_{\mu}^*\psi=\partial_{\mu}^*\psi$ and so $\Delta_{\mu}^*\psi=\nabla_{\mu}^*\psi$. When acting on an ob-
1510.06699_F00493	22	■ 1.000	$\Gamma^{\mathbf{\sigma}}_{\mathbf{\rho} \mathbf{\mu}}$	$\Gamma^{\sigma}_{\rho\mu}$	connection coefficients $\Gamma^{\sigma}_{\rho\mu}$ are themselves dynamical
1510.06699_F00494	22	■ 1.000	$J^{\mathbf{a}}+\delta J^{\mathbf{a}}$	$J^a+\delta J^a$	ported from x to $x+\delta x$, be equal to $J^a+\delta J^a$, i.e.
1510.06699_F00495	22	■ 1.000	Γ	Γ	which relates A and Γ (and B); in particular, we note that
1510.06699_F00496	22	■ 1.000	$w(\Gamma^{\mathbf{\sigma}}_{\mathbf{\nu} \mathbf{\mu}})=0$	$w(\Gamma^{\sigma}_{\nu\mu})=0$	$w(\Gamma^{\sigma}_{\nu\mu})=0$. The relation (121) is sometimes known as
1510.06699_F00497	22	■ 1.000	$\nabla_{\mathbf{\sigma}}^*g_{\mathbf{\mu} \mathbf{\nu}}=0$	$\nabla_{\sigma}^*g_{\mu\nu}=0$	Using (115) and (121), one finds that $\nabla_{\sigma}^*g_{\mu\nu}=0$, and
1510.06699_F00498	22	■ 1.000	$T^{*\mathbf{a}}_{\mathbf{\mu} \mathbf{\nu}}$	$T^{*a}_{\mu\nu}$	the gauge field strengths $R^{ab}_{\mu\nu}$ and $T^{*a}_{\mu\nu}$, one finds that
1510.06699_F00499	22	■ 1.000	$R^{\mathbf{\rho}}_{\mathbf{\sigma} \mathbf{\mu} \mathbf{\nu}}=e_{\mathbf{a}}^{\mathbf{\rho}}e^{\mathbf{b}}_{\mathbf{\sigma}}R^{\mathbf{a}}_{\mathbf{b} \mathbf{\mu} \mathbf{\nu}}$ $\sigma \mathrm{~}\mu \mathrm{~}\nu \mathrm{~}\mathbf{e}^{\mathbf{b}}_{\mathbf{\sigma}} \mathrm{~}\mathbf{e}^{\mathbf{b}}_{\mathbf{\sigma}}$	$R^{\rho}_{\sigma\mu\nu}=e_a{}^{\rho}e^b{}_{\sigma}R^a{}_{b\mu\nu}$	where $R^{\rho}_{\sigma\mu\nu}=e_a{}^{\rho}e^b{}_{\sigma}R^a{}_{b\mu\nu}$ and $T^{*\lambda}_{\mu\nu}=e_a{}^{\lambda}T^{*a}_{\mu\nu}$.
1510.06699_F00500	22	■ 1.000	$T^{*\mathbf{\lambda}}_{\mathbf{\mu} \mathbf{\nu}}=e_{\mathbf{a}}^{\mathbf{\lambda}}T^{*\mathbf{a}}_{\mathbf{\mu} \mathbf{\nu}}$	$T^{*\lambda}_{\mu\nu}=e_a{}^{\lambda}T^{*a}_{\mu\nu}$	where $R^{\rho}_{\sigma\mu\nu}=e_a{}^{\rho}e^b{}_{\sigma}R^a{}_{b\mu\nu}$ and $T^{*\lambda}_{\mu\nu}=e_a{}^{\lambda}T^{*a}_{\mu\nu}$.
1510.06699_F00501	22	■ 1.000	$T^{*\mathbf{\lambda}}_{\mathbf{\mu} \mathbf{\nu}}$	$T^{*\lambda}_{\mu\nu}$	Thus, although we recognise $T^{*\lambda}_{\mu\nu}$ as (minus) the tor-
1510.06699_F00502	22	■ 1.000	$R^{\mathbf{\rho}}_{\mathbf{\sigma} \mathbf{\mu} \mathbf{\nu}}$	$R^{\rho}_{\sigma\mu\nu}$	sion tensor of the Y_4 spacetime, we see that $R^{\rho}_{\sigma\mu\nu}$ is not
1510.06699_F00503	22	■ 1.000	$\widetilde{R}_{\mathbf{\rho} \mathbf{\sigma} \mathbf{\mu} \mathbf{\nu}}$	$\widetilde{R}_{\rho\sigma\mu\nu}$	One should note that, although $\widetilde{R}_{\rho\sigma\mu\nu}$ is antisymmet-
1510.06699_F00504	22	■ 0.774	μ, ν	μ, ν	ric in (μ, ν) , it is no longer antisymmetric in (ρ, σ) (in-
1510.06699_F00505	22	■ 0.774	ρ, σ	ρ, σ	ric in (μ, ν) , it is no longer antisymmetric in (ρ, σ) (in-
1510.06699_F00506	22	■ 0.764	$\widetilde{R}_{(\mathbf{\rho} \mathbf{\sigma}) \mathbf{\mu} \mathbf{\nu}}=g_{\mathbf{\rho} \mathbf{\sigma}}H_{\mathbf{\mu} \mathbf{\nu}}$ $\rho \mathrm{~}\sigma \mathrm{~}\mu \mathrm{~}\nu \mathrm{~}H_{\mathbf{\mu} \mathbf{\nu}}$	$\widetilde{R}_{(\rho\sigma)\mu\nu}=g_{\rho\sigma}H_{\mu\nu}$	deed $\widetilde{R}_{(\rho\sigma)\mu\nu}=g_{\rho\sigma}H_{\mu\nu}$) and does not satisfy the fa-
1510.06699_F00507	22	■ 1.000	$V_{\mathbf{4}}$	V_4	sor in a Riemannian V_4 spacetime. One may also show
1510.06699_F00508	22	■ 1.000	$\widetilde{R}^{\mathbf{\rho}}_{\mathbf{\sigma} \mathbf{\mu} \mathbf{\nu}}$	$\widetilde{R}^{\rho}_{\sigma\mu\nu}$	that, with the given arrangements of indices, both $\widetilde{R}^{\rho}_{\sigma\mu\nu}$
1510.06699_F00509	22	■ 1.000	$\widetilde{R}_{\mathbf{\mu} \mathbf{\nu}}\equiv\widetilde{R}_{\mathbf{\mu} \mathbf{\lambda} \mathbf{\nu}}^{\mathbf{\lambda}}=R_{\mathbf{\mu} \mathbf{\nu}}-H_{\mathbf{\mu} \mathbf{\nu}}$ $\mu \mathrm{~}\nu \mathrm{~}\lambda \mathrm{~}R_{\mathbf{\mu} \mathbf{\nu}} \mathrm{~}H_{\mathbf{\mu} \mathbf{\nu}}$	$\widetilde{R}_{\mu\nu}\equiv\widetilde{R}_{\mu\lambda\nu}{}^{\lambda}=R_{\mu\nu}-H_{\mu\nu}$	$\widetilde{R}_{\mu\nu}\equiv\widetilde{R}_{\mu\lambda\nu}{}^{\lambda}=R_{\mu\nu}-H_{\mu\nu}$ and $\widetilde{R}\equiv\widetilde{R}^{\mu}_{\mu}=R$. As one
1510.06699_F00510	22	■ 1.000	$\widetilde{R}\equiv\widetilde{R}^{\mathbf{\mu}}_{\mathbf{\mu}}=R$	$\widetilde{R}\equiv\widetilde{R}^{\mu}_{\mu}=R$	$\widetilde{R}_{\mu\nu}\equiv\widetilde{R}_{\mu\lambda\nu}{}^{\lambda}=R_{\mu\nu}-H_{\mu\nu}$ and $\widetilde{R}\equiv\widetilde{R}^{\mu}_{\mu}=R$. As one
1510.06699_F00511	23	■ 1.000	$J^{\mathbf{\rho}}$	J^{ρ}	operators acting on a vector J^{ρ} (say) of Weyl weight w ,
1510.06699_F00512	23	■ 1.000	$\{ \}^{\mathbf{0}}\Gamma^{\mathbf{\lambda}}_{\mathbf{\mu} \mathbf{\nu}}\equiv\frac{1}{2}g^{\mathbf{\lambda} \mathbf{\rho}}(\partial_{\mathbf{\mu}}g_{\mathbf{\nu} \mathbf{\rho}}+\partial_{\mathbf{\nu}}g_{\mathbf{\mu} \mathbf{\rho}}-\partial_{\mathbf{\rho}}g_{\mathbf{\mu} \mathbf{\nu}})$ $\lambda \mathrm{~}\rho \mathrm{~}\left(\partial_{\mathbf{\mu}}g_{\mathbf{\nu} \mathbf{\rho}}+\partial_{\mathbf{\nu}}g_{\mathbf{\mu} \mathbf{\rho}}-\partial_{\mathbf{\rho}}g_{\mathbf{\mu} \mathbf{\nu}}\right)$	${}^0\Gamma^{\lambda}_{\mu\nu}\equiv\frac{1}{2}g^{\lambda\rho}(\partial_{\mu}g_{\nu\rho}+\partial_{\nu}g_{\mu\rho}-\partial_{\rho}g_{\mu\nu})$	in which ${}^0\Gamma^{\lambda}_{\mu\nu}\equiv\frac{1}{2}g^{\lambda\rho}(\partial_{\mu}g_{\nu\rho}+\partial_{\nu}g_{\mu\rho}-\partial_{\rho}g_{\mu\nu})$ is the
1510.06699_F00513	23	■ 1.000	$K^{*\mathbf{\lambda}}_{\mathbf{\mu} \mathbf{\nu}}$	$K^{*\lambda}_{\mu\nu}$	standard metric (Christoffel) connection and $K^{*\lambda}_{\mu\nu}$ is
1510.06699_F00514	23	■ 1.000	$K_{\mathbf{\lambda} \mathbf{\mu} \mathbf{\nu}}^*=-K_{\mathbf{\mu} \mathbf{\lambda} \mathbf{\nu}}^*$	$K^*_{\lambda\mu\nu}=-K^*_{\mu\lambda\nu}$	for which it is worth noting that $K^*_{\lambda\mu\nu}=-K^*_{\mu\lambda\nu}$.
1510.06699_F00515	23	■ 1.000	$\{ \}^{\mathbf{0}}\Gamma^{*\mathbf{\sigma}}_{\mathbf{\rho} \mathbf{\mu}}$	${}^0\Gamma^{*\sigma}_{\rho\mu}$	connection ${}^0\Gamma^{*\sigma}_{\rho\mu}$ represent the same geometrical object
1510.06699_F00516	23	■ 0.661	$\Gamma^{\mathbf{\sigma}}_{\mathbf{\nu} \mathbf{\mu}}\rightarrow{}^0\Gamma^{*\mathbf{\sigma}}_{\mathbf{\nu} \mathbf{\mu}}$	$\Gamma^{\sigma}_{\nu\mu}\rightarrow{}^0\Gamma^{*\sigma}_{\nu\mu}$	again hold with the replacements $\Gamma^{\sigma}_{\nu\mu}\rightarrow{}^0\Gamma^{*\sigma}_{\nu\mu}$ and
1510.06699_F00517	23	■ 1.000	$A^{\mathbf{a}}_{\mathbf{b} \mathbf{\mu}}\rightarrow{}^0A^{*\mathbf{a}}_{\mathbf{b} \mathbf{\mu}}$	$A^a{}_{b\mu}\rightarrow{}^0A^{*a}{}_{b\mu}$	$A^a{}_{b\mu}\rightarrow{}^0A^{*a}{}_{b\mu}$, from which one can directly derive (131).
1510.06699_F00518	23	■ 1.000	$\{ \}^{\mathbf{0}}\nabla_{\mathbf{\sigma}}^*g_{\mathbf{\mu} \mathbf{\nu}}=0$	${}^0\nabla_{\sigma}^*g_{\mu\nu}=0$	One also obtains the metricity condition ${}^0\nabla_{\sigma}^*g_{\mu\nu}=0$. Fi-
1510.06699_F00519	23	■ 0.983	$R^{\mathbf{\rho}}_{\mathbf{\sigma} \mathbf{\mu} \mathbf{\nu}}\rightarrow{}^0R^{*\mathbf{\rho}}_{\mathbf{\sigma} \mathbf{\mu} \mathbf{\nu}}$ $\rho \mathrm{~}\sigma \mathrm{~}\mu \mathrm{~}\nu \mathrm{~}R^{*\mathbf{\rho}}_{\mathbf{\sigma} \mathbf{\mu} \mathbf{\nu}}$	$R^{\rho}_{\sigma\mu\nu}\rightarrow{}^0R^{*\rho}_{\sigma\mu\nu}$	but with $R^{\rho}_{\sigma\mu\nu}\rightarrow{}^0R^{*\rho}_{\sigma\mu\nu}$ and $\Gamma^{\sigma}_{\nu\mu}\rightarrow{}^0\Gamma^{*\sigma}_{\nu\mu}$, whereas

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1510.06699_F00520	23	■ 0.983	$\Gamma^{\sigma}_{\nu\mu} \rightarrow {}^0\Gamma^{*\sigma}_{\nu\mu}$	$\Gamma^{\sigma}_{\nu\mu} \rightarrow {}^0\Gamma^{*\sigma}_{\nu\mu}$	but with $R^{\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}_{\sigma\mu\nu}$ and $\Gamma^{\sigma}_{\nu\mu} \rightarrow {}^0\Gamma^{*\sigma}_{\nu\mu}$, whereas
1510.06699_F00521	23	■ 1.000	$\{ \}^{\{0\}} T^{\{*\} \lambda}_{\mu\nu} = 0$	${}^0T^{*\lambda}_{\mu\nu} = 0$	(126) becomes simply ${}^0T^{*\lambda}_{\mu\nu} = 0$, indicating the ab-
1510.06699_F00522	23	■ 0.941	$\nabla_{\mu}^{\{*\}} \rightarrow \{ \}^{\{0\}} \nabla_{\mu}^{\{*\}}, R^{\{*\} \rho}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}_{\sigma\mu\nu}$	$\nabla_{\mu}^{\{*\}} \rightarrow {}^0\nabla_{\mu}^{\{*\}}, R^{\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}_{\sigma\mu\nu}$	also valid, but with $\nabla_{\mu}^{\{*\}} \rightarrow {}^0\nabla_{\mu}^{\{*\}}, R^{\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}_{\sigma\mu\nu}$ and
1510.06699_F00523	23	■ 0.976	$T^{\{*\} \sigma}_{\mu\nu} \rightarrow \{ \}^{\{0\}} T^{\{*\} \sigma}_{\mu\nu} = 0$	$T^{*\sigma}_{\mu\nu} \rightarrow {}^0T^{*\sigma}_{\mu\nu} = 0$	$T^{*\sigma}_{\mu\nu} \rightarrow {}^0T^{*\sigma}_{\mu\nu} = 0$.
1510.06699_F00524	23	■ 1.000	$P_{\nu} \equiv \partial_{\nu} \rho, \mathcal{P}_a \equiv h_a{}^{\nu} P_{\nu}$	$P_{\nu} \equiv \partial_{\nu} \rho, \mathcal{P}_a \equiv h_a{}^{\nu} P_{\nu}$	where $P_{\nu} \equiv \partial_{\nu} \rho$, $\mathcal{P}_a \equiv h_a{}^{\nu} P_{\nu}$ and θ is an arbitrary pa-
1510.06699_F00525	23	■ 1.000	θ	θ	where $P_{\nu} \equiv \partial_{\nu} \rho$, $\mathcal{P}_a \equiv h_a{}^{\nu} P_{\nu}$ and θ is an arbitrary pa-
1510.06699_F00526	23	■ 1.000	$w = -2$	$w = -2$	forms covariantly with weight $w = -2$ and also acts as
1510.06699_F00527	24	■ 1.000	$\mathcal{T}^a_{bc} = 0$	$\mathcal{T}^a_{bc} = 0$	field originally has vanishing PGT torsion $\mathcal{T}^a_{bc} = 0$, the
1510.06699_F00528	24	■ 1.000	$\mathcal{T}'^a_{bc} \neq 0$	$\mathcal{T}'^a_{bc} \neq 0$	means that the local dilation leads to $\mathcal{T}'^a_{bc} \neq 0$.
1510.06699_F00529	24	■ 1.000	$\mathcal{T}^{\dagger a}_{bc}$	$\mathcal{T}^{\dagger a}_{bc}$	Thus, for general values of θ , neither $\mathcal{R}^{ab}_{cd}$ nor $\mathcal{T}^{\dagger a}_{bc}$
1510.06699_F00530	24	■ 1.000	$\theta = 0$	$\theta = 0$	transforms covariantly, but for $\theta = 0$ one recovers the
1510.06699_F00531	24	■ 1.000	$\theta = 1$	$\theta = 1$	curvature under local dilations, whereas for $\theta = 1$ one
1510.06699_F00532	24	■ 1.000	$\{ \}^{\{0\}} \mathcal{R}^{ab}_{cd}$	${}^0\mathcal{R}^{ab}_{cd}$	${}^0\mathcal{R}^{ab}_{cd}$ and its contractions (see Appendix B) are given
1510.06699_F00533	24	■ 0.971	$\{ \}^{\{0\}} \mathcal{T}^a_{bc}$	${}^0\mathcal{T}^a_{bc}$	in Section VB; the ‘reduced’ PGT torsion ${}^0\mathcal{T}^a_{bc}$ vanishes
1510.06699_F00534	24	■ 1.000	$D_{\mu}^{\dagger} \varphi$	$D_{\mu}^{\dagger} \varphi$	tive $D_{\mu}^{\dagger} \varphi$ by introducing a dilation vector gauge field
1510.06699_F00535	24	■ 1.000	V_{μ}	V_{μ}	(called V_{μ} to distinguish it from the dilation gauge field
1510.06699_F00536	24	■ 0.999	$\mathcal{V}_a = h_a{}^{\mu} V_{\mu}$	$\mathcal{V}_a = h_a{}^{\mu} V_{\mu}$	where $\mathcal{V}_a = h_a{}^{\mu} V_{\mu}$. Clearly, by construction, $A^{\dagger ab}_{\mu}$ is
1510.06699_F00537	24	■ 0.999	$A^{\dagger ab}_{\mu}$	$A^{\dagger ab}_{\mu}$	where $\mathcal{V}_a = h_a{}^{\mu} V_{\mu}$. Clearly, by construction, $A^{\dagger ab}_{\mu}$ is
1510.06699_F00538	24	■ 1.000	$\partial_{\mu} \varphi'$	$\partial_{\mu} \varphi'$	covariant derivative to cancel terms arising from $\partial_{\mu} \varphi'$
1510.06699_F00539	24	■ 1.000	$T_{\mu} =$	$T_{\mu} =$	in which w is the Weyl weight of the field φ and $T_{\mu} =$
1510.06699_F00540	24	■ 0.999	$b^a{}_{\mu} \mathcal{T}_a$	$b^a{}_{\mu} \mathcal{T}_a$	$b^a{}_{\mu} \mathcal{T}_a$, where $\mathcal{T}_a$ is the trace of the PGT torsion (see
1510.06699_F00541	24	■ 1.000	T_{μ}	T_{μ}	equation (66). Here we consider T_{μ} merely as a short-
1510.06699_F00542	24	■ 1.000	$T'_{\mu} = T_{\mu} + 3(1 - \theta)P_{\mu}$	$T'_{\mu} = T_{\mu} + 3(1 - \theta)P_{\mu}$	$T'_{\mu} = T_{\mu} + 3(1 - \theta)P_{\mu}$ and so (143) does indeed trans-
1510.06699_F00543	24	■ 0.898	$A^{\dagger ab}_{\mu} = A^{\dagger ab}_{\mu}$	$A^{\dagger ab}_{\mu} = A^{\dagger ab}_{\mu}$	itions (136)–(137), i.e. $A^{\dagger' ab}_{\mu} = A^{\dagger ab}_{\mu}$ and so it has a
1510.06699_F00544	24	■ 1.000	V	V	covariant derivative (143) and the V -field transformation
1510.06699_F00545	25	■ 1.000	$V_{\mu} = 0$	$V_{\mu} = 0$	that one can covariantly set $V_{\mu} = 0$, so that $A^{\dagger ab}_{\mu} = A^{ab}_{\mu}$
1510.06699_F00546	25	■ 1.000	$A^{\dagger ab}_{\mu} = A^{ab}_{\mu}$	$A^{\dagger ab}_{\mu} = A^{ab}_{\mu}$	that one can covariantly set $V_{\mu} = 0$, so that $A^{\dagger ab}_{\mu} = A^{ab}_{\mu}$
1510.06699_F00547	25	■ 1.000	$D_{\mu}^{\dagger} \varphi _{\theta=0} = \left(D_{\mu} - \frac{1}{3} w T_{\mu} \right) \varphi$	$D_{\mu}^{\dagger} \varphi _{\theta=0} = \left(D_{\mu} - \frac{1}{3} w T_{\mu} \right) \varphi$	as $D_{\mu}^{\dagger} \varphi _{\theta=0} = (D_{\mu} - \frac{1}{3} w T_{\mu}) \varphi$, in which the role played
1510.06699_F00548	25	■ 1.000	$T_{\mu} = 0$	$T_{\mu} = 0$	variantly set the PGT torsion to zero, and hence $T_{\mu} = 0$,
1510.06699_F00549	25	■ 1.000	$D_{\mu}^{\dagger} \varphi _{\theta=1} \equiv \left(\partial_{\mu} + \frac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab} - w V_{\mu} \right) \varphi$	$D_{\mu}^{\dagger} \varphi _{\theta=1} \equiv \left(\partial_{\mu} + \frac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab} - w V_{\mu} \right) \varphi$	$D_{\mu}^{\dagger} \varphi _{\theta=1} \equiv (\partial_{\mu} + \frac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab} - w V_{\mu}) \varphi$.
1510.06699_F00550	25	■ 1.000	$\partial_{\mu}^{\dagger} \varphi(x)$	$\partial_{\mu}^{\dagger} \varphi(x)$	(51) Thus the new minimally-coupled matter Lagrangian is a
1510.06699_F00551	25	■ 1.000	$w - 1$	$w - 1$	which transforms covariantly with weight $w - 1$, as de-
1510.06699_F00552	25	■ 0.999	$\mathcal{L}_{\mathbf{M}} \equiv h^{-1} L_{\mathbf{M}}(\varphi, \mathcal{D}^{\dagger} \varphi)$	$\mathcal{L}_{\mathbf{M}} \equiv h^{-1} L_{\mathbf{M}}(\varphi, \mathcal{D}^{\dagger} \varphi)$	action by the Lagrangian density $\mathcal{L}_{\mathbf{M}} \equiv h^{-1} L_{\mathbf{M}}(\varphi, \mathcal{D}^{\dagger} \varphi)$.
1510.06699_F00553	25	■ 1.000	$\phi^2 \mathcal{R}^{\dagger}$	$\phi^2 \mathcal{R}^{\dagger}$	tional to $\phi^2 \mathcal{R}^{\dagger}$ or $\phi^2 L_{\mathcal{T}^{\dagger 2}}$ are extended Weyl-covariant
1510.06699_F00554	25	■ 1.000	$\phi^2 L_{\mathcal{T}^{\dagger 2}}$	$\phi^2 L_{\mathcal{T}^{\dagger 2}}$	tional to $\phi^2 \mathcal{R}^{\dagger}$ or $\phi^2 L_{\mathcal{T}^{\dagger 2}}$ are extended Weyl-covariant
1510.06699_F00555	25	■ 0.968	$\text{operatorname{SU}}(2, 2)$	$\text{SU}(2, 2)$	of $\text{SU}(2, 2)$, to which $\text{O}(4, 2)$ is locally isomorphic. In
1510.06699_F00556	25	■ 0.968	$\text{O}(4, 2)$	$\text{O}(4, 2)$	of $\text{SU}(2, 2)$, to which $\text{O}(4, 2)$ is locally isomorphic. In
1510.06699_F00557	26	■ 1.000	$e_{b\mu} b_a - e_{a\mu} b_b$	$e_{b\mu} b_a - e_{a\mu} b_b$	in which a piece $e_{b\mu} b_a - e_{a\mu} b_b$ is added to the spin connec-
1510.06699_F00558	26	■ 1.000	$\omega_{\mu ab}$	$\omega_{\mu ab}$	tion $\omega_{\mu ab}$ (their b -field is the ‘dilatation field’, which in
1510.06699_F00559	26	■ 0.999	$\mathcal{V}$	$\mathcal{V}$	our case corresponds to $\mathcal{V}$, while their $e_{a\mu}$ is a version of

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1510.06699_F00560	26	■ 0.999	$e_{\{a \ \mu\}}$	$e_{a\mu}$	our case corresponds to $\mathcal{V}$, while their $e_{a\mu}$ is a version of
1510.06699_F00561	26	■ 1.000	$c_{\{b \ \mu\}} x_{\{a\}} - c_{\{a \ \mu\}} x_{\{b\}}$	$c_{b\mu}x_a - c_{a\mu}x_b$	$c_{b\mu}x_a - c_{a\mu}x_b$, where here $c_{a\mu}$ is the extra ‘vierbein-type’
1510.06699_F00562	26	■ 1.000	$c_{\{a \ \mu\}}$	$c_{a\mu}$	$c_{b\mu}x_a - c_{a\mu}x_b$, where here $c_{a\mu}$ is the extra ‘vierbein-type’
1510.06699_F00563	26	■ 1.000	$x_{\{a\}}$	x_a	tions. Here x_a is a global surrogate for $\mathcal{V}_a$ (noting the fact
1510.06699_F00564	26	■ 1.000	$\mathcal{V}_{\{a\}}$	$\mathcal{V}_a$	tions. Here x_a is a global surrogate for $\mathcal{V}_a$ (noting the fact
1510.06699_F00565	26	■ 1.000	$x \cdot \nabla$	$x \cdot \nabla$	that the generator of global dilations is $x \cdot \nabla$), and so this
1510.06699_F00566	26	■ 1.000	$\{ \}^{\{0\}} \nabla_{\{\mu\}}^{\{\dagger\}} \equiv \{ \}^{\{0\}} \nabla_{\{\mu\}} - w \left(V_{\{\mu\}} + \frac{1}{3} T_{\{\mu\}} \right), \{ \}^{\{0\}} \Gamma^{\{\lambda\}}_{\{\mu \ \nu\}} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} +$	${}^0\nabla_{\mu}^{\dagger} \equiv {}^0\nabla_{\mu} - w \left(V_{\mu} + \frac{1}{3} T_{\mu} \right), {}^0\Gamma^{\lambda}_{\mu\nu} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} +$	where ${}^0\nabla_{\mu}^{\dagger} \equiv {}^0\nabla_{\mu} - w(V_{\mu} + \frac{1}{3}T_{\mu})$, ${}^0\Gamma^{\lambda}_{\mu\nu} \equiv \frac{1}{2}\gamma^{\lambda\rho}(\partial_{\mu}\gamma_{\nu\rho} +$
1510.06699_F00567	26	■ 0.817	$\partial_{\{\nu\}} \gamma_{\{\mu \ \rho\}} - \partial_{\{\rho\}} \gamma_{\{\mu \ \nu\}}$	$\partial_{\nu}\gamma_{\mu\rho} - \partial_{\rho}\gamma_{\mu\nu}$	$\partial_{\nu}\gamma_{\mu\rho} - \partial_{\rho}\gamma_{\mu\nu}$) is the metric connection corresponding to
1510.06699_F00568	26	■ 0.801	$D_{\{\mu\}}$	D_{μ}	$D_{\mu}^{\dagger}$ is applied, and w is its Weyl weight.
1510.06699_F00569	26	■ 1.000	$T^{\{a\}\{ \}_{\{\mu \ \nu\}}}$	$T^a_{\mu\nu}$	vature) in PGT (and WGT), $T^a_{\mu\nu}$ is the PGT transla-
1510.06699_F00570	26	■ 1.000	$D_{\{\mu\}}$	D_{μ}	tional gauge field (torsion), and D_{μ} is the PGT covariant
1510.06699_F00571	26	■ 1.000	$R^{\{\dagger a \ b\}\{ \}_{\{\mu \ \nu\}}}$	$R^{\dagger ab}_{\mu\nu}$	As in WGT, both $R^{\dagger ab}_{\mu\nu}$ and $H^{\dagger}_{\mu\nu}$ transform covariantly
1510.06699_F00572	26	■ 1.000	$H_{\{\mu \ \nu\}}^{\{\dagger\}}$	$H^{\dagger}_{\mu\nu}$	As in WGT, both $R^{\dagger ab}_{\mu\nu}$ and $H^{\dagger}_{\mu\nu}$ transform covariantly
1510.06699_F00573	26	■ 1.000	$\mathcal{R}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}} = h_{\{c\}\{ \}^{\{\mu\}} h_{\{d\}\{ \}^{\{\nu\}} R^{\{\dagger a \ b\}\{ \}_{\{\mu \ \nu\}}}$	$\mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu\nu}$	where $\mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu\nu}$ and $\mathcal{H}_{cd}^{\dagger} = h_c^{\mu} h_d^{\nu} H^{\dagger}_{\mu\nu}$,
1510.06699_F00574	26	■ 1.000	$\mathcal{H}_{\{c \ d\}}^{\{\dagger\}} = h_{\{c\}\{ \}^{\{\mu\}} h_{\{d\}\{ \}^{\{\nu\}} H_{\{\mu \ \nu\}}^{\{\dagger\}}$	$\mathcal{H}_{cd}^{\dagger} = h_c^{\mu} h_d^{\nu} H^{\dagger}_{\mu\nu}$	where $\mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu\nu}$ and $\mathcal{H}_{cd}^{\dagger} = h_c^{\mu} h_d^{\nu} H^{\dagger}_{\mu\nu}$,
1510.06699_F00575	27	■ 1.000	$\mathcal{R}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}}, \mathcal{H}_{\{c \ d\}}^{\{\dagger\}}$	$\mathcal{R}^{\dagger ab}_{cd}, \mathcal{H}_{cd}^{\dagger}$	It is easy to show that, similarly to WGT, $\mathcal{R}^{\dagger ab}_{cd}, \mathcal{H}_{cd}^{\dagger}$
1510.06699_F00576	27	■ 1.000	$\mathcal{T}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}}}$	$\mathcal{T}^{\dagger a}_{cd}$	and $\mathcal{T}^{\dagger a}_{cd}$ transform covariantly under extended local
1510.06699_F00577	27	■ 1.000	$w \left(\mathcal{R}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}} \right) = w \left(\mathcal{H}_{cd}^{\{\dagger\}} \right) = -2$	$w \left(\mathcal{R}^{\dagger ab}_{cd} \right) = w \left(\mathcal{H}_{cd}^{\dagger} \right) = -2$	dilations with weights $w(\mathcal{R}^{\dagger ab}_{cd}) = w(\mathcal{H}_{cd}^{\dagger}) = -2$ and
1510.06699_F00578	27	■ 0.999	$w \left(\mathcal{T}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}} \right) = -1$	$w \left(\mathcal{T}^{\dagger a}_{cd} \right) = -1$	$w(\mathcal{T}^{\dagger a}_{cd}) = -1$ respectively.
1510.06699_F00579	27	■ 1.000	$\mathcal{R}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}}}$	$\mathcal{R}^{\dagger ab}_{cd}$	plicit forms of $\mathcal{R}^{\dagger ab}_{cd}$ and its contractions are given in
1510.06699_F00580	27	■ 1.000	$\mathcal{H}_{\{a \ b\}}^{\{\dagger\}}$	$\mathcal{H}_{ab}^{\dagger}$	strength $\mathcal{H}_{ab}^{\dagger}$ is antisymmetric in a and b and hence also
1510.06699_F00581	27	■ 0.725	$\mathcal{R}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}}(h, A, \partial A, V, \partial V), \mathcal{T}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}}(h, \partial h, A)$	$\mathcal{R}^{\dagger ab}_{cd}(h, A, \partial A, V, \partial V), \mathcal{T}^{\dagger a}_{bc}(h, \partial h, A)$	tives are $\mathcal{R}^{\dagger ab}_{cd}(h, A, \partial A, V, \partial V), \mathcal{T}^{\dagger a}_{bc}(h, \partial h, A)$ and
1510.06699_F00582	27	■ 1.000	$\mathcal{H}_{\{a \ b\}}^{\{\dagger\}} \left(h, \partial h, \partial^2 h, A, \partial A, \partial V \right)$	$\mathcal{H}_{ab}^{\dagger}(h, \partial h, \partial^2 h, A, \partial A, \partial V)$	$\mathcal{H}_{ab}^{\dagger}(h, \partial h, \partial^2 h, A, \partial A, \partial V)$. It is important to note that
1510.06699_F00583	27	■ 1.000	$\partial h, \partial^2 h, A$	$\partial h, \partial^2 h, A$	strength $\mathcal{H}_{ab}^{\dagger}$, which depends on $\partial h, \partial^2 h, A$ and ∂A , in
1510.06699_F00584	27	■ 1.000	$\partial^2 h$	$\partial^2 h$	tive $\partial^2 h$ is a particularly unusual feature as compared
1510.06699_F00585	27	■ 1.000	$\{ \}^{\{0\}} \mathcal{A}_{\{a \ b \ c\}}^{\{\dagger\}} \equiv \frac{1}{2} \left(c_{\{a \ b \ c\}}^{\{\dagger\}} +$	${}^0\mathcal{A}_{abc}^{\dagger} \equiv \frac{1}{2} \left(c_{abc}^{\dagger} +$	that both ${}^0\mathcal{A}_{abc}^*$ and $\mathcal{K}_{abc}^*$ are antisymmetric in their first
1510.06699_F00586	27	■ 0.973	$\left(c_{\{b \ c \ a\}}^{\{\dagger\}} - c_{\{c \ a \ b\}}^{\{\dagger\}} \right)$	$c_{bca}^{\dagger} - c_{cab}^{\dagger}$	two indices ⁹⁹ .
1510.06699_F00587	27	■ 0.973	$c^{\{\dagger c\}}_{\{a \ b\}} \equiv h_a^{\mu} h_b^{\nu} \left(\partial_{\mu}^{\dagger} b^c_{\nu} - \partial_{\nu}^{\dagger} b^c_{\mu} \right)$	$c^{\dagger c}_{ab} \equiv h_a^{\mu} h_b^{\nu} \left(\partial_{\mu}^{\dagger} b^c_{\nu} - \partial_{\nu}^{\dagger} b^c_{\mu} \right)$	two indices ⁹⁹ .
1510.06699_F00588	27	■ 0.981	$\mathcal{K}_{\{a \ b \ c\}}^{\{\dagger\}} \equiv -\frac{1}{2} \left(\mathcal{T}_{\{a \ b \ c\}}^{\{\dagger\}} + \mathcal{T}_{\{b \ c \ a\}}^{\{\dagger\}} - \mathcal{T}_{\{c \ a \ b\}}^{\{\dagger\}} \right)$	$\mathcal{K}_{abc}^{\dagger} \equiv -\frac{1}{2} \left(\mathcal{T}_{abc}^{\dagger} + \mathcal{T}_{bca}^{\dagger} - \mathcal{T}_{cab}^{\dagger} \right)$	Under a local weyl transformation, the quantities ${}^0\mathcal{A}^*$, transform in the same way as $\mathcal{A}^*$, whereas $\mathcal{K}^*$,
1510.06699_F00589	27	■ 0.981	$\{ \}^{\{0\}} \mathcal{A}_{\{a \ b \ c\}}^{\{\dagger\}}$	${}^0\mathcal{A}_{abc}^{\dagger}$	Under a local weyl transformation, the quantities ${}^0\mathcal{A}^*$, transform in the same way as $\mathcal{A}^*$, whereas $\mathcal{K}^*$,
1510.06699_F00590	27	■ 0.840	$\mathcal{K}_{\{a \ b \ c\}}^{\{\dagger\}}$	$\mathcal{K}_{abc}^{\dagger}$	$\mathcal{A}_{abc}$ transform in the same way as $\mathcal{A}_{abc}$, whereas $\mathcal{K}_{abc}$ transform as the components of a local tensor with weight
1510.06699_F00591	27	■ 0.568	$\{ \}^{\{0\}} \mathcal{A}_{\{\text{abc}\}}^{\{\dagger\}}$	${}^0\mathcal{A}_{abc}^{\dagger}$	well as $\mathcal{A}_{abc}^*$ in the construction of a covariant derivative.

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1510.06699_F00592	27	■ 0.551	$\backslash\mathrm{mathcal}\{K\}_{\backslash\mathrm{text}\{abc\}}^{\backslash\mathrm{dagger}}$	$\mathcal{K}_{abc}^\dagger$	We may therefore construct the ‘reduced’ (Λ, ρ) -covariant
1510.06699_F00593	27	■ 0.992	$\backslash\mathrm{mathcal}\{A\}_{a\ b\ c}^{\backslash\mathrm{dagger}}$	$\mathcal{A}_{abc}^\dagger$	
1510.06699_F00594	27	■ 1.000	$\{ \ }^{\backslash\mathrm{dagger}}\{ \}_0\ \backslash\mathrm{D}_{\backslash\mathrm{mu}}^{\backslash\mathrm{dagger}}\ \backslash\mathrm{varphi}$	${}^0D_\mu^\dagger\varphi$	in eWGT, however, is that ${}^0D_\mu^\dagger\varphi$ still depends on the
1510.06699_F00595	27	■ 0.523	$\{ \ }^{\backslash\mathrm{dagger}}\{ \}_0\ \backslash\mathrm{mathcal}\{D\}_{a}^{\backslash\mathrm{dagger}}\ \backslash\mathrm{equiv}\ h_{a}{}^\mu\{ \ }^{\backslash\mathrm{dagger}}\{ \}_0\ \backslash\mathrm{D}_{\backslash\mathrm{mu}}^{\backslash\mathrm{dagger}}$	${}^0\mathcal{D}_a^\dagger \equiv h_a{}^\mu {}^0D_\mu^\dagger$	${}^0\mathcal{D}_a^\dagger \equiv h_a{}^\mu {}^0D_\mu^\dagger$, which consequently transforms in the
1510.06699_F00596	27	■ 1.000	$\backslash\mathrm{mathcal}\{D\}_{a}^{\backslash\mathrm{dagger}}$	$\mathcal{D}_a^\dagger$	same way as $\mathcal{D}_a^\dagger$, as required. The ‘full’ generalised co-
1510.06699_F00597	27	■ 1.000	$\{ \ }^{\backslash\mathrm{dagger}}\{ \}_0\ \backslash\mathrm{mathcal}\{D\}_{a}^{\backslash\mathrm{dagger}}$	${}^0\mathcal{D}_a^\dagger$	to ${}^0\mathcal{D}_a^\dagger$.
1510.06699_F00598	27	■ 1.000	$\backslash\mathrm{mathcal}\{R\}^{\backslash\mathrm{dagger}}_{a\ b}\{ \ }_{c\ d},\ \backslash\mathrm{mathcal}\{T\}^{\backslash\mathrm{dagger}}_{a\ b\ c}$	$\mathcal{R}^{\dagger ab}_{cd}, \mathcal{T}^{\dagger a}_{bc}$	the gravitational gauge field strengths $\mathcal{R}^{\dagger ab}_{cd}, \mathcal{T}^{\dagger a}_{bc}$ and
1510.06699_F00599	28	■ 0.999	$\backslash\mathrm{mathcal}\{T\}_{a\ b\ c}^{\backslash\mathrm{dagger}}=\backslash\mathrm{epsilon}_{a\ b\ c\ d}\ t^d$	$\mathcal{T}_{abc}^\dagger = \epsilon_{abcd}t^d$	sion’ is totally antisymmetric, such that $\mathcal{T}_{abc}^\dagger = \epsilon_{abcd}t^d$
1510.06699_F00600	28	■ 0.800	$\backslash\mathrm{mathcal}\{R\}_{a\ b\ c\ d}^{\backslash\mathrm{dagger}}$	$\mathcal{R}_{abcd}^\dagger$	and discussed in Section II G. In principle, from $\mathcal{R}_{abcd}^\dagger$,
1510.06699_F00601	28	■ 0.899	$\backslash\mathrm{mathcal}\{T\}_{a\ b\ c}^{\backslash\mathrm{dagger}}$	$\mathcal{T}_{abc}^\dagger$	$\mathcal{T}_{abc}^\dagger$ and $\mathcal{H}_{ab}^\dagger$, one can construct a free gravitational ac-
1510.06699_F00602	28	■ 1.000	$\backslash\mathrm{mathcal}\{R\}^{\backslash\mathrm{dagger}}\ \backslash\mathrm{equiv}$	$\mathcal{R}^\dagger \equiv$	$\mathcal{H}_{ab}$, but terms linear in $\mathcal{R} \equiv \mathcal{R}_{ab}$ or quadratic in $\mathcal{R}_{ab}$ are not allowed. Similarly, higher-order terms in $\mathcal{R}_{abcd}$
1510.06699_F00603	28	■ 1.000	$\backslash\mathrm{mathcal}\{R\}^{\backslash\mathrm{dagger}}_{a\ b}\{ \ }_{a\ b}$	$\mathcal{R}^{\dagger ab}_{ab}$	and $\mathcal{H}_{ab}$ are forbidden; in principle one could include
1510.06699_F00604	28	■ 1.000	$L_{\backslash\mathrm{mathcal}\{R\}^{\backslash\mathrm{dagger}}_2}$	$L_{\mathcal{R}^{\dagger 2}}$	where the expressions for $L_{\mathcal{R}^{\dagger 2}}$ and $L_{\mathcal{H}^{\dagger 2}}$ are given in
1510.06699_F00605	28	■ 1.000	$L_{\backslash\mathrm{mathcal}\{H\}^{\backslash\mathrm{dagger}}_2}$	$L_{\mathcal{H}^{\dagger 2}}$	where the expressions for $L_{\mathcal{R}^{\dagger 2}}$ and $L_{\mathcal{H}^{\dagger 2}}$ are given in
1510.06699_F00606	28	■ 1.000	$\backslash\mathrm{mathcal}\{R\}_{a\ b\ c\ d}\ \backslash\mathrm{rightarrow}\ \backslash\mathrm{mathcal}\{R\}_{a\ b\ c\ d}^{\backslash\mathrm{dagger}}$	$\mathcal{R}_{abcd} \rightarrow \mathcal{R}_{abcd}^\dagger$	i.e. making the replacements $\mathcal{R}_{abcd} \rightarrow \mathcal{R}_{abcd}^\dagger$ and $\mathcal{H}_{ab} \rightarrow$
1510.06699_F00607	28	■ 1.000	$\backslash\mathrm{mathcal}\{H\}_{a\ b}\ \backslash\mathrm{rightarrow}$	$\mathcal{H}_{ab} \rightarrow$	i.e. making the replacements $\mathcal{R}_{abcd} \rightarrow \mathcal{R}_{abcd}^\dagger$ and $\mathcal{H}_{ab} \rightarrow$
1510.06699_F00608	28	■ 0.829	$\{ \ }^{\backslash\mathrm{dagger}}\{ \}_{\backslash\mathrm{underline}\{71\}}$	$\overline{71}$	$\mathcal{H}_{ab}^\dagger$. Performing a similar derivation to the WGT case ⁷¹ ,
1510.06699_F00609	28	■ 0.556	$\phi^2\ L_{\backslash\mathrm{mathcal}\{T\}^{\backslash\mathrm{dagger}}_2}$	$\phi^2 L_{\mathcal{T}^{\dagger 2}}$	form $\phi^2\mathcal{R}^\dagger$ and $\phi^2 L_{\mathcal{T}^{\dagger 2}}$, where $L_{\mathcal{T}^{\dagger 2}}$ is given in (3) with
1510.06699_F00610	28	■ 0.556	$L_{\backslash\mathrm{mathcal}\{T\}^{\backslash\mathrm{dagger}}_2}$	$L_{\mathcal{T}^{\dagger 2}}$	form $\phi^2\mathcal{R}^\dagger$ and $\phi^2 L_{\mathcal{T}^{\dagger 2}}$, where $L_{\mathcal{T}^{\dagger 2}}$ is given in (3) with
1510.06699_F00611	28	■ 0.999	$\backslash\mathrm{mathcal}\{T\}_{a\ b\ c}\ \backslash\mathrm{rightarrow}\ \backslash\mathrm{mathcal}\{T\}_{a\ b\ c}^{\backslash\mathrm{dagger}}$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^\dagger$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^\dagger$ (but with no term proportional to β_3 ,
1510.06699_F00612	28	■ 1.000	$\backslash\mathrm{mathcal}\{H\}_{a\ b}^{\backslash\mathrm{dagger}}\ \backslash\mathrm{mathcal}\{H\}_{a\ b}^{\backslash\mathrm{dagger}}$	$\mathcal{H}_{ab}^\dagger\mathcal{H}^{\dagger ab}$	tional to $\mathcal{H}_{ab}^\dagger\mathcal{H}^{\dagger ab}$ in (166). In the matter sector, covari-
1510.06699_F00613	28	■ 1.000	$\backslash\mathrm{varphi},\ \backslash\mathrm{partial}\ \backslash\mathrm{varphi},\ h,\ \backslash\mathrm{partial}\ h,\ A$	$\varphi, \partial\varphi, h, \partial h, A$	on $\varphi, \partial\varphi, h, \partial h, A$ and V ; the dependence on ∂h is new
1510.06699_F00614	28	■ 0.998	$\backslash\mathrm{mathcal}\{D\}_{a}^{\backslash\mathrm{dagger}}\ \backslash\mathrm{varphi}$	$\mathcal{D}_a^\dagger\varphi$	$\mathcal{T}_a$ in $\mathcal{D}_a^\dagger\varphi$. We will also consider here the inclusion of
1510.06699_F00615	28	■ 1.000	$\backslash\mathrm{partial}\ V$	∂V	tional dependence on ∂A and ∂V ; the latter is new com-
1510.06699_F00616	29	■ 0.626	$\backslash\mathrm{mathcal}\{R\}_{a\ b\ c\ d}^{\backslash\mathrm{dagger}}(h,\ A,\ \backslash\mathrm{partial}\ A,\ V,\ \backslash\mathrm{partial}\ V),\ \backslash\mathrm{mathcal}\{T\}_{a\ b\ c}^{\backslash\mathrm{dagger}}(h,\ \backslash\mathrm{partial}\ h,\ A)$	$\mathcal{R}_{abcd}^\dagger(h, A, \partial A, V, \partial V), \mathcal{T}_{abc}^\dagger(h, \partial h, A)$	from Ostrogradsky’s instability. It is worth noting, how-
1510.06699_F00617	29	■ 1.000	$\backslash\mathrm{mathcal}\{H\}^{\backslash\mathrm{dagger}}_{a\ b}\ \backslash\mathrm{mathcal}\{H\}_{a\ b}^{\backslash\mathrm{dagger}}$	$\mathcal{H}^{\dagger ab}_{ab}\mathcal{H}_{ab}^\dagger$	ever, that if one substitutes for $\mathcal{A}_{abc}$ using (47), one ob-
1510.06699_F00618	29	■ 1.000	$t^a{}_{\backslash\mathrm{mu}}\ \backslash\mathrm{equiv}\ \backslash\mathrm{delta}\ \backslash\mathrm{mathcal}\{L\}_{\backslash\mathrm{mathrm}\{G\}}\ /\ \backslash\mathrm{delta}\ h_{a}{}_{\backslash\mathrm{mu}}\{ \ }^{\backslash\mathrm{dagger}}_{\backslash\mathrm{mu}},\ s_{a\ b}{}_{\backslash\mathrm{mu}}\{ \ }^{\backslash\mathrm{dagger}}_{\backslash\mathrm{mu}}\ \backslash\mathrm{equiv}\ \backslash\mathrm{delta}\ \backslash\mathrm{mathcal}\{L\}_{\backslash\mathrm{mathrm}\{G\}}\ /\ \backslash\mathrm{delta}\ A^a{}_{\backslash\mathrm{mu}}\{ \ }_{\backslash\mathrm{mu}}$	$t^a{}_\mu \equiv \delta\mathcal{L}_G/\delta h_a{}^\mu, s_{ab}{}^\mu \equiv \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	Variation of S_T with respect to the gauge fields $h_a{}^\mu$,
1510.06699_F00619	29	■ 1.000	$j^{\backslash\mathrm{mu}}\ \backslash\mathrm{equiv}$	$j^\mu \equiv$	accordance with their respective index structures, and
1510.06699_F00620	29	■ 1.000	$\backslash\mathrm{delta}\ \backslash\mathrm{mathcal}\{L\}_{\backslash\mathrm{mathrm}\{G\}}\ /\ \backslash\mathrm{delta}\ V_{\backslash\mathrm{mu}}$	$\delta\mathcal{L}_G/\delta V_\mu$	are also covariant under local dilations with Weyl weights
1510.06699_F00621	29	■ 1.000	$\backslash\mathrm{zeta}^{\backslash\mathrm{mu}}\ \backslash\mathrm{equiv}\ \backslash\mathrm{delta}\ \backslash\mathrm{mathcal}\{L\}_{\backslash\mathrm{mathrm}\{M\}}\ /\ \backslash\mathrm{delta}\ V_{\backslash\mathrm{mu}}$	$\zeta^\mu \equiv \delta\mathcal{L}_M/\delta V_\mu$	convenient to work with quantities carrying only Latin
1510.06699_F00622	29	■ 0.998	$t^a{}_{\backslash\mathrm{mu}}\{ \ }_{\backslash\mathrm{mu}}$	$t^a{}_b$	similarly for the matter sector. All the quantities in
1510.06699_F00623	29	■ 1.000	$t^a{}_{\backslash\mathrm{mu}}\{ \ }_{\backslash\mathrm{mu}}$	$t^a{}_a$	(62a)–(62c) are clearly invariant under GCT transform
1510.06699_F00624	29	■ 1.000	$\tau^a{}_{\backslash\mathrm{mu}}\{ \ }_{\backslash\mathrm{mu}}$	$\tau^a{}_a$	although it is worth noting that the traces $t^a{}_a$ and $\tau^a{}_a$
1510.06699_F00625	29	■ 1.000	$t_{a\ b}$	t_{ab}	although it is worth noting that the traces $t^a{}_a$ and $\tau^a{}_a$
1510.06699_F00626	29	■ 1.000	$\tau^{\backslash\mathrm{dagger}}_{a\ b}\{ \ }_{\backslash\mathrm{mu}}=\tau^{\backslash\mathrm{dagger}}_{a\ b}\{ \ }_{\backslash\mathrm{mu}}$	$\tau^{\dagger a}{}_a = \tau^a{}_a$	momentum tensor. Nonetheless, although neither t_{ab} nor
1510.06699_F00627	29	■ 0.999	$t_{a\ b}^{\backslash\mathrm{dagger}}$	$t_{ab}^\dagger$	is invariant). It is also worth noting that $\tau^{\dagger a}{}_a = \tau^a{}_a$.
1510.06699_F00628	29	■ 1.000	$\backslash\mathrm{mathcal}\{T\}_{a}^{\backslash\mathrm{dagger}}\ \backslash\mathrm{equiv}\ 0$	$\mathcal{T}_a^\dagger \equiv 0$	One may construct a covariant $t_{ab}^\dagger$ in a similar way. In-

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1510.06699_F00629	29	■ 1.000	$\bar{\partial}\mathrm{L}_{\mathrm{M}}/\partial\varphi\equiv\left[\partial L_{\mathrm{M}}\left(\varphi,\mathcal{D}_a^\dagger u,\phi,\mathcal{D}_a^\dagger\phi,\dots\right)/\partial\varphi\right]_{u=\varphi}$	$\bar{\partial}L_{\mathrm{M}}/\partial\varphi\equiv\left[\partial L_{\mathrm{M}}\left(\varphi,\mathcal{D}_a^\dagger u,\phi,\mathcal{D}_a^\dagger\phi,\dots\right)/\partial\varphi\right]_{u=\varphi}$	where $\bar{\partial}L_{\mathrm{M}}/\partial\varphi\equiv\left[\partial L_{\mathrm{M}}\left(\varphi,\mathcal{D}_a^\dagger u,\phi,\mathcal{D}_a^\dagger\phi,\dots\right)/\partial\varphi\right]_{u=\varphi}$, so
1510.06699_F00630	30	■ 1.000	$H_{\{a\}b}^\dagger$	$H_{ab}^\dagger$	strength tensor $H_{ab}^\dagger$ of the vector gauge field (152), but
1510.06699_F00631	30	■ 1.000	$H_{\{a\}b}=h_{\{a\}\{\ \}^\mu}h_{b\{\ \}^\nu}\mathrm{left}(\partial_{\mu}V_{\nu}-\partial_{\nu}V_{\mu})$	$H_{ab}=h_a{}^\mu h_b{}^\nu(\partial_\mu V_\nu-\partial_\nu V_\mu)$	instead the simpler object $H_{ab}=h_a{}^\mu h_b{}^\nu(\partial_\mu V_\nu-\partial_\nu V_\mu)$,
1510.06699_F00632	30	■ 1.000	$j^{\dagger a}$	$j^{\dagger a}$	by $j^{\dagger a}$ rather than simply by j^a . The largest difference
1510.06699_F00633	30	■ 1.000	$j^{\{a\}}$	j^a	by $j^{\dagger a}$ rather than simply by j^a . The largest difference
1510.06699_F00634	30	■ 1.000	$\varphi,\phi,h_{\{a\}^\mu},A^{\{a\}b\}_{\mu}},V_{\mu}$	$\varphi,\phi,h_a^\mu,A^{ab}{}_\mu,V_\mu$	from the set $\varphi,\phi,h_a^\mu,A^{ab}{}_\mu,V_\mu$ and their derivatives,
1510.06699_F00635	30	■ 1.000	φ,ϕ	φ,ϕ	with which we have worked so far, to the new set φ,ϕ ,
1510.06699_F00636	30	■ 1.000	$h_{\{a\}^\mu},A^{\dagger ab}{}_{\mu},V_{\mu}$	$h_a^\mu,A^{\dagger ab}{}_\mu,V_\mu$	$h_a^\mu,A^{\dagger ab}{}_\mu,V_\mu$ and their derivatives (i.e. we replace $A^{ab}{}_\mu$
1510.06699_F00637	30	■ 1.000	$h_{\{a\}^\mu},A^{\{a\}b\}_{\mu}},V_{\mu}$	$h_a^\mu,A^{ab}{}_\mu,V_\mu$	of $h_a^\mu,A^{ab}{}_\mu,V_\mu$ by its defining relationship (142), rather
1510.06699_F00638	30	■ 1.000	$\mathrm{mathcal{T}}_{\{b\}}^{\natural}\equiv\mathrm{mathcal{T}}^{\natural a}{}_{ba}$	$\mathcal{T}_b^\natural\equiv\mathcal{T}^\natural a{}_{ba}$	and the corresponding trace $\mathcal{T}_b^\natural\equiv\mathcal{T}^\natural a{}_{ba}$. It is straight-
1510.06699_F00639	30	■ 1.000	$\mathrm{mathcal{T}}_{\{b\}}^{\natural}=\mathrm{mathcal{T}}_{\{b\}}+3\mathcal{V}_b$	$\mathcal{T}_b^\natural=\mathcal{T}_b+3\mathcal{V}_b$	forward to show that $\mathcal{T}_b^\natural=\mathcal{T}_b+3\mathcal{V}_b$, where $\mathcal{T}_b$ is the
1510.06699_F00640	31	■ 0.997	$h,\partial h$	$h,\partial h$	which is expressible wholly in terms of the fields $h,\partial h$
1510.06699_F00641	31	■ 1.000	$A^{\dagger}$	$A^\dagger$	and $A^\dagger$ (suppressing indices). Since (182) does not de-
1510.06699_F00642	31	■ 0.566	$\mathrm{quad}\,\mathrm{mathcal{R}}_{\{a\}b\,c\,d}^{\dagger}\mathrm{left}(h,A^{\dagger},\partial A^{\dagger}),\,\mathrm{quad}\,\mathrm{mathcal{T}}_{\{a\}b\,c}^{\dagger}\mathrm{left}(h,\partial h,A^{\dagger})\,\mathrm{quad}$	$\mathcal{R}_{abcd}^\dagger(h,A^\dagger,\partial A^\dagger),\quad\mathcal{T}_{abc}^\dagger(h,\partial h,A^\dagger)$	tensors $\mathcal{R}_{abcd}^\dagger(h,A^\dagger,\partial A^\dagger),\quad\mathcal{T}_{abc}^\dagger(h,\partial h,A^\dagger)$ and
1510.06699_F00643	31	■ 1.000	$\mathrm{mathcal{H}}_{\{a\}b}^{\dagger}\mathrm{left}(h,\partial h,\partial^2 h,A^{\dagger},\partial A^{\dagger})$	$\mathcal{H}_{ab}^\dagger(h,\partial h,\partial^2 h,A^\dagger,\partial A^\dagger)$	$\mathcal{H}_{ab}^\dagger(h,\partial h,\partial^2 h,A^\dagger,\partial A^\dagger)$, where once again the last
1510.06699_F00644	31	■ 1.000	$\mathcal{L}$	$\mathcal{L}$	derivative of a Lagrangian $\mathcal{L}$ with respect to any one
1510.06699_F00645	31	■ 1.000	$(\delta\mathcal{L}/\delta\chi)_{\dagger}$	$(\delta\mathcal{L}/\delta\chi)_{\dagger}$	of the fields χ by $(\delta\mathcal{L}/\delta\chi)_{\dagger}$ to distinguish it from the
1510.06699_F00646	31	■ 1.000	$\delta\mathcal{L}/\delta\chi$	$\delta\mathcal{L}/\delta\chi$	variational derivative $\delta\mathcal{L}/\delta\chi$ obtained previously using
1510.06699_F00647	31	■ 1.000	$\sigma_{ab}{}^\mu$	$\sigma_{ab}{}^\mu$	ther that the spin-angular momentum tensor $\sigma_{ab}{}^\mu$ and
1510.06699_F00648	31	■ 1.000	$\zeta^{\dagger\mu}=0$	$\zeta^{\dagger\mu}=0$	gauge field V_μ or its derivative, then $\zeta^{\dagger\mu}=0$ and so
1510.06699_F00649	31	■ 1.000	$\phi,h_{\{a\}\ \}^\mu},A^{\{a\}b\}_{\mu}},V_{\mu}$	$\phi,h_a{}^\mu,A^{ab}{}_\mu,V_\mu$	$\phi,h_a{}^\mu,A^{ab}{}_\mu,V_\mu$ and their derivatives, but we will use the
1510.06699_F00650	32	■ 1.000	$\mathcal{D}_{\{a\}}^*\rightarrow\mathcal{D}_{\{a\}}^\dagger$	$\mathcal{D}_a^*\rightarrow\mathcal{D}_a^\dagger$	with $\mathcal{D}_a^*\rightarrow\mathcal{D}_a^\dagger$. Similar to WGT, since both ψ and $\bar{\psi}$ have
1510.06699_F00651	32	■ 1.000	$\gamma^a\mathrm{left}(\mathrm{mathcal{D}}_{\{a\}}+\frac{1}{2}\mathcal{T}_{\{a\}})\psi=\gamma^a\mathcal{D}_{\{a\}}^{\dagger}\psi$	$\gamma^a\left(\mathcal{D}_a+\frac{1}{2}\mathcal{T}_a\right)\psi=\gamma^a\mathcal{D}_a^\dagger\psi$	opments. Following our discussion in Section III, we begin with the special-relativistic Lagrangian (14) for a
1510.06699_F00652	32	■ 1.000	$\gamma^a\Sigma_{ba}=\frac{1}{4}\gamma^a\left[\gamma_b,\gamma_a\right]=-\frac{3}{2}\gamma_b$	$\gamma^a\Sigma_{ba}=\frac{1}{4}\gamma^a\left[\gamma_b,\gamma_a\right]=-\frac{3}{2}\gamma_b$	field ϕ . For the moment, however, we will consider only
1510.06699_F00653	32	■ 0.997	$\mathcal{T}_{[ab\,c]}=0$	$\mathcal{T}_{[abc]}=0$	$w=-\frac{3}{5}$ and $w=-1$, respectively; we note that the
1510.06699_F00654	32	■ 1.000	$\{ \}^{\{0\}}\mathrm{mathcal{D}}_{\{a\}}\rightarrow{}^0\mathcal{D}_{\{a\}}^{\dagger}$	${}^0\mathcal{D}_a\rightarrow{}^0\mathcal{D}_a^\dagger$	
1510.06699_F00655	32	■ 1.000	$\mathcal{T}_{[ab\,c]}\rightarrow\mathcal{T}_{[ab\,c]}^{\dagger}$	$\mathcal{T}_{[abc]}\rightarrow\mathcal{T}_{[abc]}^\dagger$	$\mathcal{L}_{\mathrm{D}}=h^{-1}L_{\mathrm{D}}=h^{-1}(\tfrac{1}{5}i\bar{\psi}\gamma^a\overleftrightarrow{\mathcal{D}}_a^*\psi-\mu\phi\bar{\psi}\psi),\qquad(71)$
1510.06699_F00656	32	■ 1.000	$\sigma_{ab}{}^c$	$\sigma_{ab}{}^c$	global phase transformation. The first point to note is
1510.06699_F00657	32	■ 1.000	$\sigma_{ab}{}^b=0$	$\sigma_{ab}{}^b=0$	Recalling that $\sigma_{ab}{}^b=0$ for Dirac matter, the transfor-
1510.06699_F00658	32	■ 0.444	β_1	β_1	unnecessary for the Dirac field. is therefore to convert
1510.06699_F00659	32	■ 0.444	β_2	β_2	unnecessary for the Dirac field. is therefore to convert
1510.06699_F00660	32	■ 0.444	$L_{\mathrm{\mathcal{T}+\epsilon}}$	$L_{\mathcal{T}+\epsilon}$	unnecessary for the Dirac field. is therefore to convert

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1510.06699_F00661	32	■ 0.998	$\widehat{\mathcal{F}}_{ab}^* \rightarrow \widehat{\mathcal{F}}_{ab}^\dagger, \mathcal{T}^{*c}_{ab} \rightarrow \mathcal{T}^{\dagger c}_{ab}$ $\widehat{\mathcal{F}}_{ab}^* \rightarrow \widehat{\mathcal{F}}_{ab}^\dagger, \mathcal{T}^{*c}_{ab} \rightarrow \mathcal{T}^{\dagger c}_{ab}$ and $\mathcal{D}_a^* \rightarrow \mathcal{D}_a^\dagger$, and recalling		
1510.06699_F00662	33	■ 1.000	$\tau^{\dagger a}_b = \tau^a_b$		deed, from (170), one sees immediately that $\tau^{\dagger a}_b = \tau^a_b$,
1510.06699_F00663	33	■ 1.000	φ, ϕ, h_a^μ		set of field variables φ, ϕ, h_a^μ and $A^{\dagger ab}_\mu$ discussed in
1510.06699_F00664	33	■ 0.607	$\widehat{\mathcal{A}}^{\dagger ab}_c \equiv \widehat{h}_a^\mu \widehat{A}^{\dagger ab}_\mu$		(60) may be written as (su
1510.06699_F00665	33	■ 1.000	$\mathcal{D}_a^\dagger \chi = \left(\mathcal{D}_a^\natural - \frac{1}{3} w \mathcal{T}_a^\natural \right) \chi$		In other words, when written in terms of the scale-
1510.06699_F00666	33	■ 1.000	$\mathcal{D}_a^\natural$		In other words, when written in terms of the scale-
1510.06699_F00667	33	■ 1.000	$\mathcal{T}_a^\natural$		invariant field variables, the Lagrangian density $\mathcal{L}_T$ (in-
1510.06699_F00668	33	■ 1.000	$\widehat{\mathcal{D}}_a^\natural$		deed each term separately) has the <i>same</i> functional form
1510.06699_F00669	33	■ 1.000	$\widehat{\mathcal{T}}_a^\natural$		as it does in terms of the original variables with $\phi = \phi_0$.
1510.06699_F00670	33	■ 1.000	$\widehat{A}^{\dagger ab}_\mu$		mediately conclude that (each term in) the equation of
1510.06699_F00671	33	■ 1.000	$\widehat{\mathcal{D}}_a^\natural = \widehat{h}_a^\mu \mathcal{D}_\mu^\natural$		mediately conclude that (each term in) the equation of
1510.06699_F00672	33	■ 1.000	$\mathcal{D}_a^\dagger \phi = \frac{1}{3} (\phi^2 / \phi_0) \widehat{\mathcal{T}}_a^\natural$		For the ϕ -field equation, however, the situation is more subtle, since (92) does not depend explicitly on ϕ .
1510.06699_F00673	33	■ 0.996	$\widehat{\mathcal{D}}_a^\dagger = \widehat{\mathcal{D}}_a^\natural - \frac{1}{3} w \widehat{\mathcal{T}}_a^\natural$		law (69c), which may be straightforwardly rewritten as
1510.06699_F00674	33	■ 1.000	$\widehat{\mathcal{D}}_a^\dagger \widehat{\chi}$		$\delta \phi _{\phi=\phi_0} = \delta h_a^\mu _{\phi=\phi_0} \setminus \delta B_\mu _{\phi=\phi_0}$
1510.06699_F00675	33	■ 1.000	$\widehat{\chi}, \widehat{h}_a^\mu$		(94)
1510.06699_F00676	33	■ 1.000	$\mathcal{D}_a^\dagger \chi$		where we have made use of the φ -field equation $\delta \mathcal{L}_T / \delta \varphi = \delta \mathcal{L}_M / \delta \varphi = 0$ and set $\phi = \phi_0$ in the resulting expression. Turning now to the expression (90) for $\mathcal{L}_T$ in terms of scale-invariant variables and applying the
1510.06699_F00677	33	■ 0.999	$\mathcal{R}^{\dagger ab}_{cd} = (\phi / \phi_0)^2 \widehat{\mathcal{R}}^{\dagger ab}_{cd}$		chain rule for partial derivatives, one may show after a
1510.06699_F00678	33	■ 1.000	$\mathcal{T}^{\dagger a}_{bc} = (\phi / \phi_0) \widehat{\mathcal{T}}^{\dagger a}_{bc}$		chain rule for partial derivatives, one may show after a
1510.06699_F00679	33	■ 1.000	$\mathcal{H}^{\dagger}_{ab} = (\phi / \phi_0)^2 \widehat{\mathcal{H}}^{\dagger}_{ab}$		
1510.06699_F00680	33	■ 1.000	φ, h_a^μ		if χ represents φ, h_a^μ , or $A^{\dagger ab}_\mu$, one may immediately
1510.06699_F00681	34	■ 1.000	φ, h		for the φ, h and A (or $A^\dagger$) fields are the only three <i>in-</i>
1510.06699_F00682	34	■ 1.000	$\varphi, \phi, h_a^\mu, A^{ab}_\mu$		ables $\varphi, \phi, h_a^\mu, A^{ab}_\mu$ and V_μ , rather than the alternative
1510.06699_F00683	34	■ 1.000	$\widehat{V}_\mu = \frac{1}{3} \widehat{\mathcal{T}}_\mu^\natural$		where it is clear from the last definition that $\widehat{V}_\mu = \frac{1}{3} \widehat{\mathcal{T}}_\mu^\natural$.
1510.06699_F00684	34	■ 1.000	$\widehat{\mathcal{V}}_a \equiv \widehat{h}_a^\mu \widehat{V}_\mu$		variables $\widehat{\mathcal{A}}^{\dagger ab}_c \equiv \widehat{h}_a^\mu \widehat{A}^{\dagger ab}_\mu$ and $\widehat{\mathcal{V}}_a \equiv \widehat{h}_a^\mu \widehat{V}_\mu$. The situa-
1510.06699_F00685	34	■ 1.000	φ, h_a^μ		Specifically, the equations of motion in terms of φ, h_a^μ ,
1510.06699_F00686	34	■ 1.000	$V_\mu + \frac{1}{3} T_\mu$		tions $A^{\dagger ab}_\mu$ and $V_\mu + \frac{1}{3} T_\mu$ (which will always be possible).
1510.06699_F00687	34	■ 1.000	$\widehat{\varphi}, \widehat{h}_a^\mu, \widehat{A}^{\dagger ab}_\mu$		$\widehat{\varphi}, \widehat{h}_a^\mu, \widehat{A}^{\dagger ab}_\mu$ and $\widehat{V}_\mu$, respectively.
1510.06699_F00688	34	■ 1.000	$\widehat{V}_\mu$		$\widehat{\varphi}, \widehat{h}_a^\mu, \widehat{A}^{\dagger ab}_\mu$ and $\widehat{V}_\mu$, respectively.
1510.06699_F00689	34	■ 1.000	$v^c (\mathcal{D}_c^\dagger p_a -$		in a manifestly eWGT-covariant manner as $v^c (\mathcal{D}_c^\dagger p_a -$
1510.06699_F00690	34	■ 0.971	$\mathcal{T}_{cab}^\dagger p^b) = e \mu^2 \phi \mathcal{D}_a^\dagger \phi$		$\mathcal{T}_{cab}^\dagger p^b) = e \mu^2 \phi \mathcal{D}_a^\dagger \phi$. By analogy with our treatment in

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1510.06699_F00691	34	■ 0.997	$v^{\{c\}\}\wedge\{0\}}\mathcal{D}_{c}^{\dagger}p_a=e\mu^2\phi^0\mathcal{D}_a^{\dagger}\phi$ $\phi^{\{0\}}\mathcal{D}_{\{a}^{\dagger}\}\wedge\{0\}}\phi$	$v^{c0}\mathcal{D}_c^{\dagger}p_a=e\mu^2\phi^0\mathcal{D}_a^{\dagger}\phi$	$v^c{}^0\mathcal{D}_c^{\dagger}p_a=e\mu^2\phi^0\mathcal{D}_a^{\dagger}\phi$, which is again manifestly eWGT-
1510.06699_F00692	34	■ 1.000	$\{ \}^{\{0\}}D_{\{\mu\}}^{\dagger}$	${}^0D_{\mu}^{\dagger}$	covariant derivative operator ${}^0D_{\mu}^{\dagger}$ in (159), to which the
1510.06699_F00693	34	■ 0.718	$\mathcal{T}_{\{a\}b\}c}^{\dagger}\equiv 0$	$\mathcal{T}_{abc}^{\dagger}\equiv 0$	with $\mathcal{T}_{abc}^{\dagger}\equiv 0$ (and hence $\mathcal{K}_{abc}^{\dagger}\equiv 0$). In contrast to
1510.06699_F00694	34	■ 0.718	$\left.\mathcal{K}_{\{a\}b\}c}^{\dagger}\right)\equiv 0$	$\mathcal{K}_{abc}^{\dagger}\equiv 0$	with $\mathcal{T}_{abc}^{\dagger}\equiv 0$ (and hence $\mathcal{K}_{abc}^{\dagger}\equiv 0$). In contrast to
1510.06699_F00695	35	■ 1.000	$\{ \}^{\{0\}}R^{\dagger}a\}b\{ \}_\{\mu\}\nu\left(h,\partial h,\partial^2h,A,\partial A,V,\partial V\right)$	${}^0R^{\dagger ab}{}_{\mu\nu}(h,\partial h,\partial^2h,A,\partial A,V,\partial V)$	where we have introduced the new scale-invariant ein-
1510.06699_F00696	35	■ 1.000	$\{ \}^{\{0\}}A^{\dagger}a\}b\{ \}_\{\mu\}$	${}^0A^{\dagger ab}{}_{\mu}$	bein $\widehat{e}=[1-\lambda(dx^{\mu}/d\lambda)\partial_{\mu}\ln(\phi/\phi_0)]\widehat{e}$. Thus, in terms
1510.06699_F00697	35	■ 1.000	$\{ \}^{\{0\}}R^{\dagger}a\}b\{ \}_\{\mu\}\nu$	${}^0R^{\dagger ab}{}_{\mu\nu}$	of the scale-invariant variables $\widehat{\lambda}$, $\widehat{p}_a$, $\widehat{v}^a$ and $\widehat{e}$, respec-
1510.06699_F00698	35	■ 1.000	$R^{\dagger}a\}b\{ \}_\{\mu\}\nu(h,A,\partial A,V,\partial V)$	$R^{\dagger ab}{}_{\mu\nu}(h,A,\partial A,V,\partial V)$	invariant variables (105), each occurrence of the gravita-
1510.06699_F00699	35	■ 1.000	$\{ \}^{\{0\}}\mathcal{R}^{\dagger}a\}b\{ \}_\{c\}d=h_{\{c\}\}\wedge\{\mu\}h_{\{d\}\}\wedge\{\mu\}\{ \}^{\{0\}}R^{\dagger}a\}b\{ \}_\{\mu\}\nu$	${}^0\mathcal{R}^{\dagger ab}{}_{cd}=h_c{}^{\mu}h_d{}^{\mu}{}^0R^{\dagger ab}{}_{\mu\nu}$	where ${}^0\mathcal{R}^{\dagger ab}{}_{cd}=h_c{}^{\mu}h_d{}^{\mu}{}^0R^{\dagger ab}{}_{\mu\nu}$, and the commuta-
1510.06699_F00700	35	■ 1.000	$\{ \}^{\{0\}}\mathcal{T}^{\dagger}a\}b\{ \}_\{c\}\equiv$	${}^0\mathcal{T}^{\dagger a}{}_{bc}\equiv$	strength of the h gauge field (or ‘torsion’), since ${}^0\mathcal{T}^{\dagger a}{}_{bc}\equiv$
1510.06699_F00701	35	■ 1.000	$h_{\{b\}\}\wedge\{\mu\}h_{\{c\}\}\wedge\{\nu\}\left(\{ \}^{\{0\}}D_{\{\mu\}}^{\dagger}b^a{}_{\nu}-{}^0D_{\nu}^{\dagger}b^a{}_{\mu}\right)=0$	$h_b{}^{\mu}h_c{}^{\nu}\left({}^0D_{\mu}^{\dagger}b^a{}_{\nu}-{}^0D_{\nu}^{\dagger}b^a{}_{\mu}\right)=0$	$h_b{}^{\mu}h_c{}^{\nu}\left({}^0D_{\mu}^{\dagger}b^a{}_{\nu}-{}^0D_{\nu}^{\dagger}b^a{}_{\mu}\right)=0$. As one might expect, the
1510.06699_F00702	35	■ 0.785	$\mathcal{D}_{\{a\}}^{\dagger}\rightarrow{}^0\mathcal{D}_{\{a\}}^{\dagger},\mathcal{R}_{abcd}\rightarrow{}^0\mathcal{R}_{abcd}^{\dagger}$ $\mathcal{D}_{\{a\}}^{\dagger},\mathcal{R}_{\{a\}b\}c\}d\rightarrow\{ \}^{\{0\}}\mathcal{R}_{\{a\}b\}c\}d^{\dagger}$	$\mathcal{D}_a^{\dagger}\rightarrow{}^0\mathcal{D}_a^{\dagger},\mathcal{R}_{abcd}\rightarrow{}^0\mathcal{R}_{abcd}^{\dagger}$	with the replacements $\mathcal{D}_a^{\dagger}\rightarrow{}^0\mathcal{D}_a^{\dagger}$, $\mathcal{R}_{abcd}\rightarrow{}^0\mathcal{R}_{abcd}^{\dagger}$ and
1510.06699_F00703	35	■ 0.831	$\mathcal{T}_{\{a\}b\}c}^{\dagger}\rightarrow\{ \}^{\{0\}}\mathcal{T}_{\{a\}b\}c}^{\dagger}=0$	$\mathcal{T}_{abc}^{\dagger}\rightarrow{}^0\mathcal{T}_{abc}^{\dagger}=0$	$\mathcal{T}_{abc}^{\dagger}\rightarrow{}^0\mathcal{T}_{abc}^{\dagger}=0$.
1510.06699_F00704	35	■ 1.000	$\{ \}^{\{0\}}\mathcal{R}^{\dagger}a\}b\{ \}_\{c\}d\rightarrow$	${}^0\mathcal{R}^{\dagger ab}{}_{cd}\rightarrow$	(110c), but with the replacements on the RHS ${}^0\mathcal{R}^{\dagger ab}{}_{cd}\rightarrow$
1510.06699_F00705	35	■ 1.000	$\{ \}^{\{0\}}\mathcal{R}^{\dagger}a\}b\{ \}_\{c\}d,\{ \}^{\{0\}}\mathcal{D}_{\{a\}}^{\dagger}\rightarrow\{ \}^{\{0\}}\mathcal{D}_{\{a\}}^{\dagger},\mathcal{K}_{\{a\}b\}c}^{\dagger}\rightarrow\mathcal{K}_{\{a\}b\}c}^{\dagger}$ $\mathcal{D}_{\{a\}}^{\dagger},\mathcal{K}_{\{a\}b\}c}^{\dagger}\rightarrow\{ \}^{\{0\}}\mathcal{D}_{\{a\}}^{\dagger},\mathcal{K}_{\{a\}b\}c}^{\dagger}\rightarrow\{ \}^{\{0\}}\mathcal{K}_{\{a\}b\}c}^{\dagger}$	${}^0\mathcal{R}^{\dagger ab}{}_{cd},{}^0\mathcal{D}_a^{\dagger}\rightarrow{}^0\mathcal{D}_a^{\dagger},\mathcal{K}_{abc}\rightarrow\mathcal{K}_{abc}^{\dagger}$	${}^0\mathcal{R}^{\dagger ab}{}_{cd},{}^0\mathcal{D}_a^{\dagger}\rightarrow{}^0\mathcal{D}_a^{\dagger},\mathcal{K}_{abc}\rightarrow\mathcal{K}_{abc}^{\dagger}$ and $\mathcal{T}_{abc}\rightarrow\mathcal{T}_{abc}^{\dagger}$, and
1510.06699_F00706	35	■ 1.000	$\mathcal{R}^{\dagger}a\}b\{ \}_\{c\}d\rightarrow\{ \}^{\{0\}}\mathcal{R}^{\dagger}a\}b\{ \}_\{c\}d$	$\mathcal{R}^{\dagger ab}{}_{cd}\rightarrow{}^0\mathcal{R}^{\dagger ab}{}_{cd}$	gauge fields: although one can substitute for the rota-
1510.06699_F00707	35	■ 1.000	$\{ \}^{\{0\}}\mathcal{R}^{\dagger}a\}b\{ \}_\{a\}b\}$	${}^0\mathcal{R}^{\dagger ab}$	gauge field to obtain equations that contain higher-
1510.06699_F00708	35	■ 0.999	$\{ \}^{\{0\}}\mathcal{R}_{\{a\}b\}c\}d^{\dagger}$	${}^0\mathcal{R}_{abcd}^{\dagger}$	order dervivatives, the theory does not suffer from Os-
1510.06699_F00709	35	■ 1.000	$\mathcal{R}_{\{a\}b\}c\}d\rightarrow\{ \}^{\{0\}}\mathcal{R}_{\{a\}b\}c\}d^{\dagger}$	$\mathcal{R}_{abcd}\rightarrow{}^0\mathcal{R}_{abcd}^{\dagger}$	correspond to ‘second-order’ and ‘first-order’ variational
1510.06699_F00710	35	■ 0.535	$\mathcal{D}_{\{a\}}^{\dagger}\rightarrow{}^0\mathcal{D}_{\{a\}}^{\dagger}$	$\mathcal{D}_a^{\dagger}\rightarrow{}^0\mathcal{D}_a^{\dagger}$	ment $\mathcal{D}_a^{\dagger}\rightarrow{}^0\mathcal{D}_a^{\dagger}$. As in WGT, however, the gravitational
1510.06699_F00711	35	■ 0.720	$\phi^2\{ \}^{\{0\}}\mathcal{R}^{\dagger}$	$\phi^2{}^0\mathcal{R}^{\dagger}$	$\mathcal{L}_G$ and the term proportional to $\phi^2{}^0\mathcal{R}^{\dagger}$ in $\mathcal{L}_M$ depends
1510.06699_F00712	35	■ 0.806	$\mathcal{D}_{\{a\}}^{\dagger}\rightarrow{}^0\mathcal{D}_{\{a\}}^{\dagger},\mathcal{R}_{abcd}^{\dagger}\rightarrow{}^0\mathcal{R}_{abcd}^{\dagger}$ $\mathcal{D}_{\{a\}}^{\dagger},\mathcal{R}_{\{a\}b\}c\}d^{\dagger}\rightarrow\{ \}^{\{0\}}\mathcal{D}_{\{a\}}^{\dagger},\mathcal{R}_{\{a\}b\}c\}d^{\dagger}$	$\mathcal{D}_a^{\dagger}\rightarrow{}^0\mathcal{D}_a^{\dagger},\mathcal{R}_{abcd}^{\dagger}\rightarrow{}^0\mathcal{R}_{abcd}^{\dagger}$	replacements $\mathcal{D}_a^{\dagger}\rightarrow{}^0\mathcal{D}_a^{\dagger}$, $\mathcal{R}_{abcd}^{\dagger}\rightarrow{}^0\mathcal{R}_{abcd}^{\dagger}$ and $\mathcal{T}_{abc}^{\dagger}\rightarrow$
1510.06699_F00713	35	■ 0.806	$\mathcal{T}_{\{a\}b\}c}^{\dagger}\rightarrow$	$\mathcal{T}_{abc}^{\dagger}\rightarrow$	replacements $\mathcal{D}_a^{\dagger}\rightarrow{}^0\mathcal{D}_a^{\dagger}$, $\mathcal{R}_{abcd}^{\dagger}\rightarrow{}^0\mathcal{R}_{abcd}^{\dagger}$ and $\mathcal{T}_{abc}^{\dagger}\rightarrow$
1510.06699_F00714	35	■ 0.961	$\{ \}^{\{0\}}\mathcal{T}_{\{a\}b\}c}^{\dagger}\equiv 0$	${}^0\mathcal{T}_{abc}^{\dagger}\equiv 0$	${}^0\mathcal{T}_{abc}^{\dagger}\equiv 0$. Furthermore, by performing analogous calcu-
1510.06699_F00715	35	■ 0.891	$\mathcal{T}_{\{a\}b\}c}^{\dagger}=0$	$\mathcal{T}_{abc}^{\dagger}=0$	erly eWGT-covariant) condition $\mathcal{T}_{abc}^{\dagger}=0$ in the field
1510.06699_F00716	36	■ 1.000	$b^{\{b\}\}\wedge\{\nu\}$	$b^b{}_{\nu}$	notate $h_a{}^{\mu}$ and $b^b{}_{\nu}$ by $e_a{}^{\mu}$ and $e^b{}_{\nu}$, respectively.
1510.06699_F00717	36	■ 1.000	$e^{\{b\}\}\wedge\{\nu\}$	$e^b{}_{\nu}$	notate $h_a{}^{\mu}$ and $b^b{}_{\nu}$ by $e_a{}^{\mu}$ and $e^b{}_{\nu}$, respectively.
1510.06699_F00718	36	■ 1.000	$e_{\{a\}\}\wedge\{\mu\},A^{\{a\}b\}\wedge\{\mu\}$	$e_a{}^{\mu},A^{ab}{}_{\mu}$	(142) of the three gauge fields $e_a{}^{\mu}$, $A^{ab}{}_{\mu}$, and V_{μ} that
1510.06699_F00719	36	■ 1.000	$\partial_{\{\mu\}}^{\dagger}$	$\partial_{\mu}^{\dagger}$	ject of weight w , for all these derivative operators the re-
1510.06699_F00720	36	■ 1.000	$A^{\dagger}a\}b\{ \}_\{\mu\}=-A^{\dagger}b\}a\{ \}_\{\mu\}$	$A^{\dagger ab}{}_{\mu}=-A^{\dagger ba}{}_{\mu}$	weight w .

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1510.06699_F00752	37	■ 0.986	$T^{\dagger}\{\mathrm{\sigma}\}_{\{\mu\nu\}}\rightarrow{}^0T^{\dagger}\{\mathrm{\sigma}\}_{\{\mu\nu\}}=0$	$T^{\dagger\sigma}_{\mu\nu}\rightarrow{}^0T^{\dagger\sigma}_{\mu\nu}=0$	$\hat{\nabla}_{\mu}^{\dagger}\rightarrow{}^0\nabla_{\mu}^{\dagger}$, $R^{\rho}_{\sigma\mu\nu}\rightarrow{}^0R^{\dagger\rho}_{\sigma\mu\nu}$ and $T^{\dagger\sigma}_{\mu\nu}\rightarrow{}^0T^{\dagger\sigma}_{\mu\nu}=0$.
1510.06699_F00753	37	■ 1.000	$L_{\mathrm{T}}=$	$L_{\mathrm{T}}=$	we now consider in detail has a total Lagrangian $L_{\mathrm{T}}=$
1510.06699_F00754	37	■ 1.000	$L_{\mathrm{G}}+L_{\mathrm{M}}$	$L_{\mathrm{G}}+L_{\mathrm{M}}$	$L_{\mathrm{G}}+L_{\mathrm{M}}$ corresponding to the general eWGT form (167).
1510.06699_F00755	37	■ 1.000	$\{\}^{\{0\}}\Gamma^{\lambda}_{\mu\nu}$	${}^0\Gamma^{\lambda}_{\mu\nu}$	in which ${}^0\Gamma^{\lambda}_{\mu\nu}$ is the standard metric (Christoffel) con-
1510.06699_F00756	37	■ 1.000	$K^{\dagger}\{\mathrm{\lambda}\}_{\{\mu\nu\}}$	$K^{\dagger\lambda}_{\mu\nu}$	nection and $K^{\dagger\lambda}_{\mu\nu}$ is the $\widehat{Y}_4$ contortion tensor
1510.06699_F00757	37	■ 1.000	$L_{\mathrm{G}}=L_{\mathrm{mathcal{R}}^{\dagger 2}}+L_{\mathrm{mathcal{H}}^{\dagger 2 2}}$	$L_G=L_{\mathcal{R}^{\dagger 2}}+L_{\mathcal{H}^{\dagger 2}}$	namely $L_G=L_{\mathcal{R}^{\dagger 2}}+L_{\mathcal{H}^{\dagger 2}}$, where
1510.06699_F00758	37	■ 1.000	$L_{\mathrm{M}}=$	$L_{\mathrm{M}}=$	of gravity. In particular, we adopt the form $L_{\mathrm{M}}=$
1510.06699_F00759	38	■ 1.000	ν, λ, a	ν, λ, a	in which ν, λ, a and β_i are again dimensionless parame-
1510.06699_F00760	38	■ 1.000	ν, λ	ν, λ	ters (with ν, λ and a usually positive) and some factors
1510.06699_F00761	38	■ 1.000	L_{φ}	L_{φ}	the moment we allow the form of L_{φ} for the matter field
1510.06699_F00762	38	■ 1.000	$\left(\tau^{\dagger}_{\varphi}\right)^a{}_b=h_b{}^{\mu}(\delta L_{\varphi}/\delta h_a{}^{\mu})_{\dagger}$	$\left(\tau^{\dagger}_{\varphi}\right)^a{}_b=h_b{}^{\mu}(\delta L_{\varphi}/\delta h_a{}^{\mu})_{\dagger}$	and in the matter sector, in addition to $(\tau^{\dagger}_{\varphi})^a{}_b=h_b{}^{\mu}(\delta L_{\varphi}/\delta h_a{}^{\mu})_{\dagger}$, one has
1510.06699_F00763	38	■ 1.000	$\left(t^{\dagger}_{\mathcal{H}^2}\right)^a{}_b$	$\left(t^{\dagger}_{\mathcal{H}^2}\right)^a{}_b$	In particular, we note that $(t^{\dagger}_{\mathcal{H}^2})^a{}_b$ in (230) is linear
1510.06699_F00764	39	■ 1.000	$\left(\sigma^{\dagger}_{\varphi}\right)_{ab}{}^c=$	$(\sigma^{\dagger}_{\varphi})_{ab}{}^c=$	on the scalar field ϕ .
1510.06699_F00765	39	■ 1.000	$b^c{}_{\mu}\delta L_{\varphi}/\delta A^{ab}{}_{\mu}$	$b^c{}_{\mu}\delta L_{\varphi}/\delta A^{ab}{}_{\mu}$	—
1510.06699_F00766	39	■ 0.860	$(s_{\mathcal{H}^2})_{ab}{}^c$	$(s_{\mathcal{H}^2})_{ab}{}^c$	We note that $(s_{\mathcal{H}^2})_{ab}{}^c$ in (236) is linear in third-order
1510.06699_F00767	39	■ 0.554	$\mathcal{T}^{\dagger}_{cab}+2\mathcal{T}^{\dagger}_{[ab]c}-3\mathcal{T}^{\dagger}_{[cab]}=0$	$\mathcal{T}^{\dagger}_{cab}+2\mathcal{T}^{\dagger}_{[ab]c}-3\mathcal{T}^{\dagger}_{[cab]}=0$	strength satisfies the identity $\mathcal{T}^{\dagger}_{cab}+2\mathcal{T}^{\dagger}_{[ab]c}-3\mathcal{T}^{\dagger}_{[cab]}=0$.
1510.06699_F00768	39	■ 1.000	α_4	α_4	in the action (namely those in (235) proportional to α_4 ,
1510.06699_F00769	39	■ 1.000	α_5	α_5	α_5 and α_6 , respectively) have a form similar to the LHS
1510.06699_F00770	39	■ 0.606	$\overline{\mathcal{R}}^{\dagger}_{abcd}\equiv\mathcal{R}^{\dagger}_{cdab}$	$\overline{\mathcal{R}}^{\dagger}_{abcd}\equiv\mathcal{R}^{\dagger}_{cdab}$	$\overline{\mathcal{R}}^{\dagger}_{abcd}\equiv\mathcal{R}^{\dagger}_{cdab}$. Since $\overline{\mathcal{R}}^{\dagger a}{}_b=\mathcal{R}^{\dagger a}{}_b$, one also sees that
1510.06699_F00771	39	■ 0.606	$\overline{\mathcal{R}}^{\dagger a}{}_b=\mathcal{R}^{\dagger a}{}_b$	$\overline{\mathcal{R}}^{\dagger a}{}_b=\mathcal{R}^{\dagger a}{}_b$	$\overline{\mathcal{R}}^{\dagger}_{abcd}\equiv\mathcal{R}^{\dagger}_{cdab}$. Since $\overline{\mathcal{R}}^{\dagger a}{}_b=\mathcal{R}^{\dagger a}{}_b$, one also sees that
1510.06699_F00772	39	■ 1.000	α_2	α_2	terms in (235) proportional to α_2 and α_3 , respectively,
1510.06699_F00773	39	■ 1.000	α_3	α_3	terms in (235) proportional to α_2 and α_3 , respectively,
1510.06699_F00774	39	■ 1.000	α_1	α_1	proportional to α_1 , since $\overline{\mathcal{R}}^{\dagger}=\mathcal{R}^{\dagger}$. The term propor-
1510.06699_F00775	39	■ 1.000	$\overline{\mathcal{R}}^{\dagger}=\mathcal{R}^{\dagger}$	$\overline{\mathcal{R}}^{\dagger}=\mathcal{R}^{\dagger}$	proportional to α_1 , since $\overline{\mathcal{R}}^{\dagger}=\mathcal{R}^{\dagger}$. The term propor-
1510.06699_F00776	39	■ 1.000	$(\zeta_{\varphi})_a=$	$(\zeta_{\varphi})_a=$	—
1510.06699_F00777	39	■ 1.000	$b_{a\mu}\delta L_{\varphi}/\delta V_{\mu}$	$b_{a\mu}\delta L_{\varphi}/\delta V_{\mu}$	$\mathcal{L}_{\mathrm{eff}})-\delta^a\mathcal{R}^{\dagger}.\mathcal{L}$
1510.06699_F00778	39	■ 1.000	c	c	equation (234) contracted on the indices b and c , in ac-
1510.06699_F00779	39	■ 1.000	$j^{\dagger}_a$	$j^{\dagger}_a$	so $j^{\dagger}_a$ and $\zeta_a^{\dagger}$ vanish identically for each part of the La-
1510.06699_F00780	39	■ 1.000	$\zeta_a^{\dagger}$	$\zeta_a^{\dagger}$	so $j^{\dagger}_a$ and $\zeta_a^{\dagger}$ vanish identically for each part of the La-
1510.06699_F00781	39	■ 1.000	$\partial_{\phi}\equiv\partial/\partial\phi$	$\partial_{\phi}\equiv\partial/\partial\phi$	where $\partial_{\phi}\equiv\partial/\partial\phi$ and $\delta_{\phi}\equiv\delta/\delta\phi$, since only L_{ϕ} depends
1510.06699_F00782	39	■ 1.000	$\delta_{\phi}\equiv\delta/\delta\phi$	$\delta_{\phi}\equiv\delta/\delta\phi$	where $\partial_{\phi}\equiv\partial/\partial\phi$ and $\delta_{\phi}\equiv\delta/\delta\phi$, since only L_{ϕ} depends
1510.06699_F00783	39	■ 1.000	L_{ϕ}	L_{ϕ}	where $\partial_{\phi}\equiv\partial/\partial\phi$ and $\delta_{\phi}\equiv\delta/\delta\phi$, since only L_{ϕ} depends
1510.06699_F00784	40	■ 1.000	$-\phi$	$-\phi$	by $-\phi$, in accordance with the general results (177) and
1510.06699_F00785	40	■ 0.999	$\delta_{\varphi}L_{\varphi}=0$	$\delta_{\varphi}L_{\varphi}=0$	(178d), assuming the φ -equation $\delta_{\varphi}L_{\varphi}=0$ is satisfied.
1510.06699_F00786	40	■ 1.000	$\xi=0$	$\xi=0$	tional Lagrangian (by setting $\xi=0$). Indeed, one can
1510.06699_F00787	40	■ 1.000	$\mathcal{D}^{\dagger}_a\phi _{\phi=\phi_0}=\phi_0\left(\mathcal{V}_a+\frac{1}{3}\mathcal{T}_a\right)$	$\mathcal{D}^{\dagger}_a\phi _{\phi=\phi_0}=\phi_0\left(\mathcal{V}_a+\frac{1}{3}\mathcal{T}_a\right)$	the replacement $\mathcal{D}^{\dagger}_a\phi _{\phi=\phi_0}=\phi_0(\mathcal{V}_a+\frac{1}{3}\mathcal{T}_a)$ throughout,
1510.06699_F00788	40	■ 1.000	$\widehat{\varphi},\widehat{h}_{\mathrm{a}}{}^{\mu}$	$\widehat{\varphi},\widehat{h}_{\mathrm{a}}{}^{\mu}$	expressed in terms of the scale-invariant variables $\widehat{\varphi},\widehat{h}_{\mathrm{a}}{}^{\mu}$,
1510.06699_F00789	41	■ 1.000	$h^{\prime}{}_{\mathrm{a}}{}^{\mu}=e^{-\rho}h_{\mathrm{a}}{}^{\mu}$	$h^{\prime}{}_{\mathrm{a}}{}^{\mu}=e^{-\rho}h_{\mathrm{a}}{}^{\mu}$	fields are given by $h^{\prime}{}_{\mathrm{a}}{}^{\mu}=e^{-\rho}h_{\mathrm{a}}{}^{\mu}$ and $A^{\prime ab}{}_{\mu}=A^{ab}{}_{\mu}+$
1510.06699_F00790	41	■ 1.000	$A^{\prime ab}{}_{\mu}=A^{ab}{}_{\mu}+$	$A^{\prime ab}{}_{\mu}=A^{ab}{}_{\mu}+$	fields are given by $h^{\prime}{}_{\mathrm{a}}{}^{\mu}=e^{-\rho}h_{\mathrm{a}}{}^{\mu}$ and $A^{\prime ab}{}_{\mu}=A^{ab}{}_{\mu}+$

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1510.06699_F00791	41	■ 1.000	$\theta\left(b^{\{a\}}_{\{\mu\}}\mathcal{P}^{\{b\}}-b^{\{b\}}_{\{\mu\}}\mathcal{P}^{\{a\}}\right)$	$\theta\left(b^a_{\mu}\mathcal{P}^b-b^b_{\mu}\mathcal{P}^a\right)$	$\theta(b^a_{\mu}\mathcal{P}^b-\tilde{b}^b_{\mu}\mathcal{P}^a)$, respectively, where $\mathcal{P}_a=h_a^{\mu}\partial_{\mu}\rho$. Here
1510.06699_F00792	41	■ 1.000	$\mathcal{P}_{\{a\}}=h_{\{a\}}^{\{\mu\}}\partial_{\{\mu\}}\rho$	$\mathcal{P}_a=h_a^{\mu}\partial_{\mu}\rho$	$\theta(b^a_{\mu}\mathcal{P}^b-\tilde{b}^b_{\mu}\mathcal{P}^a)$, respectively, where $\mathcal{P}_a=h_a^{\mu}\partial_{\mu}\rho$. Here
1510.06699_F00793	41	■ 1.000	$\mathcal{P}^{\{2\}}\equiv\mathcal{P}_{\{a\}}\mathcal{P}^{\{a\}}$	$\mathcal{P}^2\equiv\mathcal{P}_a\mathcal{P}^a$	where $\mathcal{P}^2\equiv\mathcal{P}_a\mathcal{P}^a$, and the translational gauge field
1510.06699_F00794	41	■ 1.000	$(\theta=0)$	$(\theta=0)$	is covariant with weight $w=-2$ under ‘normal’ ($\theta=0$)
1510.06699_F00795	41	■ 1.000	$(\theta=1)$	$(\theta=1)$	$w=-1$ under ‘special’ transformations ($\theta=1$).
1510.06699_F00796	41	■ 1.000	$\theta\neq 0$	$\theta\neq 0$	extended transformation law for the A -field, with $\theta\neq 0$,
1510.06699_F00797	41	■ 1.000	$\alpha_{\{i\}}=0$	$\alpha_i=0$	forms covariantly (aside from the trivial case $\alpha_i=0$ for
1510.06699_F00798	41	■ 0.996	i	i	all i) and so L_G must vanish. Nonetheless, by contrast,
1510.06699_F00799	41	■ 0.998	$L_{\{\mathcal{T}\}^{\{2\}}}$	$L_{\mathcal{T}^2}$	ance under electromagnetic gauge
1510.06699_F00800	42	■ 0.989	$\mathcal{L}_{\{\mathcal{M}\}}\equiv\mathcal{L}_{\{\psi\}}+\mathcal{L}_{\{\phi\}}+\phi^2\mathcal{L}_{\mathcal{R}}+\phi^2\mathcal{L}_{\mathcal{T}^2}$	$\mathcal{L}_{\mathbf{M}}\equiv\mathcal{L}_{\psi}+\mathcal{L}_{\phi}+\phi^2\mathcal{L}_{\mathcal{R}}+\phi^2\mathcal{L}_{\mathcal{T}^2}$	Let us denote (256) by $\mathcal{L}_{\mathbf{M}}\equiv\mathcal{L}_{\psi}+\mathcal{L}_{\phi}+\phi^2\mathcal{L}_{\mathcal{R}}+\phi^2\mathcal{L}_{\mathcal{T}^2}$.
1510.06699_F00801	42	■ 0.990	$\mathcal{L}_{\{\psi\}}$	$\mathcal{L}_{\psi}$	As discussed in Section III K, $\mathcal{L}_{\psi}$ is invariant under ex-
1510.06699_F00802	42	■ 1.000	ν, a	ν, a	rameters ν, a and β_i ($i=1,2,3$) satisfy the conditions
1510.06699_F00803	42	■ 1.000	$\beta_{\{i\}}(i=1,2,3)$	$\beta_i(i=1,2,3)$	rameters ν, a and β_i ($i=1,2,3$) satisfy the conditions
1510.06699_F00804	42	■ 1.000	ν	ν	the parameters ν and a must be of opposite sign, which
1510.06699_F00805	42	■ 1.000	$\nu=0=2\beta_1+\beta_2+3\beta_3$	$\nu=0=2\beta_1+\beta_2+3\beta_3$	$\nu=0=2\beta_1+\beta_2+3\beta_3$, and a ‘special’ transforma-
1510.06699_F00806	42	■ 0.997	$\nu=0=a$	$\nu=0=a$	tion ($\theta=1$), for which $\nu=0=a$. Thus, in both special
1510.06699_F00807	42	■ 1.000	$\nu=\beta_1=\beta_2=\beta_3=0$	$\nu=\beta_1=\beta_2=\beta_3=0$	Obukhov ⁸⁹ , who set $\nu=\beta_1=\beta_2=\beta_3=0$. Adopting
1510.06699_F00808	42	■ 1.000	$2\beta_1+\beta_2+3\beta_3=0$	$2\beta_1+\beta_2+3\beta_3=0$	with $2\beta_1+\beta_2+3\beta_3=0$ in $L_{\mathcal{T}^{+2}}$, which is merely a special
1510.06699_F00809	42	■ 0.706	$\mathcal{D}_{\{a\}}\phi\rightarrow\left(\mathcal{D}_a+\frac{1}{3}\mathcal{T}_a\right)\phi$	$\mathcal{D}_a\phi\rightarrow\left(\mathcal{D}_a+\frac{1}{3}\mathcal{T}_a\right)\phi$	$\mathcal{D}_a\phi\rightarrow(\mathcal{D}_a+\frac{1}{3}\mathcal{T}_a)\phi$ (and similarly for $\bar{\mathcal{D}}^a\phi$) in the kinetic
1510.06699_F00810	42	■ 0.706	$\mathcal{D}^{\{a\}}\phi$	$\mathcal{D}^a\phi$	$\mathcal{D}_a\phi\rightarrow(\mathcal{D}_a+\frac{1}{3}\mathcal{T}_a)\phi$ (and similarly for $\bar{\mathcal{D}}^a\phi$) in the kinetic
1510.06699_F00811	42	■ 1.000	$\frac{1}{2}\nu+6a=0$	$\frac{1}{2}\nu+6a=0$	vided only that $\frac{1}{2}\nu+6a=0$ and $2\beta_1+\beta_2+3\beta_3=0$
1510.06699_F00812	42	■ 0.994	$2\beta_1+\beta_2=0$	$2\beta_1+\beta_2=0$	with $2\beta_1+\beta_2=0$ in $L_{\mathcal{T}^{+2}}$ (in which there is no term
1510.06699_F00813	42	■ 0.994	$L_{\{\mathcal{T}\}^{\{+2\}}}$	$L_{\mathcal{T}^{+2}}$	with $2\beta_1+\beta_2=0$ in $L_{\mathcal{T}^{+2}}$ (in which there is no term
1510.06699_F00814	43	■ 1.000	$\{ \}^{\{0\}}\mathcal{D}_{\{a\}}$	${}^0\mathcal{D}_a$	derivative ${}^0\mathcal{D}_a$, to which the full PGT covariant deriva-
1510.06699_F00815	43	■ 1.000	$\{ \}^{\{0\}}\mathcal{T}^{\{a\}}_{\{b\}c}\equiv h_{\{b\}c}^{\{a\}}\left(\{ \}^{\{0\}}\mathcal{D}_{\{a\}}b^{\{b\}}_{\{c\}}-\{ \}^{\{0\}}\mathcal{D}_{\{c\}}b^{\{b\}}_{\{a\}}\right)=0$	${}^0\mathcal{T}^a_{bc}\equiv h_b^{\mu}h_c^{\nu}\left({}^0D_{\mu}b^a_{\nu}-{}^0D_{\nu}b^a_{\mu}\right)=0$	${}^0\mathcal{T}^a_{bc}\equiv h_b^{\mu}h_c^{\nu}\left({}^0D_{\mu}b^a_{\nu}-{}^0D_{\nu}b^a_{\mu}\right)=0$. Thus, reduced
1510.06699_F00816	43	■ 1.000	$\{ \}^{\{0\}}A^{\{\prime ab\}}_{\{\mu\}}=\{ \}^{\{0\}}A^{\{a\}b}_{\{\mu\}}+b^{\{a\}}_{\{\mu\}}\mathcal{P}^b-b^{\{b\}}_{\{\mu\}}\mathcal{P}^a$	${}^0A^{\prime ab}_{\mu}={}^0A^{ab}_{\mu}+b^a_{\mu}\mathcal{P}^b-b^b_{\mu}\mathcal{P}^a$	${}^0A^{\prime ab}_{\mu}={}^0A^{ab}_{\mu}+b^a_{\mu}\mathcal{P}^b-b^b_{\mu}\mathcal{P}^a$, which is independent
1510.06699_F00817	43	■ 1.000	$\{ \}^{\{0\}}A^{\{a\}b}_{\{\mu\}},\{ \}^{\{0\}}\mathcal{D}_{\{c\}}\varphi$	${}^0A^{ab}_{\mu},{}^0\mathcal{D}_c\varphi$	${}^0\bar{A}^{ab}_{\mu},{}^0\mathcal{D}_c\varphi$ and ${}^0\mathcal{R}^{ab}_{cd}$ all transform independently of
1510.06699_F00818	43	■ 0.999	$A^{\{a\}b}_{\{\mu\}},\mathcal{D}_{\{c\}}\varphi$	$A^{ab}_{\mu},\mathcal{D}_c\varphi$	$A^{ab}_{\mu},\mathcal{D}_c\varphi$ and $\mathcal{R}^{ab}_{cd}$ do assuming the ‘special’ transfor-
1510.06699_F00819	43	■ 1.000	$\mathcal{D}_{\{c\}}\rightarrow{}^0\mathcal{D}_{\{c\}}$	$\mathcal{D}_c\rightarrow{}^0\mathcal{D}_c$	replacements $\mathcal{D}_c\rightarrow{}^0\mathcal{D}_c$ and $\mathcal{T}^a_{bc}\rightarrow{}^0\mathcal{T}^a_{bc}\equiv 0$.
1510.06699_F00820	43	■ 1.000	$\mathcal{T}^{\{a\}}_{\{b\}c}\rightarrow{}^0\mathcal{T}^a_{bc}\equiv 0$	$\mathcal{T}^a_{bc}\rightarrow{}^0\mathcal{T}^a_{bc}\equiv 0$	replacements $\mathcal{D}_c\rightarrow{}^0\mathcal{D}_c$ and $\mathcal{T}^a_{bc}\rightarrow{}^0\mathcal{T}^a_{bc}\equiv 0$.
1510.06699_F00821	43	■ 0.915	$L_{\{G\}}$	L_G	Following the discussion in Appendix B, if L_G contains
1510.06699_F00822	43	■ 1.000	$2\alpha_3+\alpha_2=0$	$2\alpha_3+\alpha_2=0$	$2\alpha_3+\alpha_2=0$ and $3\alpha_1-\alpha_3=0$. Letting $\alpha=\alpha_3$ for con-
1510.06699_F00823	43	■ 1.000	$3\alpha_1-\alpha_3=0$	$3\alpha_1-\alpha_3=0$	$2\alpha_3+\alpha_2=0$ and $3\alpha_1-\alpha_3=0$. Letting $\alpha=\alpha_3$ for con-
1510.06699_F00824	43	■ 1.000	$\alpha=\alpha_3$	$\alpha=\alpha_3$	$2\alpha_3+\alpha_2=0$ and $3\alpha_1-\alpha_3=0$. Letting $\alpha=\alpha_3$ for con-
1510.06699_F00825	43	■ 1.000	$\{ \}^{\{0\}}\mathcal{R}_{\{a\}b\,c\,d}\{ \}^{\{0\}}\mathcal{R}^{\{a\}b\,c\,d}$	${}^0\mathcal{R}_{abcd}{}^0\mathcal{R}^{abcd}$	proportional to ${}^0\mathcal{R}_{abcd}{}^0\mathcal{R}^{abcd}$ to obtain
1510.06699_F00826	43	■ 0.899	$\{ \}^{\{\underline{101}\}}$	$\underline{101}$	theory ¹⁰¹ . Indeed, as one might suspect, one can easily
1510.06699_F00827	43	■ 0.999	$L_{\{\mathcal{W}\}}=\{ \}^{\{0\}}\mathcal{C}_{\{a\}b\,c\,d}\{ \}^{\{0\}}$	$L_{\mathbf{W}}={}^0\mathcal{C}_{abcd}{}^0\mathcal{C}^{abcd}$	grangian $L_{\mathbf{W}}={}^0\mathcal{C}_{abcd}{}^0\mathcal{C}^{abcd}$, where ${}^0\mathcal{C}_{abcd}$ is the gauge
1510.06699_F00828	43	■ 0.999	$\{ \}^{\{0\}}\mathcal{C}_{\{a\}b\,c\,d}$	${}^0\mathcal{C}_{abcd}$	grangian $L_{\mathbf{W}}={}^0\mathcal{C}_{abcd}{}^0\mathcal{C}^{abcd}$, where ${}^0\mathcal{C}_{abcd}$ is the gauge
1510.06699_F00829	44	■ 1.000	$\{ \}^{\{0\}}\mathcal{R}_{\{a\}b\,c\,d}$	${}^0\mathcal{R}_{abcd}$	This tensor represents the traceless part of ${}^0\mathcal{R}_{abcd}$; it has

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1510.06699_F00830	44	■ 0.988	$\{ \}^{\{0\}} \backslash \mathrm{mathcal{C}}_{\{a \ b \ c \ d\}}^{\{\prime\}} = e^{\{-2 \ \rho\}} \{ \}^{\{0\}} \backslash \mathrm{mathcal{C}}_{\{a \ b \ c \ d\}}$	${}^0\mathcal{C}'_{abcd} = e^{-2\rho^0}\mathcal{C}_{abcd}$	zero). It transforms covariantly as ${}^0\mathcal{C}'_{abcd} = e^{-2\rho}{}^0\mathcal{C}_{abcd}$
1510.06699_F00831	44	■ 1.000	$L_{\{W\}}$	L_W	under a local change of scale, so that L_W also transforms
1510.06699_F00832	44	■ 1.000	$w\left(L_{\{W\}}\right)=-4$	$w\left(L_W\right)=-4$	covariantly with a Weyl weight $w(L_W) = -4$, and hence
1510.06699_F00833	44	■ 1.000	$\{ \}^{\{\underline{19}\}}$	$\underline{19}$	nian and survives canonical quantisation ¹⁹ ; it thus seems
1510.06699_F00834	44	■ 1.000	$\backslash \mathrm{mathcal{D}}_{\{a\}} \rightarrow \{ \}^{\{0\}} \backslash \mathrm{mathcal{D}}_{\{a\}}$	$\mathcal{D}_a \rightarrow {}^0\mathcal{D}_a$	grangian (256), but with the replacements $\mathcal{D}_a \rightarrow {}^0\mathcal{D}_a$,
1510.06699_F00835	44	■ 0.554	$\backslash \mathrm{mathcal{R}} \rightarrow \{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}$	$\mathcal{R} \rightarrow {}^0\mathcal{R}$	$\mathcal{R} \rightarrow {}^0\mathcal{R}$ and without the term proportional to $L_{\mathcal{T}^2}$,
1510.06699_F00836	44	■ 0.972	$\{ \}^{\{\underline{103}\}}$	$\underline{103}$	namely ¹⁰³
1510.06699_F00837	44	■ 1.000	$\frac{1}{2} \nu +$	$\frac{1}{2} \nu +$	resulting matter Lagrangian, obtained by setting $\frac{1}{2} \nu +$
1510.06699_F00838	44	■ 1.000	$6 \ a=0$	$6a=0$	$6a=0$ in (268), coincides with that typically assumed in
1510.06699_F00839	44	■ 0.657	$\phi^{\{20\}} \backslash \mathrm{mathcal{R}}$	$\phi^{20}\mathcal{R}$	term for ϕ and the $\phi^2{}^0\mathcal{R}$ term.
1510.06699_F00840	44	■ 1.000	$\{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}_{\{a \ b \ c \ d\}} \rightarrow \{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}_{\{a \ b \ c \ d\}}^{\{*\}}$	${}^0\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}_{abcd}^*$	placements ${}^0\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}_{abcd}^*$ or ${}^0\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}_{abcd}^\dagger$ (and
1510.06699_F00841	44	■ 1.000	$\{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}_{\{a \ b \ c \ d\}} \rightarrow \{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}_{\{a \ b \ c \ d\}}^{\{\dagger\}}$	${}^0\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}_{abcd}^\dagger$	placements ${}^0\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}_{abcd}^*$ or ${}^0\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}_{abcd}^\dagger$ (and
1510.06699_F00842	44	■ 1.000	$\frac{1}{2} \nu + 6 \ a=$	$\frac{1}{2} \nu + 6a =$	Similarly, the matter Lagrangian (268) with $\frac{1}{2} \nu + 6a =$
1510.06699_F00843	44	■ 0.996	$\backslash \mathrm{mathcal{T}}_{\{a\}} \equiv 0$	$\mathcal{T}_a \equiv 0$	$\mathcal{T}_a \equiv 0$ and either set of replacements $\mathcal{D}_a^\dagger \rightarrow {}^0\mathcal{D}_a^*$,
1510.06699_F00844	44	■ 0.996	$\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{\dagger\}} \rightarrow \{ \}^{\{0\}} \backslash \mathrm{mathcal{D}}_{\{a\}}^{\{*\}}$	$\mathcal{D}_a^\dagger \rightarrow {}^0\mathcal{D}_a^*$	$\mathcal{T}_a \equiv 0$ and either set of replacements $\mathcal{D}_a^\dagger \rightarrow {}^0\mathcal{D}_a^*$,
1510.06699_F00845	44	■ 1.000	$\backslash \mathrm{mathcal{R}}^{\{\dagger\}} \rightarrow {}^0\mathcal{R}^* \mathcal{T}_{abc}^\dagger \rightarrow {}^0\mathcal{T}_{abc}^* \equiv 0$ $\backslash \mathrm{mathcal{T}}_{\{a \ b \ c\}}^{\{\dagger\}} \rightarrow \{ \}^{\{0\}} \backslash \mathrm{mathcal{T}}_{\{a \ b \ c\}}^{\{*\}} \equiv 0$	$\mathcal{R}^\dagger \rightarrow {}^0\mathcal{R}^* \mathcal{T}_{abc}^\dagger \rightarrow {}^0\mathcal{T}_{abc}^* \equiv 0$	$\mathcal{R}^\dagger \rightarrow {}^0\mathcal{R}^* \mathcal{T}_{abc}^\dagger \rightarrow {}^0\mathcal{T}_{abc}^* \equiv 0$ or $\mathcal{D}_a^\dagger \rightarrow {}^0\mathcal{D}_a^\dagger$, $\mathcal{R}^\dagger \rightarrow {}^0\mathcal{R}^\dagger$
1510.06699_F00846	44	■ 1.000	$\backslash \mathrm{mathcal{D}}_{\{a\}}^{\{\dagger\}} \rightarrow \{ \}^{\{0\}} \backslash \mathrm{mathcal{D}}_{\{a\}}^{\{\dagger\}}, \backslash \mathrm{mathcal{R}}^{\{\dagger\}} \rightarrow \{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}^{\{\dagger\}}$	$\mathcal{D}_a^\dagger \rightarrow {}^0\mathcal{D}_a^\dagger, \mathcal{R}^\dagger \rightarrow {}^0\mathcal{R}^\dagger$	$\mathcal{R}^\dagger \rightarrow {}^0\mathcal{R}^* \mathcal{T}_{abc}^\dagger \rightarrow {}^0\mathcal{T}_{abc}^* \equiv 0$ or $\mathcal{D}_a^\dagger \rightarrow {}^0\mathcal{D}_a^\dagger$, $\mathcal{R}^\dagger \rightarrow {}^0\mathcal{R}^\dagger$
1510.06699_F00847	44	■ 0.996	$\backslash \mathrm{mathcal{T}}_{\{a \ b \ c\}}^{\{\dagger\}} \rightarrow \{ \}^{\{0\}} \backslash \mathrm{mathcal{T}}_{\{a \ b \ c\}}^{\{\dagger\}} \equiv 0$	$\mathcal{T}_{abc}^\dagger \rightarrow {}^0\mathcal{T}_{abc}^\dagger \equiv 0$	$\mathcal{T}_{abc}^\dagger \rightarrow {}^0\mathcal{T}_{abc}^\dagger \equiv 0$. Thus, (268) with $\frac{1}{2} \nu + 6a = 0$ is equiv-
1510.06699_F00848	45	■ 0.997	$w, -1$	$w, -1$	$w, -1$ and 0, respectively, under the ‘extended’ transfor-
1510.06699_F00849	45	■ 1.000	$\backslash \mathrm{mathcal{R}}_{\{a \ b \ c \ d\}}^{\{\dagger\}}(h, A, \partial A, V, \partial V)$	$\mathcal{R}_{abcd}^\dagger(h, A, \partial A, V, \partial V)$	field strength tensors $\mathcal{R}_{abcd}^\dagger(h, A, \partial A, V, \partial V)$,
1510.06699_F00850	45	■ 1.000	$\backslash \mathrm{mathcal{T}}_{\{a \ b \ c\}}^{\{\dagger\}}(h, \partial h, A) \quad \quad$	$\mathcal{T}_{abc}^\dagger(h, \partial h, A)$	$\mathcal{T}_{abc}^\dagger(h, \partial h, A)$ and $\mathcal{H}_{ab}^\dagger(h, \partial h, \partial^2 h, A, \partial A, \partial V)$,
1510.06699_F00851	45	■ 1.000	$\quad \quad \backslash \mathrm{mathcal{H}}_{\{a \ b\}}^{\{\dagger\}}\left(h, \partial h, \partial^2 h, A, \partial A, \partial V\right)$	$\mathcal{H}_{ab}^\dagger(h, \partial h, \partial^2 h, A, \partial A, \partial V)$	$\mathcal{T}_{abc}^\dagger(h, \partial h, A)$ and $\mathcal{H}_{ab}^\dagger(h, \partial h, \partial^2 h, A, \partial A, \partial V)$,
1510.06699_F00852	45	■ 1.000	$\backslash \mathrm{mathcal{T}}_{\{a\}}^{\{\dagger\}}$	$\mathcal{T}_a^\dagger$	• The trace $\mathcal{T}_a^\dagger$ of the eWGT translational gauge field
1510.06699_F00853	45	■ 0.971	$\tau_{\{a \ b\}}^{\{\dagger\}}$	$\tau_{ab}^\dagger$	versions $t_{ab}^\dagger$ and $\tau_{ab}^\dagger$ of free-gravitational and mat-
1510.06699_F00854	45	■ 1.000	$A^{\{\dagger a \ b\}}_{\{\}\{\}\{\mu\}} \equiv$	$A^{\dagger ab}{}_\mu \equiv$	variables such that $A^{ab}{}_\mu$ is replaced by $A^{\dagger ab}{}_\mu \equiv$
1510.06699_F00855	45	■ 0.932	$A^{\{a \ b\}}_{\{\}\{\}\{\mu\}} + \left(\backslash \mathrm{mathcal{V}}^{\{a\}} \ b^{\{b\}}_{\{\}\{\}\{\mu\}} - \backslash \mathrm{mathcal{V}}^{\{b\}} \ b^{\{a\}}_{\{\}\{\}\{\mu\}} \right)$	$A^{ab}{}_\mu + (\mathcal{V}^a b^b{}_\mu - \mathcal{V}^b b^a{}_\mu)$	$A^{ab}{}_\mu + (\mathcal{V}^a b^b{}_\mu - \mathcal{V}^b b^a{}_\mu)$.
1510.06699_F00856	46	■ 1.000	$t^{\{\dagger a\}}_{\{a\}}$	$t^{\dagger a}{}_a$	trace $t^{\dagger a}{}_a$ of the free-gravitational energy momen-
1510.06699_F00857	46	■ 1.000	$\tau^{\{\dagger a\}}_{\{a\}}$	$\tau^{\dagger a}{}_a$	$\tau^{\dagger a}{}_a$ of the total matter energy-momentum tensor
1510.06699_F00858	46	■ 0.948	$\{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}_{\{a \ b \ c \ d\}}^{\{\dagger\}}\left(h, \partial h, \partial^2 h, A, \partial A, V, \partial V\right)$	${}^0\mathcal{R}_{abcd}^\dagger(h, \partial h, \partial^2 h, A, \partial A, V, \partial V)$	${}^0\mathcal{R}_{abcd}^\dagger(h, \partial h, \partial^2 h, A, \partial A, V, \partial V)$, depends on all the
1510.06699_F00859	47	■ 1.000	$\not \partial \equiv \gamma^\mu \partial_\mu$	$\not{\partial} \equiv \gamma^\mu \partial_\mu$	It is worth briefly considering in an explicitly geo-
1510.06699_F00860	47	■ 0.938	$\bar{\psi} / t$	$\bar{\psi}/t$	metrically the dilation part of the Weyl trans-
1510.06699_F00861	47	■ 0.938	$J^{\{\mu\}} =$	$J^\mu =$	g of the distance part. The last part physically (re-
1510.06699_F00862	47	■ 1.000	$\bar{\psi} \gamma^\mu \psi$	$\bar{\psi} \gamma^\mu \psi$	g of the distance part. The last part physically (re-
1510.06699_F00863	47	■ 1.000	$\psi \bar{\psi}$	$\psi \bar{\psi}$	vely) dilates the Minkowski spacetime such that the distance between any two points is increased by
					The spinor outer product $\psi \bar{\psi}$ may be rewritten using the

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1510.06699_F00864	47	■ 1.000	$\gamma^5=\gamma_5=i\gamma^0\gamma^1\gamma^2\gamma^3$	$\gamma^5 = \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$	where $\gamma^5 = \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and $\sigma^{\nu\rho} = \frac{i}{2}[\gamma^\nu, \gamma^\rho]$ (and
1510.06699_F00865	47	■ 1.000	$\sigma^{\nu\rho}=\frac{i}{2}\left[\gamma^\nu,\gamma^\rho\right]$	$\sigma^{\nu\rho} = \frac{i}{2}[\gamma^\nu, \gamma^\rho]$	where $\gamma^5 = \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and $\sigma^{\nu\rho} = \frac{i}{2}[\gamma^\nu, \gamma^\rho]$ (and
1510.06699_F00866	47	■ 1.000	$\rho\equiv\bar{\psi}\psi$	$\rho \equiv \bar{\psi}\psi$	It is convenient to define the real scalar $\rho \equiv \bar{\psi}\psi$ and
1510.06699_F00867	47	■ 0.671	$\beta\equiv\bar{\psi}i\gamma^5\psi$	$\beta \equiv \bar{\psi}i\gamma^5\psi$	real pseudoscalar $\beta \equiv \bar{\psi}i\gamma^5\psi$, so that (A4) reads $\bar{\psi}\not{J} =$
1510.06699_F00868	47	■ 0.671	$\bar{\psi}/\hbar=$	$\bar{\psi}/\hbar =$	real pseudoscalar $\beta \equiv \bar{\psi}i\gamma^5\psi$, so that (A4) reads $\bar{\psi}\not{J} =$
1510.06699_F00869	47	■ 1.000	$\bar{\psi}\left(\rho+\beta i\gamma^5\right)$	$\bar{\psi}(\rho + \beta i\gamma^5)$	$\bar{\psi}(\rho + \beta i\gamma^5)$. Postmultiplying this expression by $(\rho - \beta i\gamma^5)$
1510.06699_F00870	47	■ 1.000	$\bar{\psi}\left(\rho-\beta i\gamma^5\right)$	$(\rho - \beta i\gamma^5)$	$\bar{\psi}(\rho + \beta i\gamma^5)$. Postmultiplying this expression by $(\rho - \beta i\gamma^5)$
1510.06699_F00871	47	■ 0.982	$\not{\partial}\psi$	$\not{\partial}\psi$	In particular, we note that the quantity $\not{J}\not{\partial}\psi$ appearing
1510.06699_F00872	47	■ 1.000	$J^\mu\partial_\mu\psi$	$J^\mu\partial_\mu\psi$	The quantity $J^\mu\partial_\mu\psi$ is clearly interpreted as a derivative
1510.06699_F00873	47	■ 1.000	$n^\nu\equiv-i\sigma^{\mu\nu}J_\mu$	$n^\nu \equiv -i\sigma^{\mu\nu}J_\mu$	the direction $n^\nu \equiv -i\sigma^{\mu\nu}J_\mu$, which is orthogonal to J^μ ,
1510.06699_F00874	47	■ 1.000	$J_\nu n^\nu=-i\sigma^{\mu\nu}J_\mu J_\nu=0$	$J_\nu n^\nu = -i\sigma^{\mu\nu}J_\mu J_\nu = 0$	since $J_\nu n^\nu = -i\sigma^{\mu\nu}J_\mu J_\nu = 0$. Thus, $-i\sigma^{\mu\nu}J_\mu\partial_\nu\psi$ is in-
1510.06699_F00875	47	■ 1.000	$-i\sigma^{\mu\nu}J_\mu\partial_\nu\psi$	$-i\sigma^{\mu\nu}J_\mu\partial_\nu\psi$	since $J_\nu n^\nu = -i\sigma^{\mu\nu}J_\mu J_\nu = 0$. Thus, $-i\sigma^{\mu\nu}J_\mu\partial_\nu\psi$ is in-
1510.06699_F00876	47	■ 1.000	$x^\mu(\lambda)$	$x^\mu(\lambda)$	the particle is described by a worldline $x^\mu(\lambda)$, together
1510.06699_F00877	47	■ 1.000	$R(\lambda)$	$R(\lambda)$	with a ‘rotor’ $R(\lambda)$, which is a Dirac spinor defined along
1510.06699_F00878	47	■ 1.000	$\bar{R}R=1$	$\bar{R}R = 1$	$\bar{R}R = 1$ and $\bar{R}i\gamma^5R = 0$, and contains information about
1510.06699_F00879	47	■ 1.000	$\bar{R}i\gamma^5R=0$	$\bar{R}i\gamma^5R = 0$	$\bar{R}R = 1$ and $\bar{R}i\gamma^5R = 0$, and contains information about
1510.06699_F00880	47	■ 1.000	R	R	one replaces ψ by R in (A6), restricts derivatives to lie
1510.06699_F00881	47	■ 1.000	$\dot{x}^\mu\equiv dx^\mu/d\lambda$	$\dot{x}^\mu \equiv dx^\mu/d\lambda$	locity $\dot{x}^\mu \equiv dx^\mu/d\lambda$ with $\bar{R}\gamma^\mu R$. The last identification
1510.06699_F00882	47	■ 1.000	$\bar{R}\gamma^\mu R$	$\bar{R}\gamma^\mu R$	locity $\dot{x}^\mu \equiv dx^\mu/d\lambda$ with $\bar{R}\gamma^\mu R$. The last identification
1510.06699_F00883	47	■ 1.000	$p_\mu(\lambda)$	$p_\mu(\lambda)$	multiplier $p_\mu(\lambda)$, which is identified with the momentum
1510.06699_F00884	47	■ 1.000	p_μ	p_μ	of the particle; note that, in general, p_μ and $\dot{x}^\mu$ are not
1510.06699_F00885	47	■ 1.000	$\dot{x}^\mu$	$\dot{x}^\mu$	of the particle; note that, in general, p_μ and $\dot{x}^\mu$ are not
1510.06699_F00886	47	■ 0.999	$me(\lambda)$	$me(\lambda)$	ten as $me(\lambda)$ for later convenience) to ensure reparam-
1510.06699_F00887	47	■ 1.000	$x^\mu(\lambda),R(\lambda),p^\mu(\lambda)$	$x^\mu(\lambda), R(\lambda), p^\mu(\lambda)$	to the dynamical variables $x^\mu(\lambda)$, $R(\lambda)$, $p^\mu(\lambda)$ and $e(\lambda)$
1510.06699_F00888	47	■ 1.000	p^μ/m	p^μ/m	replacing $\bar{R}\gamma^\mu R$ by p^μ/m and then neglecting the particle
1510.06699_F00889	47	■ 1.000	$e(\lambda)\rightarrow\frac{1}{2}e(\lambda)$	$e(\lambda) \rightarrow \frac{1}{2}e(\lambda)$	einbein $e(\lambda) \rightarrow \frac{1}{2}e(\lambda)$ for later convenience, this process
1510.06699_F00890	47	■ 1.000	$x^\mu(\lambda),p^\mu(\lambda)$	$x^\mu(\lambda), p^\mu(\lambda)$	variables $x^\mu(\lambda)$, $p^\mu(\lambda)$ and $e(\lambda)$ leads to the classical
1510.06699_F00891	47	■ 1.000	$\rho=0$	$\rho = 0$	ticular, by setting $\rho = 0$ in the global Weyl transforma-
1510.06699_F00892	47	■ 1.000	$\rho(x)=0$	$\rho(x) = 0$	reduce to local Poincaré transformations if $\rho(x) = 0$. In
1510.06699_F00893	48	■ 1.000	$D_\mu\varphi$	$D_\mu\varphi$	derivative $D_\mu\varphi$. Thus, all the results given in Sec-
1510.06699_F00894	48	■ 0.981	$\rho(x)=0=$	$\rho(x) = 0 =$	tions II B and II C also hold in PGT if one sets $\rho(x) = 0 =$
1510.06699_F00895	48	■ 1.000	$B_\mu=0$	$B_\mu = 0$	valid in PGT if one sets $B_\mu = 0$ and removes all asterisks.
1510.06699_F00896	48	■ 1.000	$H_{\mu\nu}=0$	$H_{\mu\nu} = 0$	prescription of setting $B_\mu = 0$ (so that $H_{\mu\nu} = 0$) and re-
1510.06699_F00897	48	■ 0.999	$\mathcal{R}\equiv\mathcal{R}^{ab}_{ab},\Lambda$	$\mathcal{R} \equiv \mathcal{R}^{ab}_{ab}, \Lambda$	where $\mathcal{R} \equiv \mathcal{R}^{ab}_{ab}$, Λ is a cosmological constant, a is
1510.06699_F00898	48	■ 1.000	$a,\Lambda,\{\alpha_i\}$	$a, \Lambda, \{\alpha_i\}$	depends on the choice of the parameters a , Λ , $\{\alpha_i\}$ and
1510.06699_F00899	48	■ 1.000	$\{\beta_i\}$	$\{\beta_i\}$	$\{\beta_i\}$ in L_G . As an illustration, in Appendix C we give
1510.06699_F00900	48	■ 1.000	$\varphi,\partial\varphi,h$	$\varphi, \partial\varphi, h$	induce a dependence on φ , $\partial\varphi$, h and A , but one typically
1510.06699_F00901	48	■ 0.546	$\mathcal{H}_{ab},j^a,\zeta^a$	$\mathcal{H}_{ab}, j^a, \zeta^a$	$\delta\varphi \quad \partial\varphi \quad \backslash \quad \partial(\mathcal{D}_a^*\varphi) \quad \backslash$
1510.06699_F00902	48	■ 1.000	$m\bar{\psi}\psi\rightarrow m\bar{\psi}\psi$	$m\bar{\psi}\psi \rightarrow m\bar{\psi}\psi$	satisfy any further required invariances that were lost in
1510.06699_F00903	48	■ 1.000	$\phi=1$	$\phi = 1$	in Section II K.
1510.06699_F00904	48	■ 0.992	${}^0\mathcal{T}^a_{bc}\equiv 0$	${}^0\mathcal{T}^a_{bc} \equiv 0$	the contributions of the free gravitational sector to the
1510.06699_F00905	48	■ 1.000	${}^0\mathcal{R}^{ab}_{cd}(h,\partial h,\partial^2h)$	${}^0\mathcal{R}^{ab}_{cd}(h, \partial h, \partial^2 h)$	gravitational field equations:
1510.06699_F00906	49	■ 1.000	${}^0\mathcal{D}_a\mathcal{T}^a=h\partial_\mu(h^{-1}h_a{}^\mu\mathcal{T}^a)$	${}^0\mathcal{D}_a\mathcal{T}^a = h\partial_\mu(h^{-1}h_a{}^\mu\mathcal{T}^a)$	terms of a total derivative as ${}^0\mathcal{D}_a\mathcal{T}^a = h\partial_\mu(h^{-1}h_a{}^\mu\mathcal{T}^a)$,
1510.06699_F00907	49	■ 0.980	$\underline{105}$	$\underline{105}$	whereas this is not possible in WGT ¹⁰⁵ .
1510.06699_F00908	49	■ 0.935	$L_0\mathcal{R}^2$	$L_0\mathcal{R}^2$	where $L_0\mathcal{R}^2$ is based on (2) with $\mathcal{R}^{ab}_{cd}$ replaced by

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1510.06699_F00909	49	■ 0.606	$L_{\{0^{\wedge}\{\mathrm{R}\}^{\wedge\{2\}}\}}$	$L_{0\mathcal{R}^2}$	tinct terms in $L_{0\mathcal{R}^2}$ is reduced to
1510.06699_F00910	49	■ 0.999	$\mathrm{R}_{\{a\ b\ c\ d\}}\rightarrow^{\{0\}}\mathrm{R}_{abcd}$	$\mathcal{R}_{abcd}\rightarrow{}^0\mathcal{R}_{abcd}$	${}^0\mathcal{R}^{\prime a}{}_c=e^{-2\rho}\{{}^0\mathcal{R}^a{}_c-2{}^0\mathcal{D}_c\mathcal{P}^a-\delta_c^a{}^0\mathcal{D}_c\mathcal{P}^a\}$
1510.06699_F00911	49	■ 1.000	$\alpha_3=0$	$\alpha_3=0$	—
1510.06699_F00912	49	■ 0.999	$\alpha_1=\alpha_2=0$	$\alpha_1=\alpha_2=0$	h , then one must set $\alpha_1=\alpha_2=0$ (at least in $D\leq 4$
1510.06699_F00913	49	■ 1.000	$a=\frac{1}{2}$	$a=\frac{1}{2}$	spacetime dimensions), and on choosing $a=\frac{1}{2}$ the re-
1510.06699_F00914	49	■ 0.999	$a=0=\Lambda,\alpha_1=-\frac{1}{3}\alpha_2$	$a=0=\Lambda,\alpha_1=-\frac{1}{3}\alpha_2$	is $a=0=\Lambda$, $\alpha_1=-\frac{1}{3}\alpha_2$, which corresponds to the free
1510.06699_F00915	49	■ 1.000	U_4	U_4	that the spacetime has, in general, a Riemann–Cartan U_4
1510.06699_F00916	49	■ 1.000	$\{ \}^{\{0\}}R^{\{\rho\}}\{ \}_{\{\sigma\mu\nu\}}$	${}^0R^{\rho}{}_{\sigma\mu\nu}$	leads one to recognise ${}^0R^{\rho}{}_{\sigma\mu\nu}$ as the standard curvature
1510.06699_F00917	49	■ 1.000	$R\equiv R^{\{\mu\nu\}}\{ \}_{\{\mu\nu\}}$	$R\equiv R^{\mu\nu}{}_{\mu\nu}$	which relates the Ricci–Cartan scalar $R\equiv R^{\mu\nu}{}_{\mu\nu}$ in
1510.06699_F00918	49	■ 0.999	$\{ \}^{\{0\}}R\equiv\{ \}^{\{0\}}R^{\{\mu\nu\}}\{ \}_{\{\mu\nu\}}$	${}^0R\equiv{}^0R^{\mu\nu}{}_{\mu\nu}$	U_4 spacetime to the Ricci scalar ${}^0R\equiv{}^0R^{\mu\nu}{}_{\mu\nu}$ in V_4
1510.06699_F00919	49	■ 1.000	$L_{\{\mathrm{G}\}}=-R/(2\kappa)$	$L_G=-R/(2\kappa)$	taken to be $L_G=-R/(2\kappa)$ and the theory is interpreted
1510.06699_F00920	49	■ 1.000	$T_{\{\mu\nu\}\sigma}$	$T_{\mu\nu\sigma}$	that includes a tensor field $T_{\mu\nu\sigma}$ (obeying the appropri-
1510.06699_F00921	49	■ 1.000	$T_{\{\mu\nu\}\lambda}$	$T_{\mu\nu\lambda}$	Lagrangian is given by the terms containing $T_{\mu\nu\lambda}$ in (B6)
1510.06699_F00922	49	■ 1.000	$\mathrm{R}=\mathrm{R}^{\{a\ b\}}\{ \}_{\{a\ b\}}$	$\mathcal{R}=\mathcal{R}^{ab}{}_{ab}$	$\mathcal{R}=\mathcal{R}^{ab}{}_{ab}$. Thus, in Kibble’s original theory, one has
1510.06699_F00923	49	■ 1.000	$\Lambda=\alpha_i=\beta_i=0$	$\Lambda=\alpha_i=\beta_i=0$	which corresponds to setting $a=\frac{1}{2}$ and $\Lambda=\alpha_i=\beta_i=0$
1510.06699_F00924	49	■ 0.983	$\{ \}^{\{\underline{106}\}}$	$\underline{106}$	theory ¹⁰⁶ , which, when re-interpreted geometrically, is a
1510.06699_F00925	50	■ 1.000	$\mathrm{T}_{\{a\}}\equiv\mathrm{T}^{\{c\}}\{ \}_{\{a\ c\}}$	$\mathcal{T}_a\equiv\mathcal{T}^c{}_{ac}$	where $\mathcal{T}_a\equiv\mathcal{T}^c{}_{ac}$, as defined in (66). In particular, by
1510.06699_F00926	50	■ 0.999	$\mathrm{A}^{\{a\ b\}}\{ \}_{\{c\}}=h_{\{c\}}\{ \}^{\{\mu\}}A^{\{a\ b\}}\{ \}_{\{\mu\}}$	$\mathcal{A}^{ab}{}_c=h_c{}^\mu A^{ab}{}_\mu$	its fully Lorentz-index counterpart $\mathcal{A}^{ab}{}_c=h_c{}^\mu A^{ab}{}_\mu$. In
1510.06699_F00927	50	■ 1.000	$\sigma_{\{a\ b\}}\{ \}^{\{c\}}\neq 0$	$\sigma_{ab}{}^c\neq 0$	isks). In the general case, in which $\sigma_{ab}{}^c\neq 0$, this is not,
1510.06699_F00928	50	■ 1.000	$\sigma_{\{a\ b\ c\}}$	σ_{abc}	derivatives of the matter field(s), then σ_{abc} is indepen-
1510.06699_F00929	50	■ 1.000	$\mathrm{T}^{\{c\}}\{ \}_{\{a\ b\}}$	$\mathcal{T}^c{}_{ab}$	field strength $\mathcal{T}^c{}_{ab}$, and so the h -field itself must satisfy
1510.06699_F00930	50	■ 0.997	$\mu\phi\equiv m$	$\mu\phi\equiv m$	Lagrangian density is given by (70) with $\mu\phi\equiv m$, sev-
1510.06699_F00931	50	■ 1.000	$\mathrm{T}^{\{c\}}\{ \}_{\{a\ b\}}=2\kappa h\sigma_{\{a\ b\}}\{ \}^{\{c\}}$	$\mathcal{T}^c{}_{ab}=2\kappa h\sigma_{ab}{}^c$	$\mathcal{T}^c{}_{ab}=2\kappa h\sigma_{ab}{}^c$. More interestingly, the condition (C9)
1510.06699_F00932	50	■ 1.000	$R_{\{\mu\nu\}}\equiv R^{\{\lambda\}}\{ \}_{\{\mu\nu\}\lambda}$	$R_{\mu\nu}\equiv R^\lambda{}_{\mu\lambda\nu}$	where $R_{\mu\nu}\equiv R^\lambda{}_{\mu\lambda\nu}$ and $R\equiv R^\mu{}_\mu$ have their usual forms,
1510.06699_F00933	50	■ 1.000	$R\equiv R^{\{\mu\}}\{ \}_{\{\mu\}}$	$R\equiv R^\mu{}_\mu$	where $R_{\mu\nu}\equiv R^\lambda{}_{\mu\lambda\nu}$ and $R\equiv R^\mu{}_\mu$ have their usual forms,
1510.06699_F00934	50	■ 1.000	$\Gamma^{\{\lambda\}}\{ \}_{\{\mu\nu\}}$	$\Gamma^\lambda{}_{\mu\nu}$	but one must remember that the affine connection $\Gamma^\lambda{}_{\mu\nu}$
1510.06699_F00935	51	■ 1.000	$\tau_{\{\mu\nu\}}=$	$\tau_{\mu\nu}=$	is not, in general, symmetric in μ and ν . Also, $\tau_{\mu\nu}=$
1510.06699_F00936	51	■ 1.000	$e^{\{a\}}\{ \}_{\{\mu\}}e^{\{b\}}\{ \}_{\{\nu\}}\tau_{\{a\ b\}}$	$e^a{}_\mu e^b{}_\nu\tau_{ab}$	$e^a{}_\mu e^b{}_\nu\tau_{ab}$ and $\sigma_{\mu\nu}{}^\lambda=e^a{}_\mu e^b{}_\nu e_c{}^\lambda\sigma_{ab}{}^c$, where the former
1510.06699_F00937	51	■ 1.000	$\sigma_{\{\mu\nu\}}\{ \}^{\{\lambda\}}=e^{\{a\}}\{ \}_{\{\mu\}}e^{\{b\}}\{ \}_{\{\nu\}}e_{\{c\}}\{ \}^{\{\lambda\}}\sigma_{\{a\ b\}}\{ \}^{\{c\}}$	$\sigma_{\mu\nu}{}^\lambda=e^a{}_\mu e^b{}_\nu e_c{}^\lambda\sigma_{ab}{}^c$	$e^a{}_\mu e^b{}_\nu\tau_{ab}$ and $\sigma_{\mu\nu}{}^\lambda=e^a{}_\mu e^b{}_\nu e_c{}^\lambda\sigma_{ab}{}^c$, where the former
1510.06699_F00938	51	■ 1.000	$\sigma_{\{\mu\nu\}}\{ \}^{\{\lambda\}}$	$\sigma_{\mu\nu}{}^\lambda$	angular-momentum $\sigma_{\mu\nu}{}^\lambda$ of the matter field vanishes,
1510.06699_F00939	51	■ 1.000	$T^{\{\lambda\}}\{ \}_{\{\mu\nu\}}$	$T^\lambda{}_{\mu\nu}$	then so too does the torsion $T^\lambda{}_{\mu\nu}$ and hence the affine
1510.06699_F00940	51	■ 1.000	$R_{\{\mu\nu\}}$	$R_{\mu\nu}$	case, $R_{\mu\nu}$ reduces to the standard symmetric Ricci ten-
1510.06699_F00941	51	■ 1.000	$\{ \}^{\{0\}}R_{\{\mu\nu\}}\equiv R_{\{\mu\nu\}}\left(V_4\right)$	${}^0R_{\mu\nu}\equiv R_{\mu\nu}(V_4)$	sor ${}^0R_{\mu\nu}\equiv R_{\mu\nu}(V_4)$ in a Riemann spacetime, expressed
1510.06699_F00942	51	■ 0.979	$\{ \}^{\{\underline{108}\}}$	$\underline{108}$	grangian suggested by Dirac has the simple form ¹⁰⁸
1510.06699_F00943	51	■ 1.000	$-\frac{1}{4}$	$-\frac{1}{4}$	sionless constant (usually positive); the factor of $-\frac{1}{4}$ is
1510.06699_F00944	51	■ 1.000	ϕ^4	ϕ^4	ϕ^4 , and a term proportional to $\phi^2\mathcal{R}$ that non-minimally
1510.06699_F00945	51	■ 1.000	$L_{\{\varphi\}}(\varphi,\partial\varphi,h,A,B,\phi)$	$L_\varphi(\varphi,\partial\varphi,h,A,B,\phi)$	for the matter field φ are $L_\varphi(\varphi,\partial\varphi,h,A,B,\phi)$; note that
1510.06699_F00946	51	■ 1.000	$\mathcal{L}_T=h^{-1}L_T$	$\mathcal{L}_T=h^{-1}L_T$	The total Lagrangian density is then $\mathcal{L}_T=h^{-1}L_T$,
1510.06699_F00947	51	■ 1.000	$L_{\{T\}}$	L_T	where L_T has the form
1510.06699_F00948	51	■ 0.993	(a,ξ,ν,λ)	(a,ξ,ν,λ)	(a,ξ,ν,λ) are dimensionless (and usually positive). In
1510.06699_F00949	51	■ 1.000	ϕ^{-2}	ϕ^{-2}	plied ϕ^{-2} should replace Newton’s gravitational constant
1510.06699_F00950	51	■ 0.967	$h_{\{a\}}\{ \}^{\{\mu\}},A^{\{a\ b\}}\{ \}_{\{\mu\}},B_{\{\mu\}}$	$h_a{}^\mu,A^{ab}{}_\mu,B_\mu$	independent fields $h_a{}^\mu$, $A^{ab}{}_\mu$, B_μ and ϕ respectively ⁵⁰ ,
1510.06699_F00951	51	■ 1.000	$B_{\{\mu\}},\phi$	B_μ,ϕ	with respect to B_μ,ϕ and the spacetime metric $g_{\mu\nu}$. Our
1510.06699_F00952	52	■ 1.000	$\left(\tau_{\{\varphi\}}\right)^{\{a\}}\{ \}_{\{b\}}$	$(\tau_\varphi)^a{}_b$	where $(\tau_\varphi)^a{}_b$ is the energy-momentum tensor of the mat-
1510.06699_F00953	52	■ 1.000	$\left(\sigma_{\{\varphi\}}\right)_{\{a\ b\}}\{ \}^{\{c\}}$	$(\sigma_\varphi)_{ab}{}^c$	where $(\sigma_\varphi)_{ab}{}^c$ is the spin-angular-momentum tensor of

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1510.06699_F00954	52	■ 1.000	$\mathcal{D}_{\{a\}}^{\{*\}} \rightarrow \mathcal{D}_{\{a\}},$ $\mathcal{T}^{\{*\}}_{\{c\}} \{ \}_{\{a\} b} \rightarrow \mathcal{T}^{\{c\}}_{\{ \}_{\{a\} b}}$	$\mathcal{D}_a^* \rightarrow \mathcal{D}_a, \mathcal{T}^{*c}_{ab} \rightarrow \mathcal{T}^c_{ab}$	if one makes the replacements $\mathcal{D}_a^* \rightarrow \mathcal{D}_a, \mathcal{T}^{*c}_{ab} \rightarrow \mathcal{T}^c_{ab}$
1510.06699_F00955	52	■ 1.000	$\mathcal{T}_{\{a\}}^{\{*\}} \rightarrow \mathcal{T}_{\{a\}}$	$\mathcal{T}_a^* \rightarrow \mathcal{T}_a$	and $\mathcal{T}_a^* \rightarrow \mathcal{T}_a$ to PGT quantities, although the resulting
1510.06699_F00956	52	■ 1.000	$\left(\zeta_{\varphi} \right)^c$	$(\zeta_{\varphi})^c$	where $(\zeta_{\varphi})^c$ is the dilation current of the φ matter field.
1510.06699_F00957	52	■ 1.000	$\left.\mathcal{D}_{\{a\}}^{\{*\}} \right _{\phi=\phi_0} \rightarrow \mathcal{D}_{\{a\}}^{\{*\}} _{\phi=\phi_0} = -\phi_0 \mathcal{B}_a$	$\mathcal{D}_a^* \phi _{\phi=\phi_0} = -\phi_0 \mathcal{B}_a$	replacement $\mathcal{D}_a^* \phi _{\phi=\phi_0} = -\phi_0 \mathcal{B}_a$ throughout. It is also
1510.06699_F00958	52	■ 1.000	$a=1/\left(\kappa\phi_0^2\right)$	$a = 1/(\kappa\phi_0^2)$	convenient to set $a = 1/(\kappa\phi_0^2)$, where κ is Einstein’s grav-
1510.06699_F00959	52	■ 1.000	$\Lambda=\lambda\phi_0^2/a \text{ / } a=\kappa\lambda$	$\Lambda = \lambda\phi_0^2/a = \kappa\lambda$	where $\Lambda = \lambda\phi_0^2/a = \kappa\lambda$ has the correct dimensions of
1510.06699_F00960	52	■ 1.000	$m^2=\nu\phi_0^2/\xi$	$m^2 = \nu\phi_0^2/\xi$	vector field (or Proca field) of mass-squared $m^2 = \nu\phi_0^2/\xi$.
1510.06699_F00961	53	■ 1.000	$\mathcal{R}^{ab}_{cd}, \mathcal{T}^a_{bc} = \mathcal{T}^{*a}_{bc} - \delta^a_c \mathcal{B}_b + \delta^a_b \mathcal{B}_c$	$\mathcal{R}^{ab}_{cd}, \mathcal{T}^a_{bc} = \mathcal{T}^{*a}_{bc} - \delta^a_c \mathcal{B}_b + \delta^a_b \mathcal{B}_c$	tensors $\mathcal{R}^{ab}_{cd}, \mathcal{T}^a_{bc} = \mathcal{T}^{*a}_{bc} - \delta^a_c \mathcal{B}_b + \delta^a_b \mathcal{B}_c$ and $\mathcal{H}_{ab}$
1510.06699_F00962	53	■ 1.000	$\left(\zeta_{\psi}\right)^{\mu}=0$	$(\zeta_{\psi})^{\mu} = 0$	does not appear in (D12), the dilation current $(\zeta_{\psi})^{\mu} = 0$
1510.06699_F00963	53	■ 1.000	$\mathcal{T}_{\{a\}}=0$	$\mathcal{T}_a = 0$	theory, from (C10), we again have $\mathcal{T}_a = 0$. Thus, the
1510.06699_F00964	53	■ 1.000	$\tau_{\psi}=h^{-1}\mu\phi\bar{\psi}\psi$	$\tau_{\psi} = h^{-1}\mu\phi\bar{\psi}\psi$	of the Dirac field is easily found to be $\tau_{\psi} = h^{-1}\mu\phi\bar{\psi}\psi$.
1510.06699_F00965	53	■ 1.000	$m^2\equiv\nu\phi_0^2/\xi$	$m^2 \equiv \nu\phi_0^2/\xi$	where we have defined $m^2 \equiv \nu\phi_0^2/\xi$ and $\sigma_{ab}{}^c$ is the spin-
1510.06699_F00966	53	■ 1.000	$\mathcal{B}_{\{a\}}$	$\mathcal{B}_a$	field $\mathcal{B}_a$ has a non-minimal coupling to the gravitational
1510.06699_F00967	53	■ 1.000	$D_{\{\mu\}}^{\{*\}}=D_{\{\mu\}}+wB_{\{\mu\}}$	$D_{\mu}^* = D_{\mu} + wB_{\mu}$	variant derivative (20), namely $D_{\mu}^* = D_{\mu} + wB_{\mu}$, the
1510.06699_F00968	54	■ 1.000	$P_{\{\mu\}}\equiv\partial_{\mu}\rho$	$P_{\mu} \equiv \partial_{\mu}\rho$	where $P_{\mu} \equiv \partial_{\mu}\rho$. This transformation for V_{μ} implies
1510.06699_F00969	54	■ 1.000	$(w=0)$	$(w = 0)$	variant ($w = 0$) under extended local dilations. One
1510.06699_F00970	54	■ 1.000	$\mathcal{R}^{\dagger ab}_{cd}\equiv h_c{}^{\mu}h_d{}^{\nu}R^{\dagger ab}{}_{\mu\nu}$	$\mathcal{R}^{\dagger ab}_{cd} \equiv h_c{}^{\mu}h_d{}^{\nu}R^{\dagger ab}{}_{\mu\nu}$	can then define $\mathcal{R}^{\dagger ab}_{cd} \equiv h_c{}^{\mu}h_d{}^{\nu}R^{\dagger ab}{}_{\mu\nu}$ as before, which
1510.06699_F00971	54	■ 0.997	$D_{\{\mu\}}^{\natural}\varphi$	$D_{\mu}^{\natural}\varphi$	ever, one quickly finds that $D_{\mu}^{\natural}\varphi$ does <i>not</i> transform co-
1510.06699_F00972	54	■ 1.000	$D_{\{\mu\}}^{\natural}$	$D_{\mu}^{\natural}$	ing our reluctance thus far to call $D_{\mu}^{\natural}$ (or $\mathcal{D}_{\mu}^{\natural}$) a covariant
1510.06699_F00973	54	■ 1.000	$\nu=\beta_1=\beta_2=0$	$\nu = \beta_1 = \beta_2 = 0$	gravitational Lagrangian L_G and $\nu = \beta_1 = \beta_2 = 0$ in the
1510.06699_F00974	54	■ 0.994	$\xi=\nu=\beta_1=\beta_2=0$	$\xi = \nu = \beta_1 = \beta_2 = 0$	in Section IV A, but with $\xi = \nu = \beta_1 = \beta_2 = 0$. These
1510.06699_F00975	54	■ 1.000	$\mathcal{D}_{\{a\}}^{\natural}\chi$	$\mathcal{D}_a^{\natural}\chi$	to $\mathcal{D}_a^{\natural}\chi$ to assemble the appropriate <i>covariant</i> derivative
1510.06699_F00976	54	■ 1.000	$\mathcal{D}_{\{a\}}^{\dagger}\chi=h_a{}^{\mu}D_{\mu}^{\dagger}\chi$	$\mathcal{D}_a^{\dagger}\chi = h_a{}^{\mu}D_{\mu}^{\dagger}\chi$	$\mathcal{D}_a^{\dagger}\chi$. In each case, one finds that $\mathcal{D}_a^{\dagger}\chi = h_a{}^{\mu}D_{\mu}^{\dagger}\chi$, where
1510.06699_F00977	54	■ 0.991	$D_{\{\mu\}}^{\dagger}$	$D_{\mu}^{\dagger}$	$D_{\mu}^{\dagger}$ has the form (143), but with the appropriate value of
1510.06699_F00978	54	■ 0.997	$\left(U_4\right)$	(U_4)	respectively, of a Riemann–Cartan (U_4) spacetime.
1510.06699_F00979	54	■ 1.000	$\delta\varphi$	$\delta\varphi$	given action, for example, then the ‘total’ variation $\delta\varphi$ and the
1510.06699_F00980	54	■ 1.000	$\left[\mathcal{D}_c,\mathcal{D}_d\right]\varphi\equiv\frac{1}{2}\mathcal{R}^{ab}{}_{cd}\Sigma_{cd}\varphi-\mathcal{T}^a{}_{cd}\mathcal{D}_a\varphi,\mathcal{R}^a{}_b\equiv\mathcal{R}^{ac}{}_{bc}$	$[\mathcal{D}_c, \mathcal{D}_d]\varphi \equiv \frac{1}{2}\mathcal{R}^{ab}{}_{cd}\Sigma_{cd}\varphi - \mathcal{T}^a{}_{cd}\mathcal{D}_a\varphi, \mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}$	signature, $[\mathcal{D}_c, \mathcal{D}_d]\varphi \equiv \frac{1}{2}\mathcal{R}^{ab}{}_{cd}\Sigma_{cd}\varphi - \mathcal{T}^a{}_{cd}\mathcal{D}_a\varphi, \mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}$
1510.06699_F00981	54	■ 1.000	$\mathcal{T}_{\{a\}}\equiv\mathcal{T}^b{}_{ab}$	$\mathcal{T}_a \equiv \mathcal{T}^b{}_{ab}$	and $\mathcal{T}_a \equiv \mathcal{T}^b{}_{ab}$.
1510.06699_F00982	55	■ 1.000	$\widehat{\mathcal{D}}_a^*$	$\widehat{\mathcal{D}}_a^*$	(Weyl weight $w = 0$) versions of the quantities listed below are denoted with a caret, for example $\widehat{h}_a{}^{\mu}$ or $\widehat{\mathcal{D}}_a^*$ or $\widehat{\mathcal{R}}^{ab}{}_{cd}$; see
1510.06699_F00983	55	■ 1.000	$\widehat{\mathcal{R}}^{ab}{}_{cd}$	$\widehat{\mathcal{R}}^{ab}{}_{cd}$	(Weyl weight $w = 0$) versions of the quantities listed below are denoted with a caret, for example $\widehat{h}_a{}^{\mu}$ or $\widehat{\mathcal{D}}_a^*$ or $\widehat{\mathcal{R}}^{ab}{}_{cd}$; see

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1510.06699_ F09984 [conf 0.000]	55	0.000	$A^{\dagger\{a\}b}_{\mu}\}_{\mu}\equiv A^{\{a\}b}_{\mu}\}_{\mu}+2\eta^{\{a\}b}_{\mu}h_{c\}^{\nu}V_{\nu}$	$A^{\dagger ab}_{\mu}\equiv A^{ab}_{\mu}+2\eta^{c[a}b^b]_{\mu}h_c{}^{\nu}V_{\nu}$	Gravitational gauge field variables $h_a{}^{\mu}$ h $b^a{}_{\mu}$ $A^{ab}{}_{\mu}$ B_{μ} V_{μ} $A^{\dagger ab}{}_{\mu}\equiv A^{ab}{}_{\mu}+2\eta^{c[a}b^{b]}{}_{\mu}h_c{}^{\nu}V_{\nu}$ translational gauge field (h-field) determinant of $h_a{}^{\mu}$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) Derivative operators $\mathcal{D}_a\equiv h_a{}^{\mu}D_{\mu}\equiv h_a{}^{\mu}(\partial_{\mu}+\frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab})$ $\partial_{\mu}^a\equiv\partial_{\mu}+wB_{\mu}$ $\mathcal{D}_{\mu}^a\equiv h_a{}^{\mu}D_{\mu}^a\equiv h_a{}^{\mu}(\partial_{\mu}^a+\frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab})$ $\partial_{\mu}^{\dagger a}\equiv\partial_{\mu}-w(V_{\mu}+\frac{1}{3}T_{\nu})$ $\mathcal{D}_{\mu}^{\dagger a}\equiv h_a{}^{\mu}D_{\mu}^{\dagger a}\equiv h_a{}^{\mu}(\partial_{\mu}^{\dagger a}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab})$ $\mathcal{D}_{\mu}^a\equiv h_a{}^{\mu}D_{\mu}^a\equiv h_a{}^{\mu}(\partial_{\mu}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab})$ PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a Gauge field strengths $\mathcal{R}^{ab}{}_{cd}=\mathcal{R}^{*ab}{}_{cd}=2h_a{}^{\mu}h_b{}^{\nu}(\partial_{[\mu}A^{ab}{}_{\nu]}+A^a{}_{c[\mu}A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd}=2h_a{}^{\mu}h_b{}^{\nu}(\partial_{[\mu}A^{\dagger ab}{}_{\nu]}+A^{\dagger a}{}_{c[\mu}A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc}=2h_b{}^{\mu}h_c{}^{\nu}D_{[\mu}b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc}=2h_b{}^{\mu}h_c{}^{\nu}D_{[\mu}^*b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc}=2h_b{}^{\mu}h_c{}^{\nu}D_{[\mu}^{\dagger}b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc}=2h_b{}^{\mu}h_c{}^{\nu}D_{[\mu}^{\mathfrak{a}}b^a{}_{\nu]}$ $\mathcal{H}_{ab}=2h_a{}^{\mu}h_b{}^{\nu}\partial_{[\mu}B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger}=2h_a{}^{\mu}h_b{}^{\nu}\partial_{[\mu}(V_{\nu}+\frac{1}{3}T_{\nu])}$ PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Reduced A-fields and related quantities $c^a{}_{bc}\equiv 2h_b{}^{\mu}h_c{}^{\nu}\partial_{[\mu}b^a{}_{\nu]}$ $c^{*a}{}_{bc}\equiv 2h_b{}^{\mu}h_c{}^{\nu}\partial_{[\mu}^*b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc}\equiv 2h_b{}^{\mu}h_c{}^{\nu}\partial_{[\mu}^{\dagger}b^a{}_{\nu]}$ ${}^0A^{ab}{}_{\mu}\equiv\frac{1}{2}b^c{}_{\mu}(c^{ab}{}_c+c^b{}_c{}^a-c_c{}^a{}^b)$ ${}^0A^{*ab}{}_{\mu}\equiv\frac{1}{2}b^c{}_{\mu}(c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c{}^{*a}{}^b)$ ${}^0A^{\dagger ab}{}_{\mu}\equiv\frac{1}{2}b^c{}_{\mu}(c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger a}{}^b)$ PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc}\equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc}\equiv 0$) eWGT reduced A[†]-field (when $\mathcal{T}^{\dagger a}{}_{bc}\equiv 0$) Currents and related quantities $\mathcal{L}\equiv h^{-1}L$ $t^a{}_b\equiv t^a{}_{\mu}h_b{}^{\mu}\equiv h_b{}^{\mu}\delta\mathcal{L}_G/\delta h_a{}^{\mu}$ $s_{ab}{}^c\equiv s_{ab}{}^{\mu}b^c{}_{\mu}\equiv b^c{}_{\mu}\delta\mathcal{L}_G/\delta A^{ab}{}_{\mu}$ $j^a\equiv j^a{}^{\mu}b^b{}_{\mu}\equiv b^b{}_{\mu}\delta\mathcal{L}_G/\delta B_{\mu}$ $t^{\dagger a}{}_b\equiv t^a{}_b+2(s_{cb}{}^aV^c-s^{ac}{}_cV_b)$ $j^{\dagger a}\equiv j^a-2s^{ab}{}_b$ $\tau^a{}_b\equiv\tau^a{}_{\mu}h_b{}^{\mu}\equiv h_b{}^{\mu}\delta\mathcal{L}_M/\delta h_a{}^{\mu}$ $\sigma_{ab}{}^c\equiv\sigma_{ab}{}^{\mu}b^c{}_{\mu}\equiv b^c{}_{\mu}\delta\mathcal{L}_M/\delta A^{ab}{}_{\mu}$ $\zeta^a\equiv\zeta^{\mu}b^a{}_{\mu}\equiv b^a{}_{\mu}\delta\mathcal{L}_M/\delta B_{\mu}$ $\tau^{\dagger a}{}_b\equiv\tau^a{}_b+2(\sigma_{cb}{}^aV^c-\sigma^{ac}{}_cV_b)$ $\zeta^{\dagger a}\equiv\zeta^a-2s^{ab}{}_b$ Lagrangian Lagrangian density gravitational sector energy-momentum tensor^e gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^e matter sector energy-momentum tensor^e matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F09985 [conf 0.000]	55	0.000	$\mathcal{D}_{\{a\}} \equiv h_{\{a\}}^{\mu} \partial_{\mu} \equiv h_a^{\mu} \partial_{\mu} + \frac{1}{2} A^{ab}_{\mu} \Sigma_{ab}$	$\mathcal{D}_a \equiv h_a^{\mu} D_{\mu} \equiv h_a^{\mu} \left(\partial_{\mu} + \frac{1}{2} A^{ab}_{\mu} \Sigma_{ab} \right)$	Gravitational gauge field variables $h_a{}^{\mu}$ h $b^a{}_{\mu}$ $A^{ab}{}_{\mu}$ B_{μ} V_{μ} $A^{\dagger ab}{}_{\mu} \equiv A^{ab}{}_{\mu} + 2\eta^{c[a} b^{\mu]}{}_{\mu} h_c{}^{\nu} V_{\nu}$ Derivative operators $\mathcal{D}_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^* \equiv \partial_{\mu} + w B_{\mu}$ $\mathcal{D}_{\mu}^* \equiv h_a{}^{\mu} D_{\mu}^* \equiv h_a{}^{\mu} (\partial_{\mu}^* + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^{\dagger} \equiv \partial_{\mu} - w(V_{\mu} + \frac{1}{3} T_{\nu})$ $\mathcal{D}_{\mu}^{\dagger} \equiv h_a{}^{\mu} D_{\mu}^{\dagger} \equiv h_a{}^{\mu} (\partial_{\mu}^{\dagger} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ $\mathcal{D}_{\mu}^{\natural} \equiv h_a{}^{\mu} D_{\mu}^{\natural} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\dagger} b^a{}_{\nu]}$ $\mathcal{T}^{\natural ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\natural} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \frac{1}{3} T_{\nu])}$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]}$ $c^{*ab}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger ab}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$ ${}^0 A^{ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0 A^{\dagger ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c{}^{\dagger ab})$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$ $s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$ $j^a \equiv j^a{}^{\mu} b^b{}_{\mu} \equiv b^b{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$ $\zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^{\mu}$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^{ab}{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*ab}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger ab}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F09986 [conf 0.000]	55	0.000	$\partial_{\mu}^{\wedge{*}} \equiv \partial_{\mu} + w B_{\mu}$	$\partial_{\mu}^* \equiv \partial_{\mu} + w B_{\mu}$	Gravitational gauge field variables $h_a{}^{\mu}$ h $b^a{}_{\mu}$ $A^{ab}{}_{\mu}$ B_{μ} V_{μ} $A^{\dagger ab}{}_{\mu} \equiv A^{ab}{}_{\mu} + 2\eta^{c[a}b^{\dagger b]}{}_{\mu} h_c{}^{\nu} V_{\nu}$ Derivative operators $\mathcal{D}_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^* \equiv \partial_{\mu} + w B_{\mu}$ $\mathcal{D}_{\mu}^* \equiv h_a{}^{\mu} D_{\mu}^* \equiv h_a{}^{\mu} (\partial_{\mu}^* + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^{\dagger} \equiv \partial_{\mu} - w(V_{\mu} + \frac{1}{3} T_{\nu})$ $\mathcal{D}_{\mu}^{\dagger} \equiv h_a{}^{\mu} D_{\mu}^{\dagger} \equiv h_a{}^{\mu} (\partial_{\mu}^{\dagger} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ $\mathcal{D}_{\mu}^{\natural} \equiv h_a{}^{\mu} D_{\mu}^{\natural} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\dagger} b^a{}_{\nu]}$ $\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\natural} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \frac{1}{3} T_{\nu])}$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]}$ $c^{*ab}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$ ${}^0 A^{ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{*ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0 A^{\dagger ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$ $s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$ $j^a \equiv j^a{}_{\mu} b^{\mu} \equiv b^{\mu}{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$ $\zeta^a \equiv \zeta^a{}_{\mu} b^{\mu} \equiv b^{\mu}{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^{\mu}$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F09987 [conf 0.000]	55	0.000	$\mathcal{D}_{\mathfrak{a}}^*\equiv h_{\mathfrak{a}}^{\mu}\mathcal{D}_{\mu}^*\equiv h_{\mathfrak{a}}^{\mu}\left(\partial_{\mu}^*+\frac{1}{2}A^{ab}_{\mu}\Sigma_{ab}\right)$ $\equiv h_{\mathfrak{a}}^{\mu}\mathcal{D}_{\mu}^*\equiv h_{\mathfrak{a}}^{\mu}\left(\partial_{\mu}^*+\frac{1}{2}A^{ab}_{\mu}\Sigma_{ab}\right)$	$\mathcal{D}_a^*\equiv h_a{}^\mu D_\mu^*\equiv h_a{}^\mu\left(\partial_\mu^*+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab}\right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu\equiv A^{ab}{}_\mu+2\eta^{c[a}b^{\dagger b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a\equiv h_a{}^\mu D_\mu\equiv h_a{}^\mu(\partial_\mu+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^*\equiv\partial_\mu+wB_\mu$ $\mathcal{D}_\mu^*\equiv h_a{}^\mu D_\mu^*\equiv h_a{}^\mu(\partial_\mu^*+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^\perp\equiv\partial_\mu-w(V_\mu+\frac{1}{3}T_\nu)$ $\mathcal{D}_a^\perp\equiv h_a{}^\mu D_\mu^\perp\equiv h_a{}^\mu(\partial_\mu^\perp+\frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ $\mathcal{D}_\mu^\perp\equiv h_a{}^\mu D_\mu^\perp\equiv h_a{}^\mu(\partial_\mu+\frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd}=\mathcal{R}^{*ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]}+A^a{}_{c[\mu}A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]}+A^{\dagger a}{}_{c[\mu}A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^*b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab}=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$ $\mathcal{H}_{ab}^\perp=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu+\frac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$ $c^{*a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^*b^a{}_{\nu]}$ $c^{\perp a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{ab}{}_c+c^b{}_c{}^a-c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c{}^{*a}{}^b)$ ${}^0A^{\dagger ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L}\equiv h^{-1}L$ $t^a{}_b\equiv t^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$ $s_{ab}{}^c\equiv s_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ $j^a\equiv j^a{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_G/\delta B_\mu$ $t^{\dagger a}{}_b\equiv t^a{}_b+2(s_{cb}{}^a\mathcal{V}^c-s^{ac}{}_c\mathcal{V}_b)$ $j^{\dagger a}\equiv j^a-2s^{ab}{}_b$ $\tau^a{}_b\equiv\tau^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$ $\sigma_{ab}{}^c\equiv\sigma_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$ $\zeta^a\equiv\zeta^a{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_M/\delta B_\mu$ $\tau^{\dagger a}{}_b\equiv\tau^a{}_b+2(\sigma_{cb}{}^a\mathcal{V}^c-\sigma^{ac}{}_c\mathcal{V}_b)$ $\zeta^{\dagger a}\equiv\zeta^a-2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc}\equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc}\equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc}\equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F09989 [conf 0.000]	55	0.000	$\mathcal{D}_a^\dagger \equiv h_a{}^\mu \mathcal{D}_\mu^\dagger \equiv h_a{}^\mu \left(\partial_\mu^\dagger + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab} \right)$	$\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu \left(\partial_\mu^\dagger + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab} \right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[ab]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0 A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{} b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{} b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^{ab}{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*ab}{}_{bc} \equiv 0$) eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F09990 [conf 0.000]	55	0.000	$\mathrm{D}_{\mathrm{a}}^{\mathrm{natural}} \equiv h_{\mathrm{a}}^{\mathrm{natural}} \mathrm{D}_{\mathrm{a}}^{\mathrm{natural}} \equiv h_{\mathrm{a}}^{\mathrm{natural}} \left(\partial_{\mu} + \frac{1}{2} A^{\dagger \mathrm{a} b}{}_{\mu} \Sigma_{ab} \right)$	$\mathcal{D}_a^{\natural} \equiv h_a{}^{\mu} D_{\mu}^{\natural} \equiv h_a{}^{\mu} \left(\partial_{\mu} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab} \right)$	Gravitational gauge field variables $h_a{}^{\mu}$ h $b^a{}_{\mu}$ $A^{ab}{}_{\mu}$ B_{μ} V_{μ} $A^{\dagger ab}{}_{\mu} \equiv A^{ab}{}_{\mu} + 2\eta^{c[a} b^b]{}_{\mu} h_c{}^{\nu} V_{\nu}$ Derivative operators $\mathcal{D}_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^a \equiv \partial_{\mu} + w B_{\mu}$ $\mathcal{D}_{\mu}^a \equiv h_a{}^{\mu} D_{\mu}^a \equiv h_a{}^{\mu} (\partial_{\mu}^a + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^{\dagger} \equiv \partial_{\mu} - w(V_{\mu} + \frac{1}{3} T_{\nu})$ $\mathcal{D}_{\mu}^{\dagger} \equiv h_a{}^{\mu} D_{\mu}^{\dagger} \equiv h_a{}^{\mu} (\partial_{\mu}^{\dagger} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ $\mathcal{D}_{\mu}^a \equiv h_a{}^{\mu} D_{\mu}^a \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ Gauge field strengths $\mathcal{R}_{cd}^{ab} = \mathcal{R}^{ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{[a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^a b^a{}_{\nu]}$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^a b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \frac{1}{3} T_{\nu])}$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$ ${}^0 A^{ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{ab}{}_{\mu} + c^b{}_{\mu}{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{*ab}{}_{\mu} + c^{*b}{}_{\mu}{}^a - c_c^{*ab})$ ${}^0 A^{\dagger ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_{\mu} + c^{\dagger b}{}_{\mu}{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$ $s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$ $j^a \equiv j^a{}_{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_{\mu} \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$ $\zeta^a \equiv \zeta^a{}_{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_{\mu} \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^{\mu}$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^{ab}{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F09991 [conf 0.000]	55	0.000	a	a	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\mu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0 A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $\hat{j}^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F09992 [conf 0.000]	55	0.000	$\backslash\mathrm{mathcal{R}}^{\{a\}b\}\{\}_{\{c\}d}=\backslash\mathrm{mathcal{R}}^{\{*\}a\}b\{\}_{\{c\}d}=2\,h_{\{a\}\{\}_{\{\backslash\mu\}}h_{\{b\}\{\}^{\backslash\nu}\}\backslash\left(\backslash\mathrm{partial}_{\{[\backslash\mu\}A^{\{a\}b\}\{\}_{\{\backslash\nu\}}\}+A^{\{a\}\{a\}\{\}_{\{c[\backslash\mu\}A^{\{c\}b\}\{\}_{\{\backslash\nu\}}\}\right)}\backslash\mathrm{mathcal{R}}^{\{\backslash\dagger a\}b\}\{\}_{\{c\}d}=2\,h_{\{a\}\{\}^{\backslash\mu\}}h_{\{b\}\{\}^{\backslash\nu}\}\backslash\left(\backslash\mathrm{partial}_{\{[\backslash\mu\}A^{\{\backslash\dagger a\}b\}\{\}_{\{\backslash\nu\}}\}+A^{\{\backslash\dagger a\}\{a\}\{\}_{\{c[\backslash\mu\}A^{\{\backslash\dagger c\}b\}\{\}_{\{\backslash\nu\}}\}\right)}\backslash\mathrm{mathcal{T}}^{\{a\}\{b\}c}=2\,h_{\{b\}\{\}^{\backslash\mu\}}h_{\{c\}\{\}^{\backslash\nu\}}D_{\{[\backslash\mu\}b^{\{a\}\}\{\}_{\{\backslash\nu\}}\}\backslash\mathrm{mathcal{T}}^{\{*\}a\}\{\}_{\{b\}c}=2\,h_{\{b\}\{\}^{\backslash\mu\}}h_{\{c\}\{\}^{\backslash\nu\}}D_{\{[\backslash\mu\}^{\{*\}b^{\{a\}\}\}\{\}_{\{\backslash\nu\}}\}\backslash\mathrm{mathcal{T}}^{\{\backslash\dagger a\}\{b\}c}=2\,h_{\{b\}\{\}^{\backslash\mu\}}h_{\{c\}\{\}^{\backslash\nu\}}D_{\{[\backslash\mu\}^{\{\backslash\dagger a\}\{b^{\{a\}\}\}\}\{\}_{\{\backslash\nu\}}\}\backslash\mathrm{mathcal{T}}^{\{\backslash\dagger a\}\{\backslash\natural a\}\{b\}c}=2\,h_{\{b\}\{\}^{\backslash\mu\}}h_{\{c\}\{\}^{\backslash\nu\}}D_{\{[\backslash\mu\}^{\{\backslash\dagger a\}\{\backslash\natural a\}\{b^{\{a\}\}\}\}\{\}_{\{\backslash\nu\}}\}\backslash\mathrm{mathcal{T}}^{\{\backslash\dagger a\}\{\backslash\natural a\}\{b\}c}=2\,h_{\{a\}\{\}^{\backslash\mu\}}h_{\{b\}\{\}^{\backslash\nu\}}h_{\{b\}\{\}^{\backslash\nu\}}\backslash\mathrm{partial}_{\{[\backslash\mu\}B_{\{\backslash\nu\}}\}\backslash\mathrm{mathcal{H}}_{\{a\}b}^{\{\backslash\dagger\}}=2\,h_{\{a\}\{\}^{\backslash\mu\}}h_{\{b\}\{\}^{\backslash\nu\}}\backslash\mathrm{partial}_{\{[\backslash\mu\}\backslash\left(V_{\{\backslash\nu\}}+\frac{1}{3}T_{\{\backslash\nu\}}\right)}$	$\mathcal{R}^{\{a\}b}_{cd}=\mathcal{R}^{*ab}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]}+A^a{}_{c[\mu}A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]}+A^{\dagger a}{}_{c[\mu}A^{\dagger cb}{}_{\nu]})$ $T^a{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$ $T^{*a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$ $T^{\dagger a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D^{\dagger}_{[\mu}b^a{}_{\nu]}$ $T^{\natural a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D^{\natural}_{[\mu}b^a{}_{\nu]}$ $\mathcal{H}_{ab}=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$ $\mathcal{H}^{\dagger}_{ab}=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_{\nu}+\tfrac{1}{3}T_{\nu]})$ <i>Reduced A-fields and related quantities</i> $c^a{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$ $c^{*a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial^{\dagger}_{[\mu}b^a{}_{\nu]}$ $c^{\natural a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial^{\natural}_{[\mu}b^a{}_{\nu]}$ ${}^0A^{ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu}(c^{ab}{}_c+c^b{}_c{}^a-c_c{}^{ab})$ ${}^0A^{*ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu}(c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c^{*ab})$ ${}^0A^{\dagger ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu}(c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger ab})$ <i>Currents and related quantities</i> $\mathcal{L}\equiv h^{-1}L$ $t^a{}_b\equiv t^a{}_{\mu}h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$ $s_{ab}{}^c\equiv s_{ab}{}^\mu b^c{}_{\mu}\equiv b^c{}_{\mu}\delta\mathcal{L}_G/\delta A^{ab}{}_{\mu}$ $j^a\equiv j^ab^a{}_{\mu}\equiv b^a{}_{\mu}\delta\mathcal{L}_G/\delta B_{\mu}$ $t^{\dagger a}{}_b\equiv t^{\dagger a}{}_{\mu}b^{\mu}{}_b+2(s_{cb}{}^aV^c-s^{ac}{}_cV_b)$ $j^{\dagger a}\equiv j^a-2s^{ab}{}_b$ $\tau^a{}_b\equiv \tau^a{}_{\mu}h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$ $\sigma_{ab}{}^c\equiv \sigma_{ab}{}^\mu b^c{}_{\mu}\equiv b^c{}_{\mu}\delta\mathcal{L}_M/\delta A^{ab}{}_{\mu}$ $\zeta^a\equiv \zeta^a{}_{\mu}b^{\mu}{}_a\equiv b^a{}_{\mu}\delta\mathcal{L}_M/\delta B_{\mu}$ $\tau^{\dagger a}{}_b\equiv \tau^a{}_b+2(\sigma_{cb}{}^aV^c-\sigma^{ac}{}_cV_b)$ $\zeta^{\dagger a}\equiv \zeta^a-2s^{ab}{}_b$	translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced <i>A</i>-field (when $T^a{}_{bc}\equiv 0$) WGT reduced <i>A</i>-field (when $T^{*a}{}_{bc}\equiv 0$) eWGT reduced $A^{\dagger}$-field (when $T^{\dagger a}{}_{bc}\equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F08993 [conf 0.000]	55	0.000	$\{ \}^{\mathrm{a}}\{ \}^{\mathrm{b}}$	ab	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[ab]}{}_{\mu} h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0 A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $\hat{j}^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F09994 [conf 0.000]	55	0.000	$\{ \}^{\wedge}\{\backslash\text{text {b }}\}$	b	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\mu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathcal{A}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathcal{A}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $\hat{j}^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F08995 [conf 0.000]	55	0.000	$\mathrm{\mathfrak{b}}$	b	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0 A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $\hat{j}^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F09997 [conf 0.000]	55	0.000	$c^{\{*\ a\}\}_{\{b\ c\}} \equiv 2\ h_{\{b\}\{ \}^{\{\mu\}}}\ h_{\{c\}\{ \}^{\{\nu\}}}\ \partial_{[\mu}^{\{*\}}\ b^{\{a\}\}_{\}_{\nu]}}$	$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[ab]}{}_{[\mu} h_{c\,}{}^\nu V_{\nu]}$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\mu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2} A^{\dagger\ ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger\ ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger\ ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger\ ab}{}_{\nu]} + A^{\dagger\ a}{}_{c[\mu} A^{\dagger\ cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathcal{A}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathcal{A}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^a{}^{*b})$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{} b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F09999 [conf 0.000]	55	0.000	$\{ \}^{\{0\}} A^{\{a\} b\} \{ \}_{\mu} \equiv \frac{1}{2} b^{\{c\} \{ \}_{\mu}} \left(c^{\{a\} b\} \{ \}_{c} + c^{\{b\} \{ \}_{c}} \{ \}^{\{a\} - c\} \{ \}^{\{a\} b\} \right)$	${}^0 A^{ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} \left(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab} \right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a} b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$ ${}^0 A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}^\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}^\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01000 [conf 0.000]	55	0.000	$\{ \}^{\{0\}} A^{\{* \text{ a b}\} \{ }_{\{\mu\}} \equiv \frac{1}{2} b^{\{c\} \{ }_{\{\mu\}} \left(c^{\{* \text{ a b}\} \{ }_{\{c\}} + c^{\{b \text{ a} \} \{ }_{\{c\}} \{ \}^{\{a\} - c_{\{c\}}^{\{* \text{ a b}\} \{ }_{\{a \text{ b}\} \{ \right) }$	${}^0 A^{*ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} \left(c^{*ab}{}_c + c^{*b \text{ a}}{}_c - c^{*ab}{}_c \right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[ab]}{}_{[\mu} h_{c\text{ }\nu} V_{\nu]}$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{3a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^3 b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0 A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^{\mu}{}_\nu \equiv b^{\nu}{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced <i>A</i>-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced <i>A</i>-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01001 [conf 0.000]	55	0.000	$\{ \}^{\wedge\{0\}} A^{\{\dagger a \ b\}\{ \}_{\mu}} \equiv \frac{1}{2} b^{\{c\}\{ \}_{\mu}} \left(c^{\{\dagger a \ b\}\{ \}_{c}} + c^{\{\dagger b \ a \}_{c}} - c_c^{\{\dagger ab\}\{ \}_{c}} \right) \{ \}_{c}^{\{ \}^{\wedge\{a\}} - c_{\{c\}^{\wedge\{\dagger a \ b\}} \right)}$	${}^0A^{\dagger ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} \left(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab} \right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^b]{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}_{cd}^{ab} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A_{c[\mu}^a A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c_c^{b\ a} - c_c^{a\ b})$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c_c^{*b\ a} - c_c^{*a\ b})$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c_c^{\dagger b\ a} - c_c^{\dagger a\ b})$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01002 [conf 0.000]	55	0.000	$\mathrm{cal}^{\mathrm{T}}\{a\}\{_{b\ c}\}\equiv 0$	$\mathcal{T}^a{}_{bc}\equiv 0$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu\equiv A^{ab}{}_\mu+2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a\equiv h_a{}^\mu D_\mu\equiv h_a{}^\mu(\partial_\mu+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^*\equiv\partial_\mu+wB_\mu$ $\mathcal{D}_\mu^*\equiv h_a{}^\mu D_\mu^*\equiv h_a{}^\mu(\partial_\mu^*+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^\perp\equiv\partial_\mu-w(V_\mu+\frac{1}{3}T_\nu)$ $\mathcal{D}_a^\perp\equiv h_a{}^\mu D_\mu^\perp\equiv h_a{}^\mu(\partial_\mu^\perp+\frac{1}{2}A^{\dagger\ ab}{}_\mu\Sigma_{ab})$ $\mathcal{D}_\mu^\perp\equiv h_a{}^\mu D_\mu^\perp\equiv h_a{}^\mu(\partial_\mu+\frac{1}{2}A^{\dagger\ ab}{}_\mu\Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd}=\mathcal{R}^{*ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]}+A^a{}_{c[\mu}A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger\ ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger\ ab}{}_{\nu]}+A^{\dagger\ a}{}_{c[\mu}A^{\dagger\ cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^*b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{H}_{ab}=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$ $\mathcal{H}^{\dagger}{}_{ab}=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu+\frac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$ $c^{*a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^*b^a{}_{\nu]}$ $c^{\perp a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{ab}{}_c+c^b{}_c{}^a-c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c{}^{*a}{}^b)$ ${}^0A^{\dagger ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L}\equiv h^{-1}L$ $t^a{}_b\equiv t^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$ $s_{ab}{}^c\equiv s_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ $j^a\equiv j^a b^b{}_\mu\equiv b^b{}_\mu\delta\mathcal{L}_G/\delta B_\mu$ $t^{\dagger a}{}_b\equiv t^{\dagger a}{}_b+2(s_{cb}{}^aV^c-s^{ac}{}_cV_b)$ $j^{\dagger a}\equiv j^a-2s^{ab}{}_b$ $\tau^a{}_b\equiv\tau^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$ $\sigma_{ab}{}^c\equiv\sigma_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$ $\zeta^a\equiv\zeta^\mu b^a{}_\mu\equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$ $\tau^{\dagger a}{}_b\equiv\tau^a{}_b+2(\sigma_{cb}{}^aV^c-\sigma^{ac}{}_cV_b)$ $\zeta^{\dagger a}\equiv\zeta^a-2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc}\equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc}\equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc}\equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01003 [conf 0.000]	55	0.000	$\mathrm{cal{T}}^{\ast\,a}\{ \}_{b\,c} \equiv 0$	$\mathcal{T}^{\ast\,a}_{bc} \equiv 0$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[ab]}{}_{[\mu} h_{c\,}{}^{\nu]} V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\ast \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^\ast \equiv h_a{}^\mu D_\mu^\ast \equiv h_a{}^\mu (\partial_\mu^\ast + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\mu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \frac{1}{2} A^{\dagger\,ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger\,ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{\ast ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger\,ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger\,ab}{}_{\nu]} + A^{\dagger\,a}{}_{c[\mu} A^{\dagger\,cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{\ast a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\ast b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\ast\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\ast\dagger} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}^{\dagger}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{\ast a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\ast b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{\ast ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\ast ab}{}_c + c^{\ast b}{}_c{}^a - c_c{}^{\ast a}{}^b)$ ${}^0 A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c{}^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{} b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_{b} \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{} b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_{b} \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{\ast a}{}_{bc} \equiv 0$) eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01005 [conf 0.000]	55	0.000	L	L	Gravitational gauge field variables $h_a{}^\mu$ translational gauge field (h-field) h determinant of $h_a{}^\mu$ $b^a{}_\mu$ inverse h-field $A^{ab}{}_\mu$ rotational gauge field (A-field) B_μ WGT dilational gauge field (B-field) V_μ eWGT dilational gauge field (V-field) $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{\dagger b]}{}_\mu h_c{}^\nu V_\nu$ eWGT extended rotational gauge field ($A^{\dagger}$-field) Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ PGT (generalised) covariant derivative^a $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ WGT 'augmented' partial derivative $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ WGT (generalised) covariant derivative^a $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\nu)$ eWGT 'augmented' partial derivative $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ eWGT (generalised) covariant derivative^a $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ eWGT (generalised) semi-covariant derivative^a Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ PGT and WGT rotational gauge field strength^{a, b} $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ eWGT rotational gauge field strength^{a, b} $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ PGT translational gauge field strength^b $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ WGT translational gauge field strength^b $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ eWGT translational gauge field strength^b $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ eWGT translational gauge 'semi' field strength^{a, b} $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ WGT dilational gauge field strength $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3}T_{\nu]})$ eWGT dilational gauge field strength Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ PGT Ricci rotation coefficients $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ WGT Ricci rotation coefficients $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ eWGT Ricci rotation coefficients ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^a{}^b)$ WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c{}^a{}^b)$ eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ Lagrangian $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ Lagrangian density $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ gravitational sector energy-momentum tensor^c $j^a \equiv j^a{}_\mu b^{\mu}{}_\nu \equiv b^{\nu}{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ gravitational sector spin-angular-momentum tensor^c $t^{\dagger a}{}_b \equiv t^{\dagger a}{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ gravitational sector dilation current^c $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ gravitational sector eWGT covariant energy-momentum tensor^c $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ matter sector energy-momentum tensor^c $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ matter sector spin-angular-momentum tensor^c $\zeta^a \equiv \zeta^a{}_\mu b^{\mu}{}_\nu \equiv b^{\nu}{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ matter sector dilation current^c $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ gravitational sector eWGT covariant energy-momentum tensor^c $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01086 [conf 0.000]	55	0.000	$\mathrm{\mathcal{L}} \equiv h^{-1} L$	$\mathcal{L} \equiv h^{-1} L$	Gravitational gauge field variables $h_a{}^\mu$ translational gauge field (h-field) h determinant of $h_a{}^\mu$ $b^a{}_\mu$ inverse h-field $A^{ab}{}_\mu$ rotational gauge field (A-field) B_μ WGT dilational gauge field (B-field) V_μ eWGT dilational gauge field (V-field) $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{\dagger b]}{}_\mu h_c{}^\nu V_\nu$ eWGT extended rotational gauge field ($A^{\dagger}$-field) Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ PGT (generalised) covariant derivative^a $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ WGT 'augmented' partial derivative $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ WGT (generalised) covariant derivative^a $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ eWGT 'augmented' partial derivative $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ eWGT (generalised) covariant derivative^a $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ eWGT (generalised) semi-covariant derivative^a Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ PGT and WGT rotational gauge field strength^{a, b} $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ eWGT rotational gauge field strength^{a, b} $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ PGT translational gauge field strength^b $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ WGT translational gauge field strength^b $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ eWGT translational gauge field strength^b $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ eWGT translational gauge 'semi' field strength^{a, b} $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ WGT dilational gauge field strength $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ eWGT dilational gauge field strength Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ PGT Ricci rotation coefficients $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ WGT Ricci rotation coefficients $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ eWGT Ricci rotation coefficients ${}^0 A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) ${}^0 A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^a{}^b)$ WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) ${}^0 A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c{}^a{}^b)$ eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Currents and related quantities $\mathcal{L}$ Lagrangian $\mathcal{L} \equiv h^{-1} L$ Lagrangian density $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ gravitational sector energy-momentum tensor^c $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ gravitational sector spin-angular-momentum tensor^c $j^a \equiv j^a b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ gravitational sector dilation current^c $t^{\dagger a}{}_b \equiv t^{\dagger a}{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac} \mathcal{V}_b)$ gravitational sector eWGT covariant energy-momentum tensor^c $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ gravitational sector eWGT covariant dilation current^c $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ matter sector energy-momentum tensor^c $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ matter sector spin-angular-momentum tensor^c $\zeta^a \equiv \zeta^a b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ matter sector dilation current^c $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac} \mathcal{V}_b)$ gravitational sector eWGT covariant energy-momentum tensor^c $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01007 [conf 0.000]	55	0.000	$t^{\{a\}\{ \}_\{b\}} \equiv t^{\{a\}\{ \}_\{\mu\}} h_{\{b\}\{ \}^{\{\mu\}}} \equiv h_{\{b\}\{ \}^{\{\mu\}}} \delta \mathcal{L}_G / \delta h_{a^{\mu}}^{\{ \}}$	$t^a_b \equiv t^a_\mu h_b^\mu \equiv h_b^\mu \delta \mathcal{L}_G / \delta h_a^\mu$	Gravitational gauge field variables h_a^μ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{\dagger b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^2 b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$ ${}^0 A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b^\mu \equiv h_b^\mu \delta \mathcal{L}_G / \delta h_a^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac} \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b^\mu \equiv h_b^\mu \delta \mathcal{L}_M / \delta h_a^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced <i>A</i>-field (when $\mathcal{T}^{ab}{}_{bc} \equiv 0$) WGT reduced <i>A</i>-field (when $\mathcal{T}^{*ab}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger ab}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01088 [conf 0.000]	55	0.000	$s_{\{a\} \{b\} \{c\}} \equiv s_{\{a\} \{b\} \{\mu\}} b^{\{c\} \{ \}_\mu} \equiv b^{\{c\} \{ \}_\mu} \delta \mathcal{L}_G / \delta A^{ab}_\mu$	$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}_\mu$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^b]{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ $c^{\mathfrak{a}}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0 A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^{\mu}{}_\nu \equiv b^{\nu}{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^{\mu}{}_\nu \equiv b^{\nu}{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced <i>A</i>-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced <i>A</i>-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01009 [conf 0.000]	55	0.000	$j^{\mathbf{a}} \equiv j^{\mathbf{a}}_{\mu} \quad b^{\mathbf{a}} \equiv b^{\mathbf{a}}_{\mu} \quad \delta \mathcal{L}_G / \delta B_{\mu}$	$j^a \equiv j^{\mu} b^a_{\mu} \equiv b^a_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$	Gravitational gauge field variables $h_a{}^{\mu}$ h $b^a{}_{\mu}$ $A^{ab}{}_{\mu}$ B_{μ} V_{μ} $A^{\dagger ab}{}_{\mu} \equiv A^{ab}{}_{\mu} + 2\eta^{c[a}b^b]{}_{\mu} h_c{}^{\nu} V_{\nu}$ Derivative operators $\mathcal{D}_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \tfrac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^* \equiv \partial_{\mu} + w B_{\mu}$ $\mathcal{D}_{\mu}^* \equiv h_a{}^{\mu} D_{\mu}^* \equiv h_a{}^{\mu} (\partial_{\mu}^* + \tfrac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^{\dagger} \equiv \partial_{\mu} - w(V_{\mu} + \tfrac{1}{3} T_{\nu})$ $\mathcal{D}_{\mu}^{\dagger} \equiv h_a{}^{\mu} D_{\mu}^{\dagger} \equiv h_a{}^{\mu} (\partial_{\mu}^{\dagger} + \tfrac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ $\mathcal{D}_{\mu}^{\natural} \equiv h_a{}^{\mu} D_{\mu}^{\natural} \equiv h_a{}^{\mu} (\partial_{\mu} + \tfrac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\dagger} b^a{}_{\nu]}$ $\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\natural} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \tfrac{1}{3} T_{\nu])}$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$ ${}^0 A^{ab}{}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0 A^{\dagger ab}{}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$ $s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$ $j^a \equiv j^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_{b} \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$ $\zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_{b} \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^{\mu}$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01010 [conf 0.000]	55	0.000	$t^{\dagger}\{\backslash dagger\ a\}\{ \}_{_b}\ \equiv\ t^{\wedge}\{a\}\{ \}_{_b}+2\left(s_{\{c\ b\}\{ \}^{\wedge}\{a\}\ \backslash mathcal\{V\}^{\wedge}\{c\}\cdot s^{\wedge}\{a\ c\}\{ \}_{_c}\ \backslash mathcal\{V\}_{_b}\right)$	$t^{\dagger}{}_b \equiv t^a{}_b + 2\left(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b\right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3}T_{\nu]})$ PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\perp a}{}_{bc} \equiv 0$) Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01011 [conf 0.000]	55	0.000	$j^{\dagger a} \equiv j^a - 2s^a{}_b j^b$	$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^b]{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}_b)$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}_b)$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}_b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}^\mu b^b{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}^\mu b^b{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01012 [conf 0.000]	55	■ 0.000	$\tau^{\{a\}\}_{\{b\}} \equiv \tau^{\{a\}\}_{\{\mu\}} h_{\{b\}\} \wedge^{\{\mu\}} \\ \equiv h_{\{b\}\} \wedge^{\{\mu\}} \delta \mathrm{cal}{L}_{\mathrm{\{M\}}} / \\ \delta h_{\{a\}\} \wedge^{\{\mu\}}$	$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_\text{M}/\delta h_a{}^\mu$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{[a} b^{\dagger b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_a^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{2}T_\mu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2}A^\dagger{}^{ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_a^\ddagger \equiv h_a{}^\mu D_\mu^\ddagger \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2}A^\dagger{}^{ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $T^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $T^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $T^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $T^{\ddagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\ddagger b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}^{\dagger}{}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{2}T_{\nu])}$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \tfrac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0A^{*ab}{}_\mu \equiv \tfrac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities L $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G/\delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G/\delta A^{ab}{}_\mu$ $j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_G/\delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_\text{M}/\delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_\text{M}/\delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_\text{M}/\delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field (<i>A[†]</i>-field) PGT (generalised) covariant derivative^a WGT ‘augmented’ partial derivative WGT (generalised) covariant derivative^a eWGT ‘augmented’ partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge ‘semi’ field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $T^a{}_{bc} \equiv 0$) WGT reduced A-field (when $T^{*a}{}_{bc} \equiv 0$) eWGT reduced <i>A[†]</i>-field (when $T^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510..06699_ FOI013 [conf 0.000]	55	■0.000	\sigma_{\mathrm{a}\,\mathrm{b}}^{\mathrm{c}} \equiv \sigma_{ab}^{\mu b^c}_{\scriptsize{\mu}} \equiv \sigma^{c\,\mu}_{{\scriptsize{\mu}}} \delta {\cal L}_M / \delta A^{ab}_{\scriptsize{\mu}}	$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c_\mu \equiv b^c_\mu \delta {\cal L}_M/\delta A^{ab}_\mu$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_μ$ $B_μ$ $V_μ$ $A^{\dagger ab}{}_μ \equiv A^{ab}{}_μ + 2η^{[a}b^{b]}_μ h_c{}^ν V_ν$ Derivative operators $D_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\hat{D}^*_a \equiv \partial_\mu + w B_\mu$ $D^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu (\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\theta^{\dagger}_a \equiv \partial_\mu -w(V_\mu+\tfrac{1}{2}T_\mu)$ ${\mathcal{D}}^{\dagger}_a \equiv h_a{}^\mu D^{\dagger}_\mu \equiv h_a{}^\mu (\theta^{\dagger}_\mu + \tfrac{1}{2}A^{\dagger\;ab}{}_μ\Sigma_{ab})$ $D^{\natural}_a \equiv h_a{}^\mu D^{\natural}_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2}A^{\dagger\;ab}{}_μ\Sigma_{ab})$ Gauge field strengths $R^{ab}{}_{cd}=R^{*\;ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu \langle \partial_{[\mu}A^{ab}{}_{\nu]}+A^a{}_{c[\mu}A^{cb}{}_{\nu]}\rangle$ $R^{\dagger\;ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu \langle \partial_{(\mu}A^{\dagger\;ab}{}_{\nu)}+A^{\dagger\;a}{}_{c[\mu}A^{\dagger\;cb}{}_{\nu]} \rangle$ $T^{aa}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$ $T^{\dagger a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D^{\dagger }_{[\mu}b^a{}_{\nu]}$ $T^{\natural a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D^{\natural }_{[\mu}b^a{}_{\nu]}$ $H_{ab}=2h_a{}^\mu h_b{}^\nu \partial_{[\mu}B_{\nu]}$ $H^{\dagger }_{ab}=2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_{\nu}+\tfrac{1}{2}T_{\nu])}$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}b^a{}_{\nu]}$ $c^{\ast a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^{\ast }_{[\mu}b^a{}_{\nu]}$ $c^{\natural a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^{\natural }_{[\mu}b^a{}_{\nu]}$ ${}^oA^{ab}{}_μ \equiv \tfrac{1}{2}b^c{}_μ(c^{oa}{}_c+c^{cb}{}_ca-c^{ab}{}_c)$ ${}^oA^{†ab}{}_μ \equiv \tfrac{1}{2}b^c{}_μ(c^{†ab}{}_c+c^{cb}{}_ca- c^{ab}{}_c)$ ${}^oA^{\natural ab}{}_μ \equiv \tfrac{1}{2}b^c{}_μ(c^{\natural ab}{}_c+c^{cb}{}_ca-c^{\natural ab}{}_c)$ Currents and related quantities L $\mathcal{L} \equiv h^{-1}L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta {\cal L}_G/\delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c_\mu \equiv b^c_\mu \delta {\cal L}_G/\delta A^{ab}{}_μ$ $j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu \delta {\cal L}_G/\delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^aV^c-s^{ac}{}_cV_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau ^a{}_b \equiv \tau ^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta {\cal L}_M/\delta h_a{}^\mu$ $\sigma _{ab}{}^c \equiv \sigma _{ab}{}^\mu b^c_\mu \equiv b^c_\mu \delta {\cal L}_M/\delta A^{ab}{}_μ$ $\zeta {}^a \equiv \zeta ^\mu b^a{}_\mu \equiv b^a{}_\mu \delta {\cal L}_M/\delta B_\mu$ $\tau ^{\dagger a}{}_b \equiv \tau ^a{}_b + 2(σ_{cb}{}^aV^c − σ^{ac}{}_cV_b)$ $\zeta ^{\dagger a} \equiv \zeta {}^a − 2s^{ab}{}_b$ translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field (<i>A[†]</i>-field) PGT (generalised) covariant derivative^a WGT ‘augmented’ partial derivative WGT (generalised) covariant derivative^a eWGT ‘augmented’ partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a,b}, eWGT rotational gauge field strength^{a,b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge ‘semi’ field strength^{a,b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $T^{aa}{}_{bc} \equiv 0$) WGT reduced A-field (when $T^{aa}{}_{bc} \equiv 0$) eWGT reduced <i>A[†]</i>-field (when $T^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitaitonal sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01014 [conf 0.000]	55	0.000	$\backslash\mathrm{zeta}^{\mathrm{a}}\backslash\mathrm{equiv}\backslash\mathrm{zeta}^{\mathrm{\mu}}\backslash\mathrm{b}^{\mathrm{a}}\{\}_{\mathrm{\mu}}\backslash\mathrm{equiv}\backslash\mathrm{b}^{\mathrm{a}}\{\}_{\mathrm{\mu}}\backslash\mathrm{delta}\backslash\mathrm{mathcal}\{\mathrm{L}\}_{\mathrm{\mathrm{M}}}\backslash\mathrm{delta}\mathrm{B}_{\mathrm{\mu}}$	$\zeta^a\equiv\zeta^\mu b^a{}_\mu\equiv b^a{}_\mu\delta\mathcal{L}_\mathrm{M}/\delta B_\mu$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu\equiv A^{ab}{}_\mu+2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a\equiv h_a{}^\mu D_\mu\equiv h_a{}^\mu(\partial_\mu+\tfrac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^*\equiv\partial_\mu+wB_\mu$ $\mathcal{D}_\mu^*\equiv h_a{}^\mu D_\mu^*\equiv h_a{}^\mu(\partial_\mu^*+\tfrac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^\perp\equiv\partial_\mu-w(V_\mu+\tfrac{1}{3}T_\nu)$ $\mathcal{D}_a^\perp\equiv h_a{}^\mu D_\mu^\perp\equiv h_a{}^\mu(\partial_\mu^\perp+\tfrac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ $\mathcal{D}_\mu^\perp\equiv h_a{}^\mu D_\mu^\perp\equiv h_a{}^\mu(\partial_\mu+\tfrac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd}=\mathcal{R}^{*ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]}+A^a{}_{c[\mu}A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]}+A^{\dagger a}{}_{c[\mu}A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^*b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathrm{S}a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^\mathrm{S} b^a{}_{\nu]}$ $\mathcal{H}_{ab}=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$ $\mathcal{H}_{ab}^\perp=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu+\tfrac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$ $c^{*a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^*b^a{}_{\nu]}$ $c^{\perp a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu\equiv\tfrac{1}{2}b^c{}_\mu(c^{ab}{}_c+c^b{}_c{}^a-c_c{}^{ab})$ ${}^0A^{*ab}{}_\mu\equiv\tfrac{1}{2}b^c{}_\mu(c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c^{*ab})$ ${}^0A^{\dagger ab}{}_\mu\equiv\tfrac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L}\equiv h^{-1}L$ $t^a{}_b\equiv t^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$ $s_{ab}{}^c\equiv s_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ $j^a\equiv j^a{}_\mu b^{\mu}{}_\nu\equiv b^{\nu}{}_\mu\delta\mathcal{L}_G/\delta B_\mu$ $\hat{t}^{\dagger a}{}_b\equiv \hat{t}^a{}_b+2(s_{cb}{}^a\mathcal{V}^c-s^{ac}{}_c\mathcal{V}_b)$ $\hat{j}^{\dagger a}\equiv j^a-2s^{ab}{}_b$ $\tau^a{}_b\equiv\tau^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_\mathrm{M}/\delta h_a{}^\mu$ $\sigma_{ab}{}^c\equiv\sigma_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_\mathrm{M}/\delta A^{ab}{}_\mu$ $\zeta^a\equiv\zeta^\mu b^a{}_\mu\equiv b^a{}_\mu\delta\mathcal{L}_\mathrm{M}/\delta B_\mu$ $\tau^{\dagger a}{}_b\equiv\tau^a{}_b+2(\sigma_{cb}{}^a\mathcal{V}^c-\sigma^{ac}{}_c\mathcal{V}_b)$ $\zeta^{\dagger a}\equiv\zeta^a-2s^{ab}{}_b$ translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field (<i>A</i>[†]-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced <i>A</i>-field (when $\mathcal{T}^a{}_{bc}\equiv 0$) WGT reduced <i>A</i>-field (when $\mathcal{T}^{*a}{}_{bc}\equiv 0$) eWGT reduced <i>A</i>[†]-field (when $\mathcal{T}^{\dagger a}{}_{bc}\equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01015 [conf 0.000]	55	0.000	$\tau^{\dagger a}{}_{\{b\}}{}_{\{c\}} \equiv \tau^a{}_{\{a\}}{}_{\{b\}} + 2 \left(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_{\{c\}}{}_{\{b\}} \mathcal{V}_b \right)$	$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2 \left(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b \right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$ ${}^0 A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced A[†]-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}^\mu b^b{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}^\mu b^b{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01016 [conf 0.000]	55	0.000	$\backslash\mathrm{zeta}^{\wedge}\{\backslash\mathrm{dagger}\ a\}\ \backslash\mathrm{equiv}\ \backslash\mathrm{zeta}^{\wedge}\{\mathrm{a}\}\mathrm{-}2\ s^{\wedge}\{\mathrm{a}\ b\}\{\ \}_{\mathrm{-}\{\mathrm{b}\}}$	$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2}A^{\dagger\ ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{\dagger\ ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger\ ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger\ ab}{}_{\nu]} + A^{\dagger\ a}{}_{c[\mu} A^{\dagger\ cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathcal{A}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathcal{A}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1}L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_ F01017 [conf 0.000]	55	0.000	c	C	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\mu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0 A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $\hat{j}^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_F01018 [conf 0.000]	55	■ 0.000	$\{ \}^{\text{a}} \backslash \text{text} \{ c \}$	c	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^{\dagger} \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\mu)$ $\mathcal{D}_a^{\dagger} \equiv h_a{}^\mu D_\mu^{\dagger} \equiv h_a{}^\mu (\partial_\mu^{\dagger} + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^{\dagger} \equiv h_a{}^\mu D_\mu^{\dagger} \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_\mu A^{ab}{}_\nu - A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_\mu A^{\dagger ab}{}_\nu + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\dagger} b^a{}_{\nu]}$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\dagger} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$ ${}^0A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^{ab}{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*ab}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger ab}{}_{bc} \equiv 0$) Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_\mu \equiv t^a{}_\mu h_a{}^\mu \equiv h_a{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}^\mu b^b{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_\mu \equiv t^a{}_\mu + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_\mu \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_\mu \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}^\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_\mu \equiv \tau^a{}_\mu + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_\mu \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c
1510.06699_F01019	55	■ 0.999	$\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}, \mathcal{R} \equiv \mathcal{R}^a{}_a$	$\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}, \mathcal{R} \equiv \mathcal{R}^a{}_a$	^b Contractions are defined by $\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}$, $\mathcal{R} \equiv \mathcal{R}^a{}_a$ and $\mathcal{T}_a = \mathcal{T}^b{}_{ab}$, and similarly for other versions.
1510.06699_F01020	55	■ 0.999	$\mathcal{T}^a{}_a = \mathcal{T}^b{}_{ab}$	$\mathcal{T}_a = \mathcal{T}^b{}_{ab}$	^b Contractions are defined by $\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}$, $\mathcal{R} \equiv \mathcal{R}^a{}_a$ and $\mathcal{T}_a = \mathcal{T}^b{}_{ab}$, and similarly for other versions.
1510.06699_F01021	56	■ 1.000	$SL(2, C)$	$SL(2, C)$	Lorentz transformations by Λ , rather than $SL(2, C)$ elements.
1510.06699_F01022	56	■ 1.000	$h^{\mu}{}_a J_{\mu} \neq \mathcal{J}_a$	$h^{\mu}{}_a J_{\mu} \neq \mathcal{J}_a$	$\mathcal{J}$, so that, for example, $h^{\mu}{}_a J_{\mu} \neq \mathcal{J}_a$. We will not, however,
1510.06699_F01023	56	■ 1.000	$\{ \}^{\{0\}} \mathcal{A}_{abc}$	${}^0\mathcal{A}_{abc}$	${}^0\mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0\mathcal{A}^{*ab}{}_c =$
1510.06699_F01024	56	■ 1.000	$\mathcal{K}_{abc}$	$\mathcal{K}_{abc}$	${}^0\mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0\mathcal{A}^{*ab}{}_c =$
1510.06699_F01025	56	■ 1.000	$\{ \}^{\{0\}} \mathcal{A}^{*ab}{}_c =$	${}^0\mathcal{A}^{*ab}{}_c =$	${}^0\mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0\mathcal{A}^{*ab}{}_c =$
1510.06699_F01026	56	■ 1.000	$\{ \}^{\{0\}} \mathcal{A}^{ab}{}_c + \delta_c^a \mathcal{B}^b - \delta_c^b \mathcal{B}^a$	${}^0\mathcal{A}^{ab}{}_c + \delta_c^a \mathcal{B}^b - \delta_c^b \mathcal{B}^a$	${}^0\mathcal{A}^{ab}{}_c + \delta_c^a \mathcal{B}^b - \delta_c^b \mathcal{B}^a$ and $\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c - \delta_c^a \mathcal{B}^b + \delta_c^b \mathcal{B}^a$, so that the
1510.06699_F01027	56	■ 1.000	$\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c - \delta_c^a \mathcal{B}^b + \delta_c^b \mathcal{B}^a$	$\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c - \delta_c^a \mathcal{B}^b + \delta_c^b \mathcal{B}^a$	${}^0\mathcal{A}^{ab}{}_c + \delta_c^a \mathcal{B}^b - \delta_c^b \mathcal{B}^a$ and $\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c - \delta_c^a \mathcal{B}^b + \delta_c^b \mathcal{B}^a$, so that the
1510.06699_F01028	56	■ 1.000	$t^a{}_\mu \equiv h \delta \mathcal{L}_G / \delta h_a{}^\mu, s_{ab}{}^\mu \equiv$	$t^a{}_\mu \equiv h \delta \mathcal{L}_G / \delta h_a{}^\mu, s_{ab}{}^\mu \equiv$	⁷² One could alternatively define $t^a{}_\mu \equiv h \delta \mathcal{L}_G / \delta h_a{}^\mu$, $s_{ab}{}^\mu \equiv$
1510.06699_F01029	56	■ 1.000	$h \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$	$h \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$	$h \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$, and $j^\mu \equiv h \delta \mathcal{L}_G / \delta B_\mu$, and similarly for the mat-
1510.06699_F01030	56	■ 1.000	$j^\mu \equiv h \delta \mathcal{L}_G / \delta B_\mu$	$j^\mu \equiv h \delta \mathcal{L}_G / \delta B_\mu$	$h \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$, and $j^\mu \equiv h \delta \mathcal{L}_G / \delta B_\mu$, and similarly for the mat-
1510.06699_F01031	56	■ 1.000	$\dot{R} \equiv$	$\dot{R} \equiv$	point particle one must, in addition, make the replacement $\dot{R} \equiv$
1510.06699_F01032	56	■ 1.000	$\dot{x}^\mu \partial_\mu R \rightarrow \dot{v}^a \mathcal{D}_a^* R$	$\dot{x}^\mu \partial_\mu R \rightarrow \dot{v}^a \mathcal{D}_a^* R$	$\dot{x}^\mu \partial_\mu R \rightarrow \dot{v}^a \mathcal{D}_a^* R$.
1510.06699_F01033	56	■ 0.994	$\{ \}^{\{77,78\}}$	77,78	theory ^{77,78} . In the standard interpretation, the scalar field in
1510.06699_F01034	56	■ 0.999	$\bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a^* \psi = \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi$	$\bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a^* \psi = \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi$	⁸¹ It is easily shown that $\bar{\psi} \gamma^a {}^0\mathcal{D}_a^* \psi = \bar{\psi} \gamma^a {}^0\mathcal{D}_a \psi$ (which does not
1510.06699_F01035	56	■ 1.000	$\theta(x)$	$\theta(x)$	a field $\theta(x)$. Provided it has Weyl weight $w = 0$, one would ob-

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1510.06699_F01036	57	■ 1.000	$D_{\mu}^{\dagger} \varphi =$	$D_{\mu}^{\dagger} \varphi =$	ticular, we note that the eWGT covariant derivative $D_{\mu}^{\dagger} \varphi =$
1510.06699_F01037	57	■ 1.000	$\left[D_{\mu} + \frac{1}{2} (\mathcal{V}^a b^b{}_{\mu} - \mathcal{V}^b b^a{}_{\mu}) \Sigma_{ab} - w V_{\mu} - \frac{1}{3} w T_{\mu}\right] \varphi$	$\left[D_{\mu} + \frac{1}{2} (\mathcal{V}^a b^b{}_{\mu} - \mathcal{V}^b b^a{}_{\mu}) \Sigma_{ab} - w V_{\mu} - \frac{1}{3} w T_{\mu}\right] \varphi$	$[D_{\mu} + \frac{1}{2} (\mathcal{V}^a b^b{}_{\mu} - \mathcal{V}^b b^a{}_{\mu}) \Sigma_{ab} - w V_{\mu} - \frac{1}{3} w T_{\mu}] \varphi$ represents a far
1510.06699_F01038	57	■ 1.000	$D_{\mu}^{\ast} \varphi = \left(D_{\mu} + w B_{\mu}\right) \varphi$	$D_{\mu}^{\ast} \varphi = (D_{\mu} + w B_{\mu}) \varphi$	derivative $D_{\mu}^{\ast} \varphi = (D_{\mu} + w B_{\mu}) \varphi$.
1510.06699_F01039	57	■ 0.995	$\{ \}^{\{0\}} \mathcal{M}^{\{A\}^{\dagger} a b\} \} _{c} = \{ \}^{\{0\}} \mathcal{M}^{\{A\}^{\{a b\} \} _{c}} - \delta _{c}^{\{a\}} \left(\mathcal{M}^{\{V\}^{\{b\} + \frac{1}{3} \{3\}} \mathcal{M}^{\{T\}^{\{b\} \right) + \delta _{c}^{\{b\}} \left(\mathcal{M}^{\{V\}^{\{a\} + \right.}$	${}^0\mathcal{A}^{\dagger ab}{}_c = {}^0\mathcal{A}^{ab}{}_c - \delta_c^a \left(\mathcal{V}^b + \frac{1}{3} \mathcal{T}^b \right) + \delta_c^b (\mathcal{V}^a +$	pendix B), one has ${}^0\mathcal{A}^{\dagger ab}{}_c = {}^0\mathcal{A}^{ab}{}_c - \delta_c^a (\mathcal{V}^b + \frac{1}{3} \mathcal{T}^b) + \delta_c^b (\mathcal{V}^a +$
1510.06699_F01040	57	■ 0.600	$\frac{1}{3} \mathcal{T}^a$	$\frac{1}{3} \mathcal{T}^a$	$\frac{1}{3} \mathcal{T}^a)$ and $\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$, so confirming that
1510.06699_F01041	57	■ 0.600	$\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$	$\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$	$\frac{1}{3} \mathcal{T}^a)$ and $\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$, so confirming that
1510.06699_F01042	57	■ 0.965	$\bar{\psi} \gamma^a \mathcal{D}_a^{\dagger} \psi = \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi$	$\bar{\psi} \gamma^{a0} \mathcal{D}_a^{\dagger} \psi = \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi$	⁹⁴ It is easily shown that $\bar{\psi} \gamma^a {}^0 \mathcal{D}_a^{\dagger} \psi = \bar{\psi} \gamma^a {}^0 \overleftrightarrow{\mathcal{D}}_a \psi$ (which does not
1510.06699_F01043	57	■ 1.000	$\nabla_{\sigma} g_{\mu \nu} =$	$\nabla_{\sigma} g_{\mu \nu} =$	that given in (124) for WGT can both be written as $\nabla_{\sigma} g_{\mu \nu} =$
1510.06699_F01044	57	■ 1.000	$-\frac{1}{2} \left(\Gamma^{\lambda}{}_{\lambda \sigma} - {}^0 \Gamma^{\lambda}{}_{\lambda \sigma} \right) g_{\mu \nu}$	$-\frac{1}{2} \left(\Gamma^{\lambda}{}_{\lambda \sigma} - {}^0 \Gamma^{\lambda}{}_{\lambda \sigma} \right) g_{\mu \nu}$	$-\frac{1}{2} (\Gamma^{\lambda}{}_{\lambda \sigma} - {}^0 \Gamma^{\lambda}{}_{\lambda \sigma}) g_{\mu \nu}$, where ${}^0 \Gamma^{\lambda}{}_{\mu \nu}$ is the metric connection;
1510.06699_F01045	57	■ 1.000	$L_{\mathcal{R}^2} =$	$L_{\mathcal{R}^2} =$	which its symmetries are more manifest, namely $L_{\mathcal{R}^2} =$
1510.06699_F01046	57	■ 1.000	$\alpha_1 \mathcal{R}^{\dagger 2} + \alpha_2' \mathcal{R}_{(ab)}^{\dagger} \mathcal{R}^{\dagger (ab)} + \alpha_3' \mathcal{R}_{[ab]}^{\dagger} \mathcal{R}^{\dagger [ab]} + \alpha_4' \mathcal{R}_{a(bc)d}^{\dagger} \mathcal{R}^{\dagger a(bc)d} +$	$\alpha_1 \mathcal{R}^{\dagger 2} + \alpha_2' \mathcal{R}_{(ab)}^{\dagger} \mathcal{R}^{\dagger (ab)} + \alpha_3' \mathcal{R}_{[ab]}^{\dagger} \mathcal{R}^{\dagger [ab]} + \alpha_4' \mathcal{R}_{a(bc)d}^{\dagger} \mathcal{R}^{\dagger a(bc)d} +$	$\alpha_1 \mathcal{R}^{\dagger 2} + \alpha_2' \mathcal{R}_{(ab)}^{\dagger} \mathcal{R}^{\dagger (ab)} + \alpha_3' \mathcal{R}_{[ab]}^{\dagger} \mathcal{R}^{\dagger [ab]} + \alpha_4' \mathcal{R}_{a(bc)d}^{\dagger} \mathcal{R}^{\dagger a(bc)d} +$
1510.06699_F01047	57	■ 1.000	$\alpha_5' \mathcal{R}_{a[bc]d}^{\dagger} \mathcal{R}^{\dagger a[bc]d} + \alpha_6 \mathcal{R}_{abcd}^{\dagger} \mathcal{R}^{\dagger cdab}$	$\alpha_5' \mathcal{R}_{a[bc]d}^{\dagger} \mathcal{R}^{\dagger a[bc]d} + \alpha_6 \mathcal{R}_{abcd}^{\dagger} \mathcal{R}^{\dagger cdab}$	$\alpha_5' \mathcal{R}_{a[bc]d}^{\dagger} \mathcal{R}^{\dagger a[bc]d} + \alpha_6 \mathcal{R}_{abcd}^{\dagger} \mathcal{R}^{\dagger cdab}$, where the new coefficients
1510.06699_F01048	57	■ 0.994	$\alpha_2' = \alpha_2 + \alpha_3, \alpha_3' = \alpha_2 - \alpha_3, \alpha_4' = \alpha_4 + \alpha_5$	$\alpha_2' = \alpha_2 + \alpha_3, \alpha_3' = \alpha_2 - \alpha_3, \alpha_4' = \alpha_4 + \alpha_5$	are $\alpha_2' = \alpha_2 + \alpha_3$, $\alpha_3' = \alpha_2 - \alpha_3$, $\alpha_4' = \alpha_4 + \alpha_5$ and $\alpha_5' = \alpha_4 - \alpha_5$.
1510.06699_F01049	57	■ 0.994	$\alpha_5' = \alpha_4 - \alpha_5$	$\alpha_5' = \alpha_4 - \alpha_5$	are $\alpha_2' = \alpha_2 + \alpha_3$, $\alpha_3' = \alpha_2 - \alpha_3$, $\alpha_4' = \alpha_4 + \alpha_5$ and $\alpha_5' = \alpha_4 - \alpha_5$.
1510.06699_F01050	57	■ 1.000	$\alpha_2' = \alpha_3'$	$\alpha_2' = \alpha_3'$	allows one to set α_1 or α_6 to zero as before, or set $\alpha_2' = \alpha_3'$.
1510.06699_F01051	57	■ 0.999	$L_{\mathcal{T}^2} = \beta_1' \mathcal{T}_{(ab)c}^{\dagger} \mathcal{T}^{\dagger (ab)c} + \beta_2' \mathcal{T}_{[ab]c}^{\dagger} \mathcal{T}^{\dagger [ab]c}$	$L_{\mathcal{T}^2} = \beta_1' \mathcal{T}_{(ab)c}^{\dagger} \mathcal{T}^{\dagger (ab)c} + \beta_2' \mathcal{T}_{[ab]c}^{\dagger} \mathcal{T}^{\dagger [ab]c}$	symmetric form $L_{\mathcal{T}^2} = \beta_1' \mathcal{T}_{(ab)c}^{\dagger} \mathcal{T}^{\dagger (ab)c} + \beta_2' \mathcal{T}_{[ab]c}^{\dagger} \mathcal{T}^{\dagger [ab]c}$, where
1510.06699_F01052	57	■ 0.998	$\beta_1' = \beta_1 + \beta_2$	$\beta_1' = \beta_1 + \beta_2$	$\beta_1' = \beta_1 + \beta_2$ and $\beta_2' = \beta_1 - \beta_2$.
1510.06699_F01053	57	■ 0.998	$\beta_2' = \beta_1 - \beta_2$	$\beta_2' = \beta_1 - \beta_2$	$\beta_1' = \beta_1 + \beta_2$ and $\beta_2' = \beta_1 - \beta_2$.
1510.06699_F01054	57	■ 1.000	$S_{\mathrm{T}} = S_{\mathrm{G}} + S_{\mathrm{M}}$	$S_{\mathrm{T}} = S_{\mathrm{G}} + S_{\mathrm{M}}$	lations, but the total action $S_{\mathrm{T}} = S_{\mathrm{G}} + S_{\mathrm{M}}$ is so. We will not
1510.06699_F01055	57	■ 0.999	${}^0\mathcal{R}^{*ab}{}_{cd} = {}^0\mathcal{R}^{ab}{}_{cd} - 2\delta_d^{[b} ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^{a]} +$	${}^0\mathcal{R}^{*ab}{}_{cd} = {}^0\mathcal{R}^{ab}{}_{cd} - 2\delta_d^{[b} ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^{a]} +$	quantity ${}^0\mathcal{R}^{*ab}{}_{cd}$ by ${}^0\mathcal{R}^{*ab}{}_{cd} = {}^0\mathcal{R}^{ab}{}_{cd} - 2\delta_d^{[b} ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^{a]} +$
1510.06699_F01056	57	■ 1.000	$2\delta_c^{[b} ({}^0\mathcal{D}_d - \mathcal{B}_d) \mathcal{B}^{a]} - 2\mathcal{B}^e \mathcal{B}_e \delta_c^{[a} \delta_d^{b]}$	$2\delta_c^{[b} ({}^0\mathcal{D}_d - \mathcal{B}_d) \mathcal{B}^{a]} - 2\mathcal{B}^e \mathcal{B}_e \delta_c^{[a} \delta_d^{b]}$	$2\delta_c^{[b} ({}^0\mathcal{D}_d - \mathcal{B}_d) \mathcal{B}^{a]} - 2\mathcal{B}^e \mathcal{B}_e \delta_c^{[a} \delta_d^{b]}$, from which one finds that their
1510.06699_F01057	57	■ 1.000	${}^0\mathcal{R}^{*a}{}_c = {}^0\mathcal{R}^a{}_c - 2 ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^a -$	${}^0\mathcal{R}^{*a}{}_c = {}^0\mathcal{R}^a{}_c - 2 ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^a -$	contractions are related by ${}^0\overline{\mathcal{R}}^{*a}{}_c = {}^0\mathcal{R}^a{}_c - 2 ({}^0\mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^a -$
1510.06699_F01058	57	■ 1.000	$({}^0\mathcal{D}_e + 2\mathcal{B}_e) \mathcal{B}^e \delta_c^a$	$({}^0\mathcal{D}_e + 2\mathcal{B}_e) \mathcal{B}^e \delta_c^a$	$({}^0\mathcal{D}_e + 2\mathcal{B}_e) \mathcal{B}^e \delta_c^a$ and ${}^0\mathcal{R}^* = {}^0\mathcal{R} - 6 ({}^0\mathcal{D}_e + \mathcal{B}_e) \mathcal{B}^e$. Similarly,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F01059	57	■ 1.000	$\{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}^{\{*\}} = \{ \}^{\{0\}} \backslash \mathrm{mathcal{R}} - 6 \backslash \mathrm{left}(\{ \}^{\{0\}} \backslash \mathrm{mathcal{D}}_{\{e\}} + \backslash \mathrm{mathcal{B}}_{\{e\}} \backslash \mathrm{right}) \backslash \mathrm{mathcal{B}}^{\{e\}}$	${}^0\mathcal{R}^* = {}^0\mathcal{R} - 6 \left({}^0\mathcal{D}_e + \mathcal{B}_e \right) \mathcal{B}^e$	$({}^0\mathcal{D}_e + 2\mathcal{B}_e)\mathcal{B}^e\delta_c^a$ and ${}^0\mathcal{R}^* = {}^0\mathcal{R} - 6({}^0\mathcal{D}_e + \mathcal{B}_e)\mathcal{B}^e$. Similarly,
1510.06699_F01060	57	■ 0.996	$\{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}^{\{* \ a \ b\} \{ \}_{c \ d}} \backslash \mathrm{rightarrow} \{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}^{\{\dagger a \ b\} \{ \}_{c \ d}}$	${}^0\mathcal{R}^{*ab}_{cd} \rightarrow {}^0\mathcal{R}^{\dagger ab}_{cd}$	placements ${}^0\mathcal{R}^{*ab}_{cd} \rightarrow {}^0\mathcal{R}^{\dagger ab}_{cd}$ and $\mathcal{B}_a \rightarrow -(\mathcal{V}_a + \frac{1}{3}\mathcal{T}_a)$.
1510.06699_F01061	57	■ 0.996	$\backslash \mathrm{mathcal{B}}_{\{a\}} \backslash \mathrm{rightarrow} - \backslash \mathrm{left}(\backslash \mathrm{mathcal{V}}_{\{a\}} + \frac{1}{3} \backslash \mathrm{mathcal{T}}_{\{a\}} \backslash \mathrm{right})$	$\mathcal{B}_a \rightarrow - \left(\mathcal{V}_a + \frac{1}{3} \mathcal{T}_a \right)$	placements ${}^0\mathcal{R}^{*ab}_{cd} \rightarrow {}^0\mathcal{R}^{\dagger ab}_{cd}$ and $\mathcal{B}_a \rightarrow -(\mathcal{V}_a + \frac{1}{3}\mathcal{T}_a)$.
1510.06699_F01062	57	■ 1.000	$\{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}^{\{*\}}$	${}^0\mathcal{R}^*$	‘reduced’ versions ${}^0\mathcal{D}_a^*$ and ${}^0\mathcal{R}^*$, respectively, as described in